MAMMA ADA CENTRE

In Collaboration with FACM ITALY

PG 2nd Year

From Data to Decisions From Ideas to Impact

GAIO

Global AI Olympiad

Think | Explore | Create | A Better Tomorrow

Fair AI Fair Future Ownership in the AI Age

PG 2nd YEAR – 6 KEY LEARNING AREAS

1. **Learning from Data**
   - Making Machines Smarter
2. **Program with AI**
   - Data into Decisions
3. **AI for Smarter Business**
   - Engineering Solutions
4. **Pick Your AI Path**
   - Portfolio to Profession
5. **AI Data Studio**
   - Smart Automation Tools
6. **Fair AI, Fair Future**
   - Ownership in the AI Age

Knowledge | Research | Innovation | Collaboration | Global Perspective | Real-World Impact

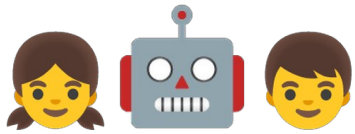
This book belongs to

Name: _____

Class & Section: _____

School: _____

📖 PREFACE



AI is transforming every profession, from engineering and medicine to commerce, law and media. As a university student, you must understand how AI works, judge its outputs critically and apply it responsibly in your discipline. This book builds exactly those capabilities.

This book has been prepared for the Global AI Olympiad (GAIO) by Mamma Ada Centre, in collaboration with FACM, Italy. It follows the GAIO six-month journey: AI Discover, AI Connect, AI Solve, AI Rise, AI Create and AI Care. Each month introduces two topics suited to the curiosity and abilities of PG 2nd Year students, with university-level depth.

Every topic opens with an in-depth "Concept Explained" chapter, followed by a Concept Deep Dive, an AI Story, a case study, a lab activity and a Python Code Lab, with worksheets, quizzes, challenge problems and university-style exam preparation.

Responsibility runs through every chapter: you will examine AI's limits, bias, errors, privacy risks and impact on work, and the ethical duties of everyone who builds or uses it.

🤝 A note for teachers and parents

Faculty can use the Concept Explained chapters for lectures, the Code Labs and lab activities for practical sessions, and the exam-prep pages for assessment. Encourage students to verify AI outputs, cite sources and reflect on ethics. An answer key is provided at the end of the book.

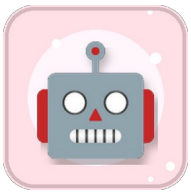
We hope this book helps you think critically, build confidently and lead responsibly in an AI-powered world. Your ideas today can change tomorrow!

With best wishes,

The GAIO Academic Team

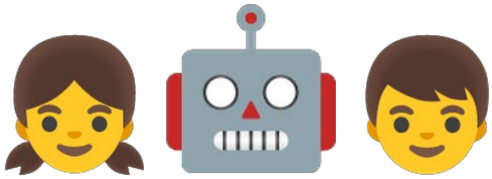
Mamma Ada Centre • In collaboration with FACM, Italy

Welcome, Innovators!



MEET BOLT!

Welcome to university-level AI! I am Bolt, your AI guide. This year you will write real Python, use AI as a coding partner, study AI in your profession and practise exam-style answers. Let's build your AI edge!



Look for these pictures in your book:

CONCEPT EXPLAINED Concept explained in depth

CONCEPT DEEP DIVE Detailed explanation & code

SCENARIO Real-life scenario

CASE STUDY Analyse a real case

LAB ACTIVITY Hands-on lab

WORKSHEET Practice exercises

PUZZLE & DEFINE Puzzles & definitions

QUIZ TIME MCQ practice

AI STORY An AI story to think about

CHALLENGE PROBLEMS Higher-order thinking

CODE LAB Python code lab

EXAM PREP University-style exam prep

My 6-Month AI Journey

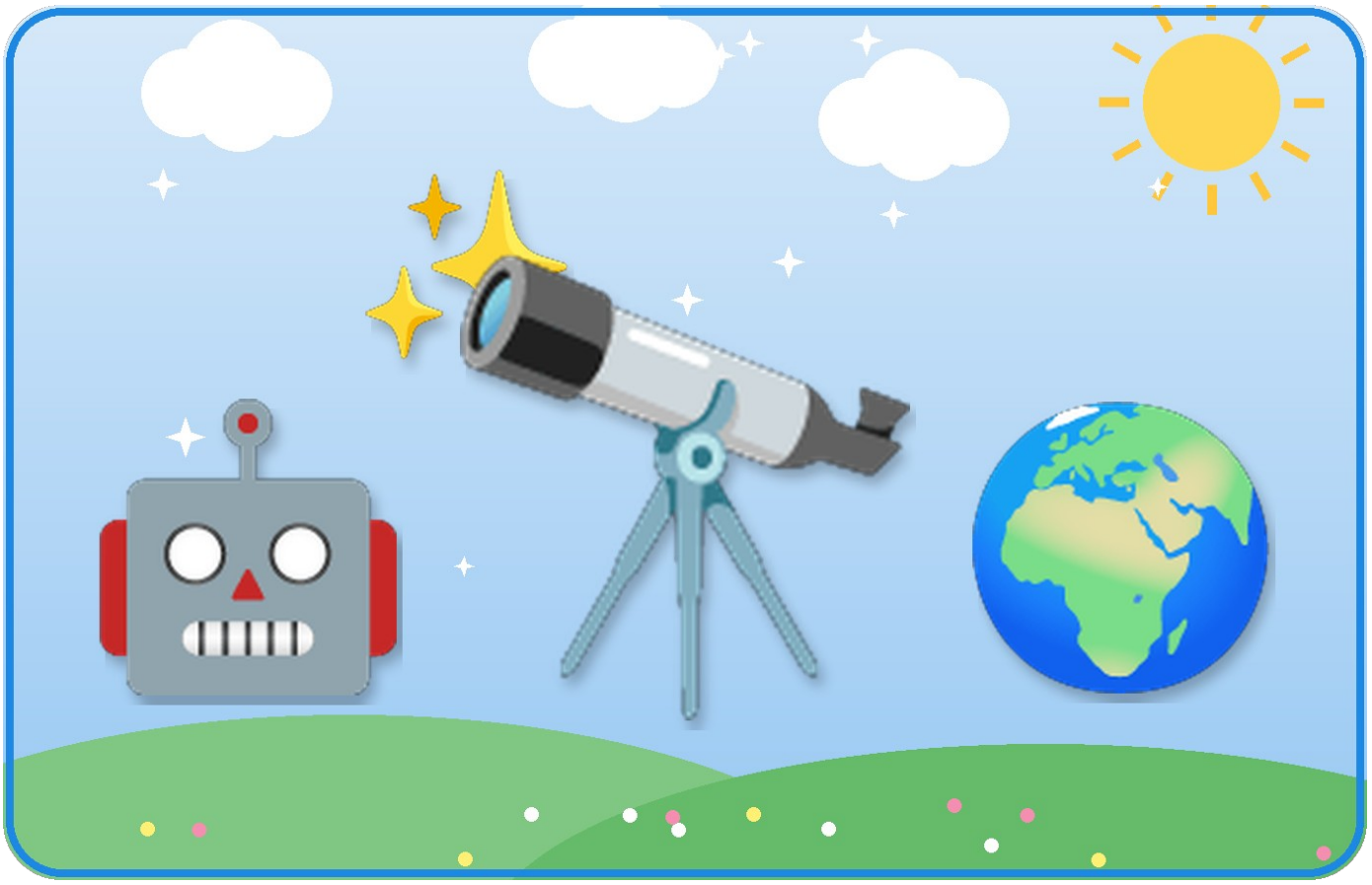
Follow the path. One month at a time!

MONTH 1 **AI DISCOVER** AI Basics	**Learning from Data • Making Machines Smarter**
MONTH 2 **AI CONNECT** Communicate with AI	**Program with AI • Data into Decisions**
MONTH 3 **AI SOLVE** AI Applications	**AI for Smarter Business • Engineering Solutions**
MONTH 4 **AI RISE** AI & Career	**Pick Your AI Path • Portfolio to Profession**
MONTH 5 **AI CREATE** AI Tools	**AI Data Studio • Smart Automation Tools**
MONTH 6 **AI CARE** Responsible AI	**Fair AI, Fair Future • Ownership in the AI Age**

MONTH 1

AI DISCOVER

See • Understand • Explore

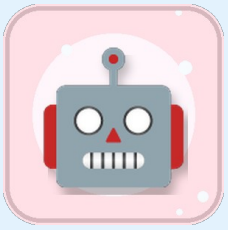


Area: AI Basics

In this unit, you will explore:

Topic 1

Learning from Data

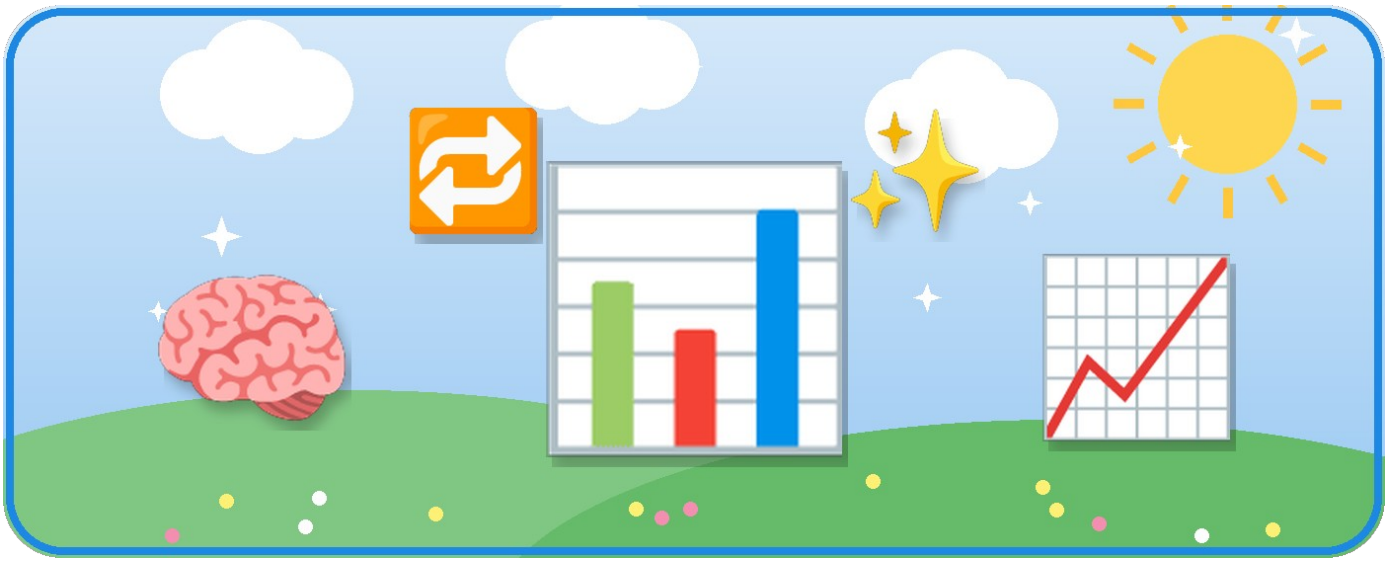


Topic 2

Making Machines Smarter

TOPIC 1 • MONTH 1 • AI BASICS

Learning from Data



LET'S LEARN

- Models learn patterns that must generalise beyond data.
- Bias–variance: too simple underfits, too complex overfits.
- Validate with held-out data and cross-validation.

Generalisation toolkit

Train/val/test

Cross-validation

Regularisation

Better data

KEY TERMS

Overfit Memorises noise

CV Rotating validation

Regularise Penalise complexity

📖 CONCEPT EXPLAINED: Learning from Data

💡 In this chapter

Every machine-learning system makes a bet: that patterns found in past data will hold for new data. Learning from data is therefore really about generalisation. At postgraduate level you need to reason precisely about why models fail on new data, how to measure that risk honestly, and which levers, from model capacity to data quality, actually reduce it.

🎯 1. Generalisation and the bias–variance trade-off

A model's expected error on new data can be split into three parts: bias, the error from wrong or overly simple assumptions; variance, the error from sensitivity to the particular training sample; and irreducible noise in the data itself. Simple models tend to have high bias and low variance; flexible models the reverse.

Learning curves make this visible. If training and validation error are both high and close together, the model underfits and needs more capacity or better features. If training error is low but validation error is much higher, it overfits and needs regularisation, simpler structure or more data. Modern very large networks can behave in surprising ways, but this diagnostic remains the first tool to reach for.

✏️ WORKED EXAMPLE

Learning-curve reading: train 0.62 / test 0.60 → underfit; train 0.85 / test 0.83 → good balance; train 0.99 / test 0.71 → overfit.

🧲 2. Regularisation and model capacity

Regularisation adds a preference for simpler solutions. L2 regularisation (ridge, weight decay) shrinks weights smoothly; L1 (lasso) pushes many weights to exactly zero, performing feature selection. In neural networks, dropout, data augmentation and early stopping act as regularisers by limiting how closely the model can fit noise.

Regularisation strength is a hyperparameter chosen on validation data, never on the test set. Plotting validation error against regularisation strength usually gives a U-shape: too little and the model overfits, too much and it underfits. The best point depends on dataset size, noise level and how different deployment data will be.

🤔 PAUSE & THINK

You have 300 rows and 200 features. Which regulariser would you try first, and what would you expect to learn from the resulting weights?

🔁 3. Validation that mirrors deployment

A validation scheme is only honest if it imitates how the model will be used. k-fold cross-validation rotates the held-out portion so every row is validated once, giving a more stable estimate on small datasets. Stratified folds keep class proportions; grouped folds keep all rows from one patient, farm or school together.

For data that unfolds over time, random splits leak the future into training; use time-based splits that always train on the past and test on the later period. When the model will be used in new regions or institutions, hold out entire regions. The final test set should be touched once, at the end, after all choices are frozen.

> ✏️ **WORKED EXAMPLE**
>
> Crop-yield model across 30 districts: GroupKFold by district answers 'how well will it work in a district we have never seen?', which is the real deployment question.

🌊 4. When the data itself moves

Most theory assumes training and future data come from the same distribution. In practice behaviour, sensors, policies and populations change. Google Flu Trends worked for several seasons, then badly overestimated flu in 2012–13 because search behaviour and the search engine itself changed, and the model had leaned on chance correlations.

Defences include preferring features with a plausible causal link to the target, monitoring input and prediction distributions after deployment, retraining on recent data, and benchmarking against simple baselines that are harder to fool. A model that beats a naive baseline only in the past is a warning sign, not a success.

> 🖐️ **TRY IT**
>
> Take a dataset with dates. Train on the first 70% of time and test on the last 30%, then repeat with a random split. Compare the scores and explain the gap.

✅ Key takeaways

- Expected error = bias² + variance + noise; capacity trades bias for variance.
- Learning curves diagnose underfitting and overfitting.
- Regularisation (L1, L2, dropout, early stopping) limits fitting to noise.
- Validation must mirror deployment: stratified, grouped or time-based splits.
- Distribution shift breaks past patterns; monitor and compare with baselines.

❓ Quick check: answer in one or two lines.

1. How would you tell underfitting from overfitting with a learning curve?

2. Why is a random split wrong for time-series data?

3. What lesson does Google Flu Trends offer about correlations?

🔬 CONCEPT DEEP DIVE: Learning from Data

🔷 Generalisation is the goal: a model is useful only if it performs well on new data from the deployment setting.

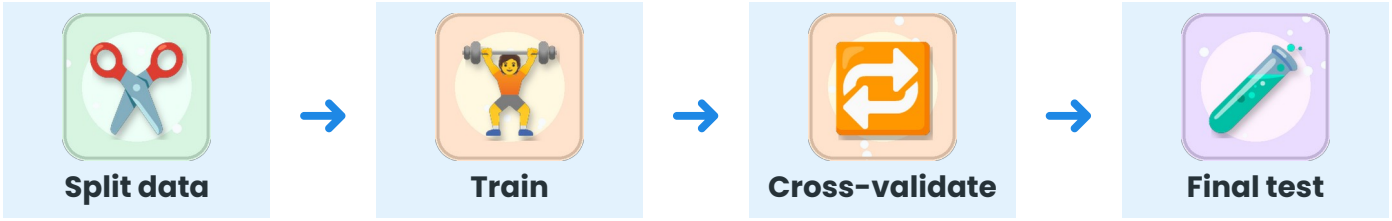
🔷 Expected error decomposes into bias², variance and irreducible noise; model capacity trades bias against variance.

🔷 Regularisation (L1, L2, dropout, early stopping) and simpler models reduce variance at a small cost in bias.

🔷 Validation must mirror deployment: stratified, grouped or time-based splits; k-fold CV for small datasets.

🔷 More and cleaner data reduces variance, but distribution shift breaks the i.i.d. assumption behind all of it.

🔄 How it works



💻 5-fold cross-validation

```
from sklearn.model_selection import cross_val_score
from sklearn.linear_model import LogisticRegression
scores = cross_val_score(LogisticRegression(), X, y, cv=5)
print(scores.mean().round(3), scores.std().round(3))
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The District That Broke the Model

Characters: *Divya (MSc Agricultural Data Science), Prof. Raman, district agriculture officer Mr. Selvam.*

SCENE 1

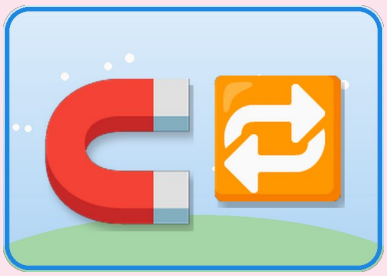
Divya's crop-yield model scored 99% on training data from 20 districts and 95% on a random test split. She proudly showed Prof. Raman, who asked one question: how would it perform in a district it had never seen?

Pause & predict: What does a random split fail to test?

SCENE 2

Mr. Selvam shared data from a coastal district. Errors tripled. The model had memorised district-specific quirks, such as one block's unusual fertiliser records, instead of learning general weather–yield relationships.

Pause & predict: Is this mainly bias, variance or shift?



SCENE 3

Divya switched to grouped cross-validation by district, added L2 regularisation and removed features that only identified places. Training accuracy fell to 86%, but held-out district accuracy rose to 81%.

Pause & predict: Why is lower training accuracy acceptable here?

SCENE 4

She reported results per district with a naive baseline for comparison. Mr. Selvam piloted the model in two new districts with monthly monitoring. Her thesis title changed to include 'generalisable'.

⏸ **Pause & predict:** Why include a naive baseline?

🌟 The big idea

A model is judged by how it performs where it will be used, not where it was trained.

💬 Discuss with your classmates

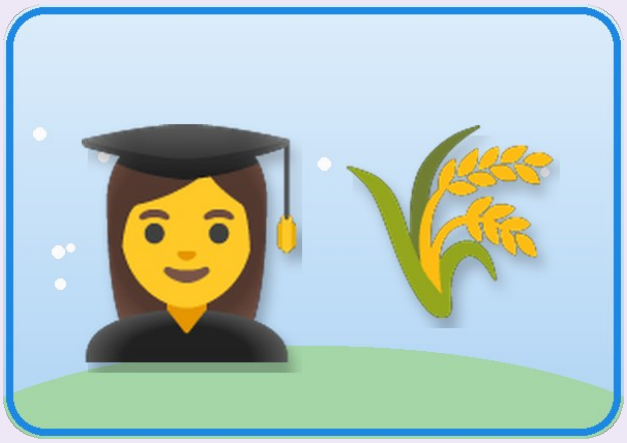
1. What 'new district' exists for models in your field?

2. How would you monitor a model after deployment?

SCENARIO: Learning from Data

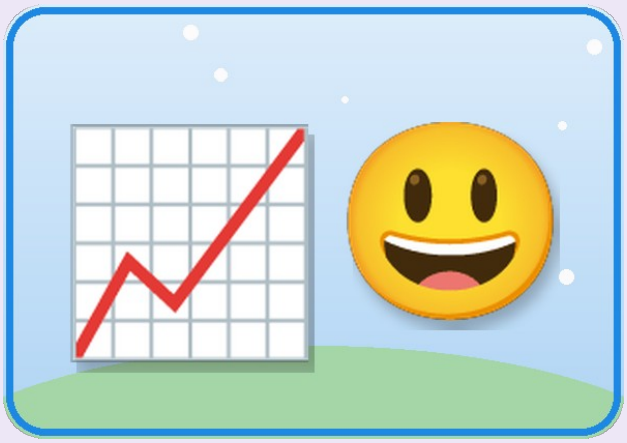
Read the scenario. Think about the choices made at each step.

1



Divya predicts crop yield for her PG project.

2



Her deep model fits training data at 99%.

3

On a new district's data, error triples.

4

Regularisation and CV give a model that generalises.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Learning from data at work

Demand forecasts

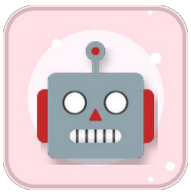
Credit scoring

Disease risk

Crop yield

Spam filters

Traffic prediction



BOLT SAYS:

I learned from yesterday's data. Test me on tomorrow's before you trust me!

FUN FACT!

Google Flu Trends, launched in 2008, overestimated US flu by over double in early 2013.

👀 **Reflect: where have YOU seen this in real life? Note it down.**

📁 CASE STUDY: The Parable of Google Flu Trends

CASE FILE

Google Flu Trends, launched in 2008, estimated flu activity from search queries. In the 2012–13 season it predicted more than double the doctor visits for flu-like illness reported by the US CDC. A 2014 Science paper by Lazer and colleagues called this 'big data hubris': the model relied on search terms linked to flu by chance, while search behaviour and Google's algorithms changed. Estimates stopped being published in 2015.

🔍 Analyse the case

1. Why did correlations with search terms fail over time?

2. Is this mainly bias, variance or distribution shift?

3. How could the model have been validated better?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Polynomial Pitfall

You need:

Graph paper

Pencils

Pairs

Steps:

1.
   Plot 8 noisy points from a straight trend.
2.
   Fit a straight line by eye.
3.
   Draw a wiggly curve through every point.
4.

   Add 4 new points; which fits better?

THINK & TALK

Why does the wiggly curve fail on new data?

Record your observations & results:

💻 CODE LAB: Bias–Variance Check

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
train = [0.62, 0.85, 0.99]   # degree 1, 3, 9
test  = [0.60, 0.83, 0.71]
gaps = [round(a - b, 2) for a, b in zip(train, ____)]
print(gaps)
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. What is printed?**

INTERPRET **T3. Which degree is best?**

EXTEND **T4. Plot train and test vs degree.**

WORKSHEET: Learning from Data (Part A)

Name: _____ Date: _____

1. Match the idea to its meaning.

Underfitting ●

● **High variance on new data**

Overfitting ●

● **Penalise complexity**

Regularisation ●

● **High bias, misses pattern**

Cross-validation ●

● **Rotate held-out folds**

2. Underfit or overfit? Tick ✓.

	Underfit	Overfit
Train 60%, test 58%		
Train 99%, test 70%		
Straight line on curved data		
Deep tree on 50 rows		

WORKSHEET: Learning from Data (Part B)

Name: _____ Date: _____

1. Cross-validation arithmetic. Use the word box.

WORD BOX: 200 ✿ 800 ✿ 5

1,000 rows, 5 folds → ____ per fold

Training rows each round = ____

Number of models trained = ____

2. Order a sound modelling workflow. Write 1–4.

Final test, once

☐

Split train/val/test

☐

Tune with CV

☐

Train a baseline

☐

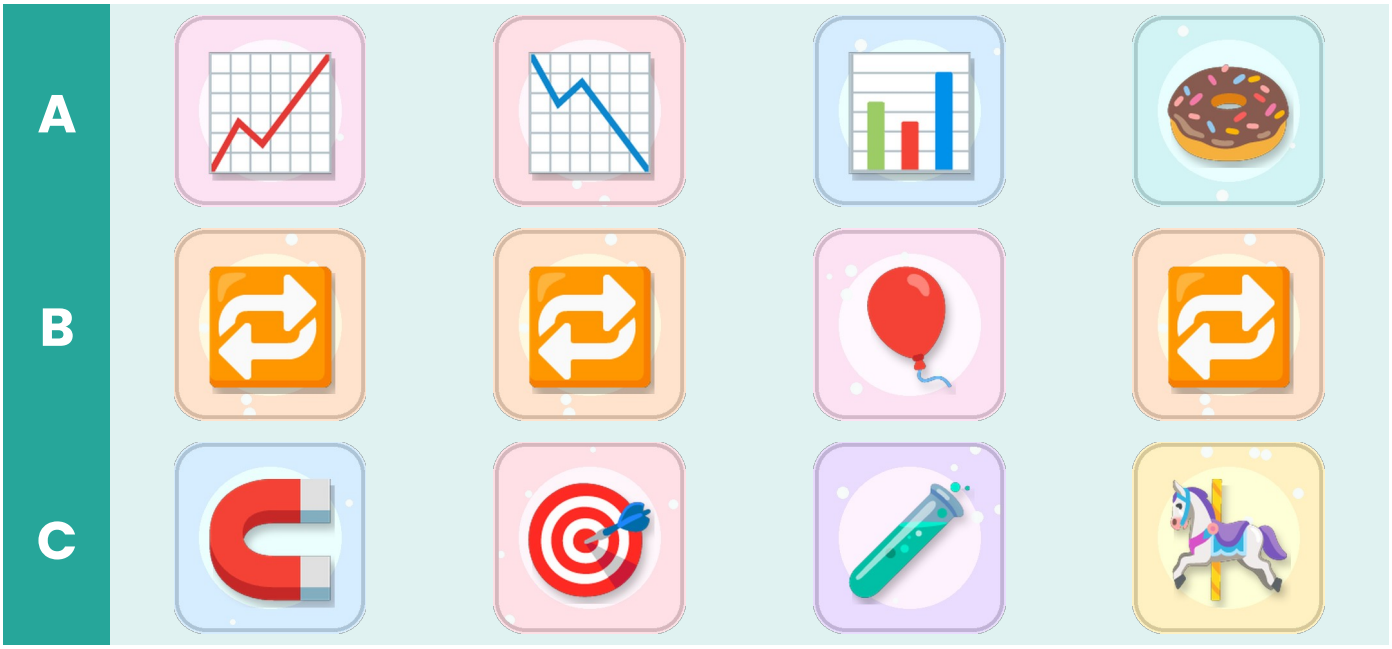
3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

PUZZLE & DEFINE: Learning from Data

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.



2. Define it! Write each term in your own words with an example.

Overfit

Meaning: _____

Example: _____

CV

Meaning: _____

Example: _____

Regularise

Meaning: _____

Example: _____

? QUIZ TIME: Learning from Data

Circle the right answer!

Q1. Train 99%, test 70% suggests...

a) Overfitting

b) Underfitting

c) Perfect fit

Q2. k-fold CV rotates the...

a) Colours

b) Validation fold

c) Volume

Q3. L2 regularisation penalises...

a) Seeds

b) Data size

c) Large weights

Q4. A simple model on complex data...

a) Underfits

b) Overfits

c) Is perfect

Q5. Google Flu Trends overestimated...

a) Rain

b) Flu levels

c) Traffic

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Learning from Data

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
📈	High train and low test accuracy suggests overfitting.		
🧲	Regularisation makes models more complex.		
🔁	In k-fold CV each row is validated once.		
⏳	Random splits are always fine for time series.		
📚	More data usually reduces variance.		
🤧	Google Flu Trends overestimated flu in 2012–13.		

HOME CONNECT

Find a news claim based on 'data shows'. Ask: what data, what test, and would it generalise elsewhere?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Learning from Data

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why does a very flexible model overfit small data?**

COMPARE **Q2. L1 vs L2 regularisation.**

APPLY **Q3. 5-fold CV on 1,000 rows: rows per fold?**

EVALUATE **Q4. Random split for time-series sales data?**

CREATE **Q5. Validation plan for a multi-city dataset.**

📝 EXAM PREP: Learning from Data

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Define overfitting.

2. State the bias–variance trade-off.

3. Why stratify folds?

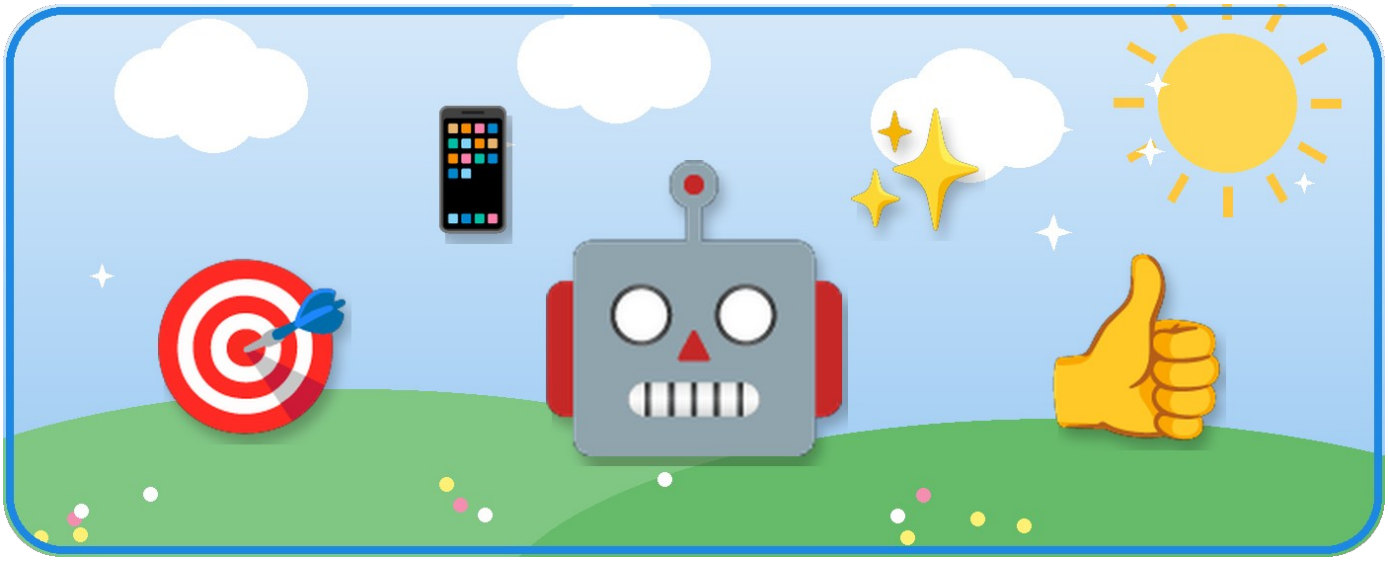
4. Name three regularisers.

PART B • Long answer (5 marks)

5. Explain generalisation and how you would validate a model for use in a new region.

TOPIC 2 • MONTH 1 • AI BASICS

Making Machines Smarter



LET'S LEARN

- Smarter models need better data, signals and feedback.
- Transfer, augmentation and active learning save labels.
- Feedback tunes models; distillation makes them small.

Ways to make models smarter

Transfer learning

Augmentation

Active learning

Human feedback

KEY TERMS

Active learn
Label the uncertain

RLHF Learning from preferences

Distillation Big teaches small

📖 CONCEPT EXPLAINED: Making Machines Smarter

💡 In this chapter

Once a baseline model exists, the question becomes how to make it better without unlimited labels, compute or money. Postgraduate practitioners improve models by reusing knowledge, adding variety to data, choosing which examples humans should label, learning from human preferences and compressing models so they run where they are needed.

🔄 1. Transfer learning and fine-tuning

Transfer learning starts from a model pre-trained on a large, general dataset and adapts it to a specific task. Early layers capture general features, such as edges in images or grammar in text, so only later layers need substantial task-specific learning. This often turns a problem needing tens of thousands of labels into one solvable with hundreds.

Options range from training only a new classification head, to fine-tuning all layers with a small learning rate, to parameter-efficient methods such as adapters or LoRA that train a few extra parameters. The closer the source domain is to yours, the more transfer helps; a model pre-trained on English news may need more adaptation for Tamil medical notes.

✏️ WORKED EXAMPLE

X-ray classifier with 300 labelled images: an ImageNet-pre-trained backbone plus a new head reaches far higher accuracy than training from scratch, which overfits almost immediately.

🎨 2. Augmentation and active learning

Augmentation creates new training examples by transforming existing ones: cropping, rotating, colour shifts and noise for images; back-translation or synonym swaps for text; speed and pitch changes for audio. The rule is that transformations must not change the correct label. Flipping a cat photo is fine; flipping a handwritten 6 may turn it into a 9.

Active learning chooses which unlabelled examples a human should label next. Uncertainty sampling picks examples where the model's predicted probability is closest to 50%; diversity sampling picks examples unlike those already labelled. Repeating train, select, label and retrain can reach target accuracy with a fraction of the labelling budget.

🤔 **PAUSE & THINK**

Your model is very confident but wrong on a group of examples it has never seen. Would uncertainty sampling find them? What other selection strategy might?

👍 3. Learning from human preferences

For generative models, 'correct' is hard to define, but people can often say which of two answers is better. Reinforcement learning from human feedback collects such comparisons, trains a reward model to predict them, and then fine-tunes the generator to score highly. Direct preference optimisation (DPO) achieves a similar effect without a separate reward model.

OpenAI's 2022 InstructGPT work showed that raters preferred a 1.3-billion-parameter model tuned this way over the 175-billion-parameter GPT-3. Preference tuning has limits: models can learn to please raters rather than be correct, and the preferences encoded reflect who the raters are and the instructions they were given.

✏️ **WORKED EXAMPLE**

Preference data row: prompt 'Explain inflation to a farmer'; answer A uses local prices and simple words; answer B uses jargon. Rater chooses A. Thousands of such choices shape the model's style.

📱 4. Making models small and fast

Many applications need models that run on phones, laptops or cheap servers. Knowledge distillation trains a small student model to match a large teacher's output probabilities, which carry richer information than hard labels. Pruning removes weights or whole channels that contribute little; quantisation stores weights in 8 or 4 bits instead of 32.

Compression trades a little accuracy for large savings in memory, latency and energy. Always measure the trade-off on your real task and hardware, including subgroups, because a small overall accuracy drop can hide a large drop for rare but important cases.

🖐️ **TRY IT**

Train or download a small text classifier. Quantise it to 8 bits, then compare size, inference time and accuracy on 200 test examples, including a subgroup you care about.

✅ Key takeaways

- Transfer learning reuses pre-trained features; fine-tune with care.
- Augmentation adds label-preserving variety; active learning picks what to label.
- Preference tuning (RLHF, DPO) shapes generative behaviour from human comparisons.
- Distillation, pruning and quantisation make models smaller and faster.
- Measure every improvement on the real task, hardware and subgroups.

❓ Quick check: answer in one or two lines.

1. Why does transfer learning reduce labelling needs?

2. Give one augmentation that would wrongly change a label.

3. What can go wrong when models are tuned to human preferences?

🔬 CONCEPT DEEP DIVE: Making Machines Smarter

🔷 Transfer learning starts from a pre-trained model, so only the task-specific layers need much new data.

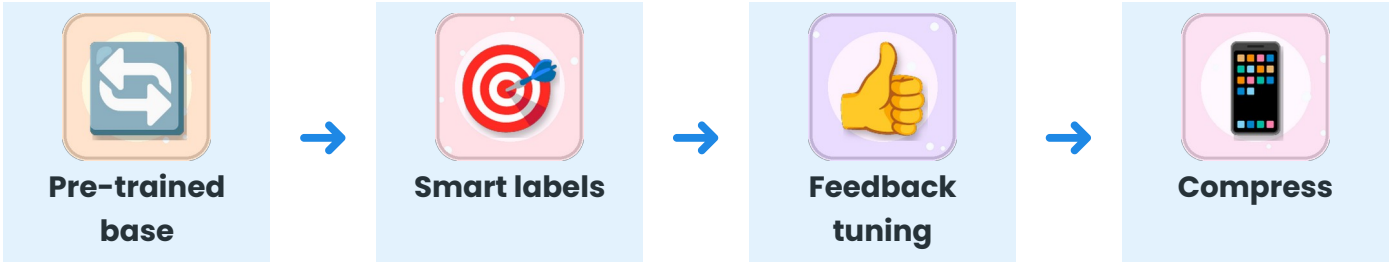
🔷 Augmentation and synthetic data add variety; transformations must preserve the label.

🔷 Active learning asks humans to label the examples the model is least sure about, cutting labelling cost.

🔷 Preference tuning (RLHF, DPO) uses human rankings of outputs to shape helpful, safe behaviour.

🔷 Distillation, pruning and quantisation compress models to run faster and cheaper on small devices.

🔄 How it works



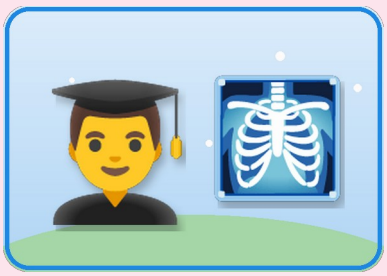
💻 Uncertainty sampling

```
probs = {'img1': 0.51, 'img2': 0.97, 'img3': 0.45, 'img4': 0.08}
unc = {k: 1 - abs(p - 0.5) * 2 for k, p in probs.items()}
ask = sorted(unc, key=unc.get, reverse=True)[:2]
print(ask)   # ['img1', 'img3']
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Three Hundred Labels

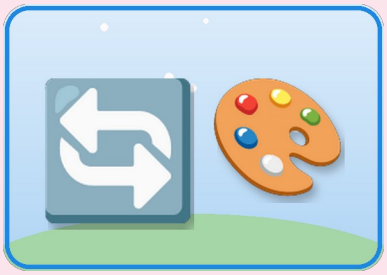
Characters: *Karan (MTech Biomedical AI), radiologist Dr. Nirmala, labmate Joel.*



SCENE 1

A rural clinic asked Karan for a chest X-ray triage tool. Dr. Nirmala could label only 300 images in her spare time. Training from scratch, his model overfitted badly and reached just 62% accuracy.

Pause & predict: Why does training from scratch fail with 300 labels?



SCENE 2

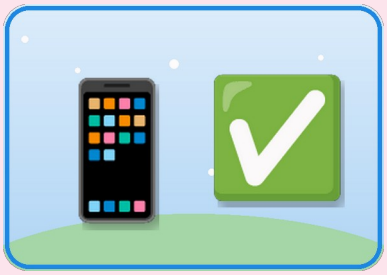
Joel suggested a pre-trained backbone and careful augmentation: small rotations and brightness changes, but no horizontal flips that would move the heart to the wrong side. Accuracy rose to 81%.

Pause & predict: Why avoid horizontal flips here?

SCENE 3

Instead of random images, Karan sent Dr. Nirmala the 50 cases his model was least sure about each week. After four rounds, accuracy reached 88% with far fewer extra labels than planned.

Pause & predict: Why do uncertain examples help most?



SCENE 4

The clinic's laptop was slow, so Karan distilled the model into a smaller student and quantised it. It ran in under a second with 86% accuracy, and every flagged case still went to a doctor.

⏸ **Pause & predict:** Was the 2-point drop worth it? Why?

🌟 The big idea

Smart use of data, labels and compression can make AI work where resources are scarce.

💬 Discuss with your classmates

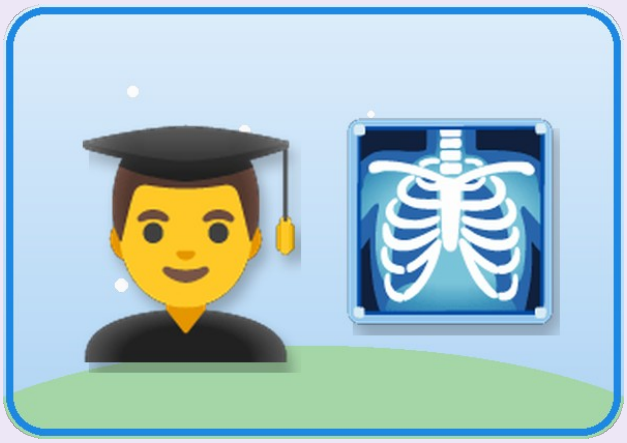
1. Where are labels scarce in your field?

2. What accuracy loss would you accept for speed?

SCENARIO: Making Machines Smarter

Read the scenario. Think about the choices made at each step.

1



Karan builds an X-ray classifier for a rural clinic.

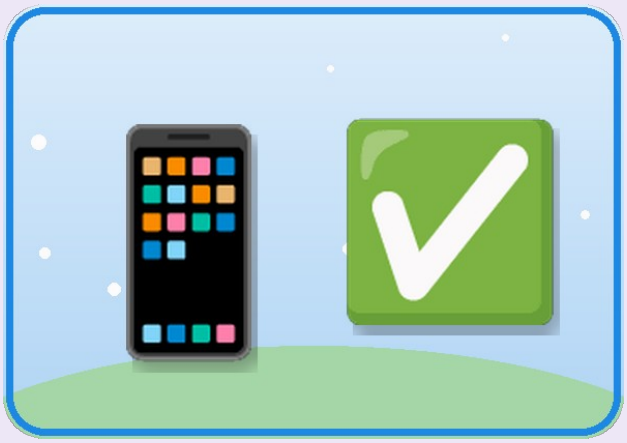
2

Radiologists can label only 300 images.

3

He uses transfer learning and active learning.

4



A distilled model runs on the clinic's laptop.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Smarter models in action

Chat assistants

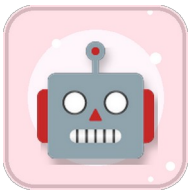
On-device models

Medical imaging

Voice typing

Recommendations

Translation



BOLT SAYS:

Show me fewer but smarter examples, and tell me which answers you prefer!

FUN FACT!

In 2022, people preferred a 1.3B InstructGPT model's answers to 175B GPT-3's.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Small Model, Preferred Answers

CASE FILE

In 2022, OpenAI fine-tuned GPT-3 with reinforcement learning from human feedback. Labellers wrote example answers and ranked model outputs; a reward model learned these preferences and guided training. Raters preferred answers from a 1.3-billion-parameter InstructGPT model over the 175-billion-parameter GPT-3, despite it being over 100 times smaller. The method shaped later chat assistants, and raised the question of whose preferences are encoded.

🔍 Analyse the case

1. Why could a much smaller model be preferred?

2. What do labellers' rankings teach the model?

3. Whose preferences should guide AI behaviour?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Teach the Classifier

You need:

Picture cards

Labels

Teams of 3

Steps:

1		One student sorts cards by a hidden rule.
2		Learner asks labels only for confusing cards.
3		Count labels needed to guess the rule.
4		Repeat with random cards; compare counts.

THINK & TALK

Why do uncertain examples teach faster?

Record your observations & results:

💻 CODE LAB: Uncertainty Sampler

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
 1  probs = [0.51, 0.97, 0.45, 0.08, 0.62]
 2  unc = [1 - abs(p - 0.5) * 2 for p in probs]
 3  top = sorted(range(len(probs)), key=lambda i: unc[i],
reverse=____)[:2]
 4  print(top)
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. Which indices?**

EXPLAIN **T3. Why these examples?**

EXTEND **T4. Add a label-retrain loop.**

WORKSHEET: Making Machines Smarter (Part A)

Name: _____ Date: _____

1. Match the technique to its idea.

Transfer learning ●

Augmentation ●

Active learning ●

Distillation ●

● Query the most useful labels

● Small model imitates large

● Reuse pre-trained features

● Create varied copies of data

2. Data fix or model fix? Tick ✓.

	Data fix	Model fix
Add rotated images		
Use a pre-trained backbone		
Relabel noisy examples		
Prune unneeded weights		

✏️ WORKSHEET: Making Machines Smarter (Part B)

Name: _____ Date: _____

1. Compression arithmetic. Use the word box.

WORD BOX: **4** ✿ **25** ✿ **2**

32-bit to 8-bit weights → ____× smaller

100 MB model ÷ 4 = ____ MB

Accuracy 88% → 86%: drop of ____ points

2. Circle label-efficient methods.

Transfer learning

Ignore errors

Active learning

Random relabelling

Augmentation

Delete hard cases

PUZZLE & DEFINE: Making Machines Smarter

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Active learn

Meaning: _____

Example: _____

RLHF

Meaning: _____

Example: _____

Distillation

Meaning: _____

Example: _____

? QUIZ TIME: Making Machines Smarter

Circle the right answer!

Q1. Reusing a pre-trained model is...

a) Transfer

b) Hard coding

c) Guessing

Q2. Flipping and cropping images is...

a) Pruning

b) Augmentation

c) Hashing

Q3. Labelling only uncertain cases:

a) Batching

b) Caching

c) Active learning

Q4. RLHF learns from human...

a) Preferences

b) Weather

c) Music

Q5. Small model copying a big one:

a) Scaling

b) Distillation

c) Mining

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Making Machines Smarter

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
Transfer learning reuses learned features.		
Augmentation always needs new human labels.		
Active learning picks informative examples.		
RLHF uses human rankings of outputs.		
Distillation makes a bigger model.		
A 1.3B InstructGPT beat 175B GPT-3 with raters.		

HOME CONNECT

Use voice typing on a phone. Note its errors. Which technique from this topic could fix each one?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Making Machines Smarter

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why does transfer learning need fewer labels?**

COMPARE **Q2. Augmentation vs synthetic data.**

APPLY **Q3. Teacher 90%, student 87% and 10× smaller: worth it?**

EVALUATE **Q4. Can augmentation hurt?**

CREATE **Q5. Active learning loop for spam labels.**

📝 EXAM PREP: Making Machines Smarter

Name: _____ Date: _____

⏱️ **Time: 30 min Max marks: 13 Answer all questions.**

PART A • Short answers (2 marks each)

1. Define transfer learning.

2. What is active learning?

3. Explain RLHF briefly.

4. Name two compression methods.

PART B • Long answer (5 marks)

5. Design a strategy to build an accurate model with only a few hundred labels and deploy it on low-cost hardware.

UNIT REVIEW: AI DISCOVER

Name: _____ Date: _____

Circle the right answer!

Q1. Good train, poor test means...

a) Overfitting

b) Underfitting

c) Success

Q2. Penalising large weights is...

a) Sampling

b) Regularisation

c) Batching

Q3. Google Flu Trends showed the risk of...

a) Bad colours

b) Slow servers

c) Shifting data

Q4. Reusing a pre-trained model:

a) Transfer

b) Deletion

c) Guessing

Q5. Labels for uncertain cases only:

a) Augment

b) Active learning

c) Distil

Q6. InstructGPT was tuned with...

a) Music

b) Weather

c) Human feedback

WORD SEARCH: AI DISCOVER

Find these words. They go across → or down ↓.

OVERFIT

CV

REGULARISE

RLHF

VARIANCE

BIAS

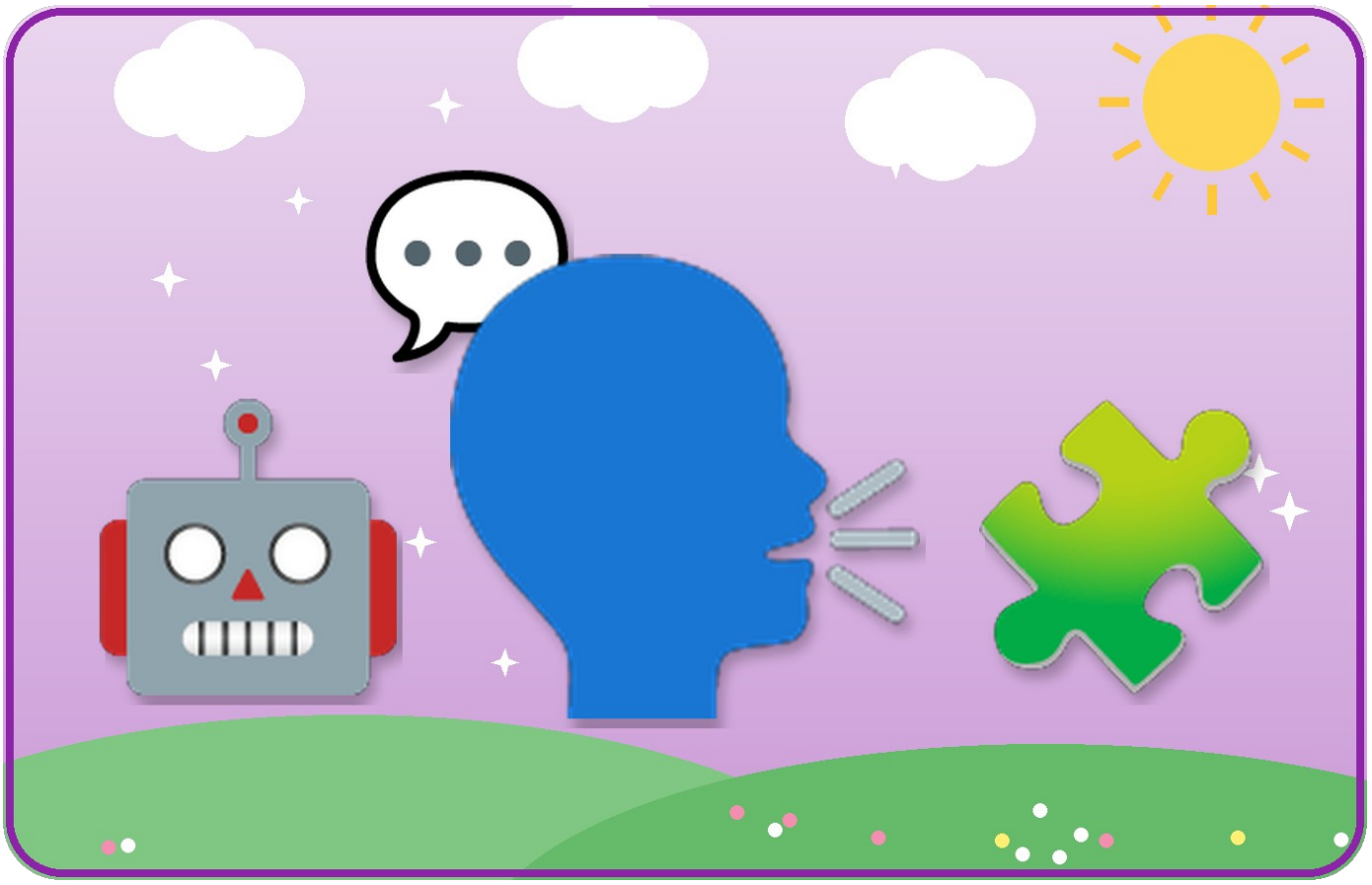
P	I	T	S	I	G	N	A	L	W
R	E	G	U	L	A	R	I	S	E
A	N	S	E	S	T	G	O	R	B
O	V	E	R	F	I	T	R	L	M
U	T	G	S	O	R	M	O	H	M
A	T	T	W	W	C	V	L	F	P
W	F	E	E	D	B	A	C	K	A
H	V	A	R	I	A	N	C	E	F
T	E	A	C	H	E	R	T	U	F
C	T	I	B	B	I	A	S	C	C

I finished AI DISCOVER! Teacher's sign: _____

MONTH 2

AI CONNECT

Think • Instruct • Code



Area: Communicate with AI

In this unit, you will explore:

Topic 3

Program with AI

Topic 4

Data into Decisions

TOPIC 3 • MONTH 2 • COMMUNICATE WITH AI

Program with AI

LET'S LEARN

- AI assistants draft, explain, test and refactor code.
- Developers stay responsible for correctness and security.
- Review, test and understand every AI-written line.

AI in the coding workflow

Generate code

Explain code

Write tests

Refactor

KEY TERMS

Unit test Checks one function

Code review Peer checks code

Vulnerability Exploitable flaw

📖 CONCEPT EXPLAINED: Program with AI

💡 In this chapter

AI coding assistants have become part of everyday software development, from autocompletion to agents that edit files, run tests and propose pull requests. For postgraduates who build AI systems, the question is not whether to use them but how to use them so that speed gains do not come at the cost of correctness, security and understanding.

🤖 1. How code assistants work

Code assistants are large language models trained on vast amounts of public code and text. In completion mode they predict the next lines from the surrounding file; in chat mode they answer questions and generate functions; in agent mode they plan steps, edit multiple files, run commands and iterate on test results.

Because they predict plausible code rather than verify it, they can produce code that looks right but contains subtle errors, uses outdated APIs or references packages that do not exist. Their quality depends heavily on context: the relevant files, interfaces, constraints and examples you provide.

✏️ WORKED EXAMPLE

Prompt with context: 'Python 3.11, pandas 2. Write clean_prices(df) that removes duplicate rows, fills missing price with the median, drops negative prices and returns a new DataFrame. Include pytest tests for empty input and all-missing prices.'

⏱️ 2. Evidence on productivity

A controlled experiment published in 2023 found developers using GitHub Copilot completed a JavaScript server task about 55.8% faster than a control group. Other studies report gains on routine tasks such as boilerplate, tests and documentation, with smaller or even negative gains on complex work in large, unfamiliar codebases.

Productivity should be measured as working, maintained software, not lines produced. Teams track cycle time, defect rates, review effort and incidents before and after adoption. Learners must also protect their own skill growth: relying on AI for everything can leave gaps that appear during debugging or interviews.

🤔 **PAUSE & THINK**

If an assistant makes you 50% faster at writing code but doubles review time for your teammates, is the team more productive? How would you measure it?

🔐 3. Security and correctness risks

A 2021 study by Pearce and colleagues found that about 40% of programs Copilot generated for security-relevant scenarios contained vulnerabilities, such as SQL injection or weak cryptography. Assistants can also reproduce insecure patterns from training data, leak secrets pasted into prompts, or suggest non-existent packages that attackers then register with malicious code.

Mitigations are standard good engineering, applied consistently: never paste secrets into external tools, pin and verify dependencies, run static analysis and dependency scanners, require code review for all changes, and treat agent-run shell commands with the same caution as code from an unknown contributor.

✏️ **WORKED EXAMPLE**

Review catch: generated login code compared passwords in plain text. Fix: store salted hashes with a vetted library such as bcrypt, and add a test that the database never contains raw passwords.

🧪 4. A test-first workflow

The most reliable pattern is to define correctness before generating code. Write or agree the specification and tests first, including edge cases such as empty inputs, zero, negative numbers, very large values and invalid types. Then let the assistant generate an implementation and run the tests and scanners automatically.

Work in small steps that a human can review in minutes. Ask the assistant to explain unfamiliar code, and rewrite anything you cannot explain yourself. In assignments and research code, disclose AI assistance according to your institution's rules, and keep ownership of design decisions.

🖐️ **TRY IT**

Write five pytest tests for a function you need, including two edge cases. Ask an assistant for the implementation, run the tests, and record how many attempts it took to pass.

✅ Key takeaways

- Assistants predict plausible code; context and constraints improve quality.
- Speed gains are real on routine tasks; measure working software, not lines.
- Risks: vulnerabilities, logic errors, fake packages, leaked secrets.
- Mitigate with reviews, scanners, pinned dependencies and no secrets in prompts.
- Test-first, small steps and understanding every line keep you in control.

❓ Quick check: answer in one or two lines.

1. Why can an assistant suggest a package that does not exist?

2. Name three engineering safeguards for AI-generated code.

3. Describe a test-first workflow with an AI assistant.

🔬 CONCEPT DEEP DIVE: Program with AI

🔷 Code assistants are language models trained on code; they complete lines, chat about code or act as agents that edit and run it.

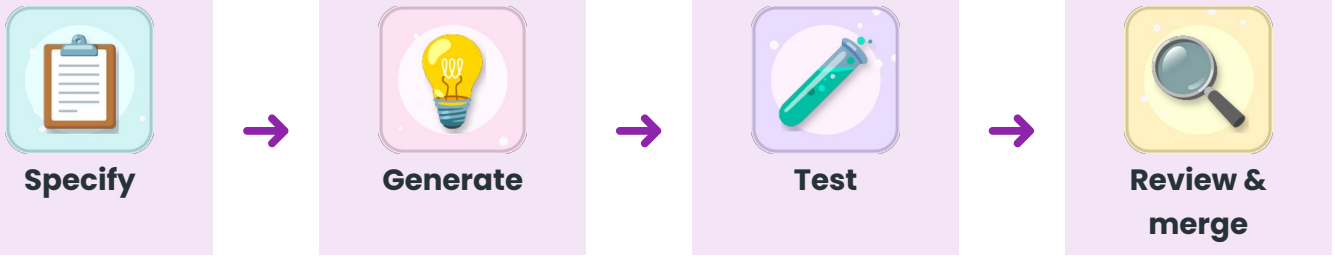
🔷 Studies show real speed gains on well-specified tasks; gains are smaller on unfamiliar, complex codebases.

🔷 Risks: insecure patterns, subtle logic errors, invented packages that attackers can register, and leaked secrets.

🔷 Test-first workflows work well: write the spec and tests, generate code, run tests and security scans, then review.

🔷 Good prompts give context, constraints, examples and expected tests; iterate in small, reviewable steps.

🔄 How it works



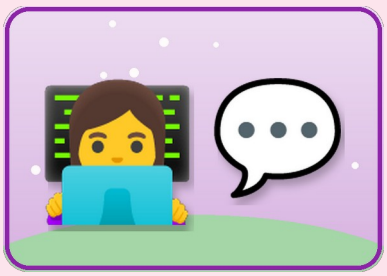
💻 Tests with edge cases

```
def is_even(n):
    return n % 2 == 0
assert is_even(4) and not is_even(7)
assert is_even(0) and is_even(-2)   # edge cases
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Login That Leaked

Characters: *Asha (MCA), team lead Mr. Prakash, security intern Zoya.*



SCENE 1

Asha needed a login API by Friday. She asked an AI assistant, and complete code appeared in seconds. It ran, the demo worked, and she opened a pull request on Thursday evening.

Pause & predict: What should happen before merging?

SCENE 2

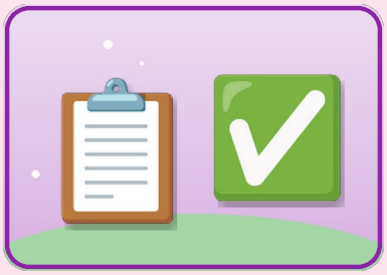
Zoya's security scan flagged two problems: passwords were stored in plain text, and user input went straight into an SQL query. The assistant had copied common but unsafe patterns.

Pause & predict: What could an attacker do with each flaw?

SCENE 3

Mr. Prakash asked Asha to write tests first: no raw passwords in the database, and injection attempts rejected. She regenerated the code with these constraints, added bcrypt hashing and parameterised queries.

Pause & predict: How do tests guide the assistant?



SCENE 4

The team adopted a rule: AI code welcome, but every pull request needs tests, a scan and a human reviewer who can explain it. Their release slipped one day but shipped safely.

⏸ **Pause & predict:** Is one day's delay worth it? Why?

🌟 The big idea

AI makes code fast; tests, scans and reviews make it safe.

💬 Discuss with your classmates

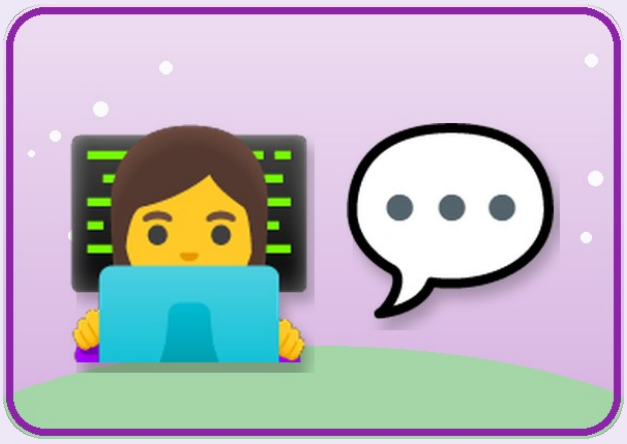
1. What rules should your team set for AI-written code?

2. How would you spot code you don't truly understand?

SCENARIO: Program with AI

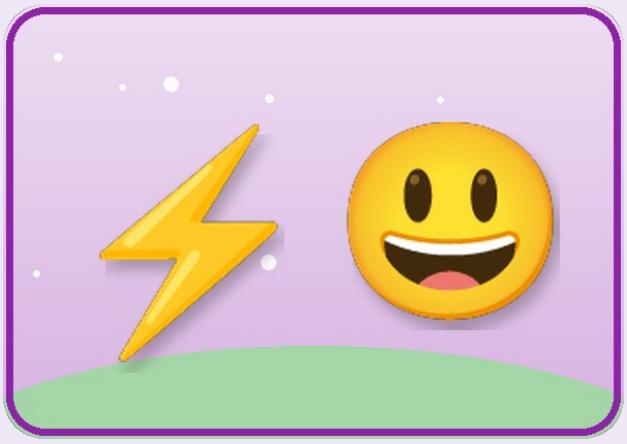
Read the scenario. Think about the choices made at each step.

1



Asha asks an AI to write a login API.

2



Working code appears in seconds.

3

Review finds passwords stored in plain text.

4

She adds hashing and tests before merging.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Where AI helps programmers

Boilerplate

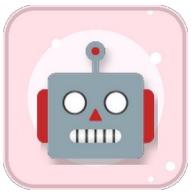
Debugging

Documentation

Test cases

Code migration

SQL queries



BOLT SAYS:

I can write code fast, but I can also write bugs fast. Please test me!

FUN FACT!

In a 2023 study, developers using Copilot finished a coding task about 55% faster.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Faster, Not Automatically Safer

CASE FILE

In a 2023 controlled experiment by Microsoft Research, GitHub and MIT researchers, developers using GitHub Copilot finished a JavaScript server task about 55.8% faster than a control group. Yet a 2021 study, 'Asleep at the Keyboard?', found that roughly 40% of Copilot programs written for security-relevant scenarios had vulnerabilities. Speed and safety are different outcomes: only review, testing and security tools make code trustworthy.

🔍 Analyse the case

1. Why might faster code still be riskier?

2. Which checks would you require before merging?

3. How should a team measure AI's net benefit?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Human Linter

You need:

Printed code

Highlighters

Pairs

Steps:

1		Get a short AI-written function on paper.
2		Mark bugs, unclear names and risky lines.
3		Write three test inputs, one an edge case.
4		Swap and compare with your partner.

THINK & TALK

What did the AI get wrong most often?

Record your observations & results:

💻 CODE LAB: Edge Case Tester

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  def safe_div(a, b):
2      return a / b if b != ____ else None
3  for a, b in [(10, 2), (5, 0), (-9, 3)]:
4      print(safe_div(a, b))
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. Three outputs?**

EXPLAIN **T3. Why return None?**

EXTEND **T4. Write assert tests.**

WORKSHEET: Program with AI (Part A)

Name: _____ Date: _____

1. Match the practice to its purpose.

Unit test ●

● **Peer reads changes**

Linting ●

● **Checks one function**

Code review ●

● **Random inputs find crashes**

Fuzzing ●

● **Flags style and errors**

2. Safe or risky? Tick ✓.

	Safe	Risky
Merge code you have tested		
Paste API keys into prompts		
Use a vetted library		
Run unknown generated commands		

WORKSHEET: Program with AI (Part B)

Name: _____ Date: _____

1. Coding maths. Use the word box.

WORD BOX: 30 ✿ 70 ✿ 6

Task 100 min, AI saves 30 min → ____% saved

Time still needed = ____ min

200 lines at 3 bugs per 100 → ____ bugs

2. Order safe AI-assisted coding. Write 1–4.

Merge after review ☐

Write spec & tests ☐

Run tests & scans ☐

Generate code ☐

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

PUZZLE & DEFINE: Program with AI

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Unit test

Meaning: _____

Example: _____

Code review

Meaning: _____

Example: _____

Vulnerability

Meaning: _____

Example: _____

? QUIZ TIME: Program with AI

Circle the right answer!

Q1. AI-generated code must be...

a) Tested

b) Blindly trusted

c) Deleted

Q2. Tests for one function are...

a) Integration

b) Unit tests

c) Mockups

Q3. A made-up package name risks...

a) Music

b) Speed

c) Malware

Q4. A coding prompt should include...

a) Context & spec

b) Emojis

c) Random words

Q5. Developers stay responsible for...

a) Logos

b) Correctness

c) Coffee

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Program with AI

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	AI-written code can contain security flaws.		
	Passing one test proves code is correct.		
	Edge cases include empty and negative inputs.		
	Secrets must never be pasted into public tools.		
	AI assistants never invent package names.		
	The Copilot study found about 55% faster completion.		

HOME CONNECT

Ask an AI to write a function you know well. Write five tests, including edge cases. How many pass?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Program with AI

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why do assistants sometimes invent package names?**

COMPARE **Q2. AI completion vs AI coding agent.**

APPLY **Q3. Buggy PRs fall from 12% to 9% of 100: change?**

EVALUATE **Q4. Should students use AI for programming work?**

CREATE **Q5. Spec for an AI-written CSV-cleaning function.**

📝 EXAM PREP: Program with AI

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What is an AI code assistant?

2. Name two risks of AI-written code.

3. What is an edge case?

4. Why write tests before generating code?

PART B • Long answer (5 marks)

5. Design a team policy for safe and productive use of AI coding assistants.

TOPIC 4 • MONTH 2 • COMMUNICATE WITH AI

Data into Decisions

LET'S LEARN

- Predictions inform decisions; decisions need causes.
- A/B tests and causal methods measure true impact.
- Weigh costs, benefits and uncertainty before acting.

From data to decision

Describe

Predict

Find causes

Decide

KEY TERMS

A/B test
Randomised comparison

Confounder
Hidden common cause

Uplift Extra effect of action

📖 CONCEPT EXPLAINED: Data into Decisions

💡 In this chapter

Organisations collect data to decide better: which customers to target, which policy to adopt, which treatment to offer. But a model that predicts well does not automatically tell you what to do. Turning data into decisions requires causal thinking, careful experiments, honest uncertainty and an explicit account of the costs and benefits of each choice.

🔮 1. Prediction is not intervention

A churn model can rank customers by their chance of leaving. That is a prediction. The decision, whether to offer a discount, depends on a different quantity: how much the offer changes each customer's behaviour. Some high-risk customers will leave regardless; others would have stayed anyway; only some are persuadable.

Uplift modelling estimates this difference directly by comparing similar customers who did and did not receive the action, ideally from an experiment. Targeting by uplift rather than risk often saves money and achieves more. The general lesson: always ask what decision a model will change, and whether the model actually measures that.

✏️ WORKED EXAMPLE

Four customer types: persuadables (stay only if offered), sure things (stay anyway), lost causes (leave anyway), sleeping dogs (leave because you contacted them). Only the first group justifies the offer.

🧪 2. Experiments: the gold standard

Randomised experiments assign treatment by chance, so on average the groups are alike in every respect, measured or not. Differences in outcomes can then be attributed to the treatment. In digital products these are called A/B tests; in medicine and policy, randomised controlled trials.

Good experiments are planned: a single primary metric, a minimum effect worth detecting, a sample size calculated to detect it, and a fixed duration. Stopping as soon as results look good, testing many metrics and reporting the best one, or ignoring effects on subgroups all produce false discoveries.

🤔 PAUSE & THINK

An A/B test shows a 30% lift with p = 0.02 after one week during Diwali. What questions would you ask before rolling it out for the whole year?

🔀 3. Causal inference without experiments

Experiments are sometimes impossible, unethical or too slow. Observational causal inference then relies on assumptions made explicit, often as a directed acyclic graph showing what causes what. Confounders, variables that affect both the treatment and the outcome, must be measured and adjusted for.

Common methods include matching or propensity scores, which compare similar treated and untreated units; difference-in-differences, which compares changes over time between groups; and instrumental variables. Simpson's paradox, as in the 1973 Berkeley admissions data, shows how aggregate comparisons can reverse once a confounder such as department is included.

✏️ **WORKED EXAMPLE**

Difference-in-differences: district A gets a new bus route; B does not. School attendance rises 8 points in A and 3 in B over the same period. Estimated effect ≈ 5 points, assuming both would otherwise have followed parallel trends.

⚖️ 4. Deciding under uncertainty

Decision analysis combines probabilities with consequences. Expected value multiplies the probability of each outcome by its payoff and subtracts costs. Thresholds for action should reflect the relative cost of errors: missing a disease is usually worse than a false alarm, so the threshold for further testing should be lower.

Communicate results as ranges, not single numbers, and show what would change the decision. After acting, monitor outcomes, because the world may respond differently than estimated. Document the reasoning so that future teams can learn from both good and bad decisions.

🖐️ **TRY IT**

Pick a decision in your department, such as whether to run extra classes. List options, outcomes, rough probabilities and payoffs, compute expected values and state what new data would change your choice.

✅ Key takeaways

- Predictions rank risk; decisions need the effect of an action (uplift).
- Randomised experiments balance confounders; plan metrics and sample size first.
- Observational causal methods need explicit assumptions and confounder adjustment.
- Simpson's paradox shows aggregates can mislead.
- Expected value and error costs set decision thresholds; monitor afterwards.

❓ Quick check: answer in one or two lines.

1. Why might targeting high-risk customers waste money?

2. Name two practices that keep A/B tests honest.

3. Explain difference-in-differences with an example.

🔬 CONCEPT DEEP DIVE: Data into Decisions

🔷 Prediction asks 'what will happen?'; decisions need 'what happens if we act?', which is a causal question.

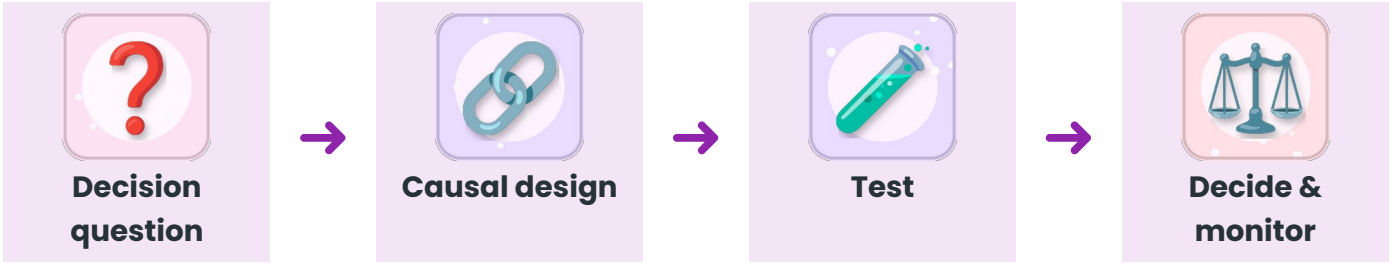
🔷 Randomised experiments (A/B tests) balance confounders; plan sample size and one primary metric in advance.

🔷 Observational methods (matching, difference-in-differences, instrumental variables) rely on stated assumptions, often drawn as DAGs.

🔷 Decision analysis weighs probabilities, payoffs and error costs; thresholds should reflect the cost of mistakes.

🔷 Communicate uncertainty with intervals and scenarios, and monitor outcomes after the decision is made.

🔄 How it works



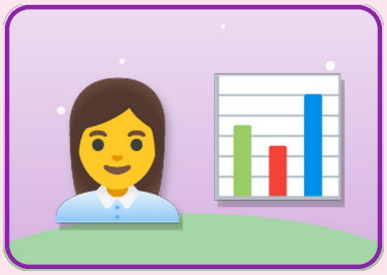
💻 Expected value of an offer

```
p_churn, value, cost = 0.30, 5000, 400
save_rate = 0.25   # share saved by the offer
ev = p_churn * save_rate * value - cost
print(ev)   # -25.0: offer not worth it
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Discount Nobody Needed

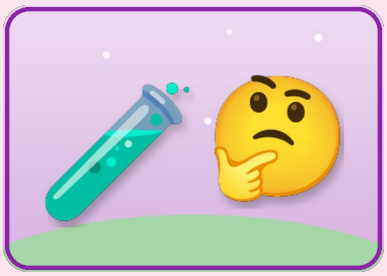
Characters: *Neha (MBA Analytics), marketing head Ms. Joseph, data scientist Rahul.*



SCENE 1

Neha's churn model identified 10,000 customers of a broadband company most likely to leave. Ms. Joseph sent all of them a ₹300 discount. Churn fell slightly, and the campaign was declared a success.

Pause & predict: What evidence is missing?



SCENE 2

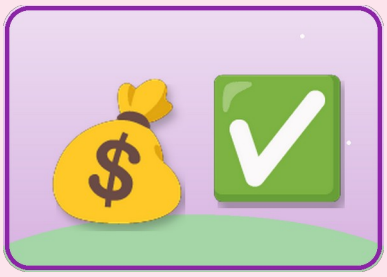
Rahul ran a proper A/B test next month: half the high-risk group received the offer, half did not. Many in both groups stayed. The discount changed behaviour for only a small segment.

Pause & predict: Why was a control group essential?

SCENE 3

Using the experiment data, Neha built an uplift model predicting who would stay because of the offer. Targeting the top 3,000 by uplift kept more customers than the original 10,000 mailing.

Pause & predict: How is uplift different from churn risk?



SCENE 4

Campaign cost fell by 70% while retention improved. Ms. Joseph now asks for an expected-value calculation and a control group before every campaign.

⏸ **Pause & predict:** Why calculate expected value before acting?

🌟 The big idea

Predictions show who is at risk; experiments show what actually helps.

💬 Discuss with your classmates

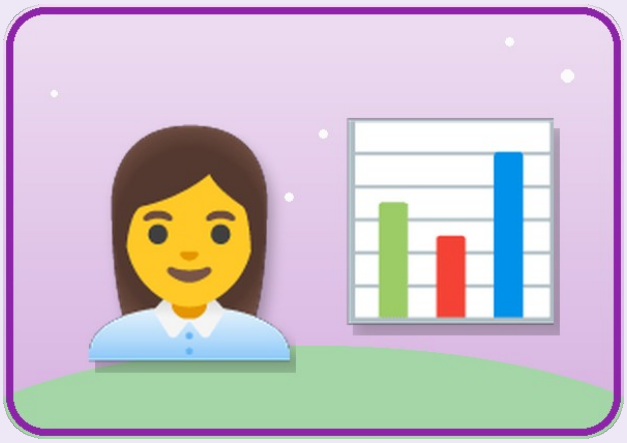
1. Which decision in your organisation needs a control group?

2. When would an experiment be unethical?

SCENARIO: Data into Decisions

Read the scenario. Think about the choices made at each step.

1

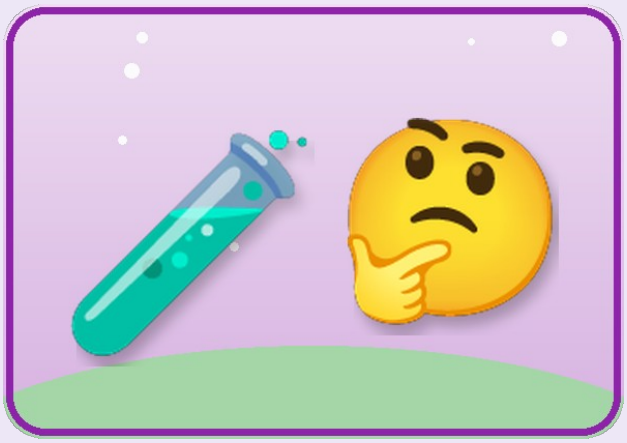


Neha's model predicts which customers will leave.

2

The team sends discounts to all high-risk users.

3



An A/B test shows many would have stayed anyway.

4

Targeting by uplift saves money and keeps more customers.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Data-driven decisions

Pricing

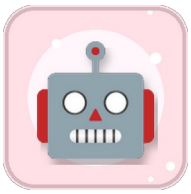
Hospital staffing

Marketing

Bus routes

Treatment choice

Loan approval



BOLT SAYS:

I can predict who will buy, but only an experiment shows what makes them buy!

FUN FACT!

Berkeley's 1973 admissions looked biased overall, yet most departments were not.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Berkeley Admissions and Simpson's Paradox

CASE FILE

In 1973, Berkeley's graduate admissions seemed to favour men: about 44% of male applicants were admitted versus 35% of female applicants. In a 1975 Science paper, Bickel, Hammel and O'Connell analysed each department and found most admitted women at similar or higher rates. Women had applied more often to highly competitive departments. The overall gap came from a confounder, department choice: a classic Simpson's paradox.

🔍 Analyse the case

1. What was the confounding variable?

2. How can aggregate data mislead decision-makers?

3. What else would you check before concluding there was no bias?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: The Coin-Flip Trial

You need:

Coins

Tally sheet

Whole class

Steps:

1.
   Flip to assign each student to A or B.
2.
   Group A gets a memory tip; B does not.
3.
   Everyone memorises 15 words in 1 minute.
4.
   Compare average recall of A and B.

THINK & TALK

Why does the coin flip make it fair?

Record your observations & results:

💻 CODE LAB: A/B Test Readout

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  a_conv, a_n = 120, 2000
2  b_conv, b_n = 156, 2000
3  lift = (b_conv / b_n) / (a_conv / a_n) - ____
4  print(round(lift * 100, 1), '% lift')
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. What is printed?**

EVALUATE **T3. Is the lift significant?**

EXTEND **T4. Run a two-proportion z-test.**

WORKSHEET: Data into Decisions (Part A)

Name: _____ Date: _____

1. Match the term to its meaning.

Correlation ●

Causation ●

Confounder ●

Randomisation ●

● **Hidden common cause**

● **Balances groups by chance**

● **Variables move together**

● **One changes the other**

2. Correlation only or causal test? Tick ✓.

	Correlation	Causal test
Ice-cream sales and drownings rise		
Randomised fertiliser trial		
Email openers buy more		
A/B test of two checkouts		

WORKSHEET: Data into Decisions (Part B)

Name: _____ Date: _____

1. Decision maths. Use the word box.

WORD BOX: 100 ✿ 60 ✿ 40

0.5 × ₹200 payoff = ₹____ expected

A converts 6 in 100 → ____ per 1,000

B 10% vs A 6% → ____ extra per 1,000

2. Match the method to its use.

A/B test ●

● **Map assumed causes**

Diff-in-diff ●

● **Randomised comparison**

Propensity score ●

● **Match similar treated & untreated**

DAG ●

● **Compare changes over time**

PUZZLE & DEFINE: Data into Decisions

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

A/B test

Meaning: _____

Example: _____

Confounder

Meaning: _____

Example: _____

Uplift

Meaning: _____

Example: _____

? QUIZ TIME: Data into Decisions

Circle the right answer!

Q1. Randomly splitting users to compare:

a) A/B test

b) Survey

c) Gamble

Q2. A hidden common cause is a...

a) Metric

b) Confounder

c) Target

Q3. Trend reverses when grouped:

a) Overfit

b) Drift

c) Simpson's

Q4. Expected value = probability ×...

a) Payoff

b) Colour

c) Time

Q5. Prediction alone doesn't show...

a) Numbers

b) Causes

c) Dates

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Data into Decisions

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Correlation always means causation.		
	Randomisation balances confounders on average.		
	Simpson's paradox can reverse a trend.		
	Expected value weighs probabilities and payoffs.		
	A churn score shows who responds to offers.		
	Most Berkeley departments did not disfavour women.		

HOME CONNECT

Find an 'X linked to Y' headline. List two possible confounders and sketch a fair test.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Data into Decisions

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why can't a churn score alone guide discounts?**

COMPARE **Q2. A/B test vs observational study.**

APPLY **Q3. p 0.4, saves 50%, value ₹2,000, cost ₹300: EV?**

EVALUATE **Q4. Sales rose after an ad. Did the ad cause it?**

CREATE **Q5. Design a test of a new tutoring app.**

📝 EXAM PREP: Data into Decisions

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Distinguish prediction and causation.

2. What is a confounder?

3. Why randomise?

4. Define uplift.

PART B • Long answer (5 marks)

5. Explain how you would decide whether to roll out a customer retention offer using data.

UNIT REVIEW: AI CONNECT

Name: _____ Date: _____

Circle the right answer!

Q1. AI-written code must be...

a) Tested | b) Trusted | c) Ignored

Q2. Copilot study: task finished...

a) 55% slower | b) 55% faster | c) Same speed

Q3. Insecure AI code needs...

a) Celebration | b) Fast merge | c) Review & scans

Q4. Randomised comparison of two versions:

a) A/B test | b) Pie chart | c) Lottery

Q5. Berkeley's paradox was driven by...

a) Fonts | b) Confounding | c) Speed

Q6. 0.4 × 0.5 × 2000 − 300 =

a) ₹400 | b) ₹700 | c) ₹100

WORD SEARCH: AI CONNECT

🔍 **Find these words. They go across → or down ↓.**

UNIT TEST

CODE REVIEW

CONFOUNDER

UPLIFT

PYTHON

DEBUG

G	C	D	B	U	B	C	I	U	H
D	O	S	P	E	I	O	H	G	B
D	D	O	Y	B	B	N	M	U	M
E	E	M	T	F	S	F	H	N	E
B	R	C	H	A	A	O	U	I	T
U	E	A	O	C	M	U	P	T	R
G	V	U	N	D	P	N	L	T	I
C	I	S	A	B	L	D	I	E	C
A	E	A	M	I	E	E	F	S	E
F	W	L	F	S	A	R	T	T	N

I finished AI CONNECT! Teacher's sign: _____

MONTH 3

AI SOLVE

Problems • Solutions • Impact

Area: AI Applications

In this unit, you will explore:

Topic 5

AI for Smarter Business

Topic 6

Engineering Solutions

TOPIC 5 • MONTH 3 • AI APPLICATIONS

AI for Smarter Business

LET'S LEARN

- Business AI must tie to revenue, cost or risk.
- Forecasting, personalisation and pricing lead adoption.
- Pilot, measure ROI and manage model risk.

Business AI value levers

Grow revenue

Cut cost

Reduce risk

Delight customers

KEY TERMS

ROI Return on investment

Forecast Predict future demand

Churn Customers leaving

📖 CONCEPT EXPLAINED: AI for Smarter Business

💡 In this chapter

Businesses adopt AI to grow revenue, cut costs, reduce risk and serve customers better. Yet many AI projects never leave the pilot stage, and some succeed technically while failing commercially. Postgraduates heading into industry, consulting or entrepreneurship need to connect models to business value, build credible cases and manage the risks of automated decisions at scale.

🎯 1. Choosing the right use cases

Strong AI use cases share four features: a clear business metric, a decision that predictions will actually change, enough relevant data, and a workflow where the output can be acted on. A simple value × feasibility matrix helps prioritise; quick wins with high value and high feasibility usually come first.

Common enterprise uses include demand forecasting, customer service automation, personalised recommendations, fraud detection, document processing and predictive maintenance. Generative AI adds drafting, summarising and search over internal knowledge. Each needs an owner in the business, not only in the data team.

✏️ WORKED EXAMPLE

Grocery chain: forecasting daily vegetable demand per store (clear metric: waste and stock-outs; data: 3 years of sales; action: order quantities) scores high on both value and feasibility.

🔮 2. Forecasting and personalisation

Forecasts drive inventory, staffing and cash planning. Accuracy is measured with metrics such as mean absolute percentage error (MAPE), but a forecast is only useful if it beats simple baselines, such as last week's sales or the same day last year, and if planners trust and use it. Festivals, weather and promotions often matter more than model complexity.

Personalisation and dynamic pricing tailor offers and prices to individuals or moments. They can raise revenue and relevance, but they also raise fairness and trust concerns: charging more when a customer is desperate, or excluding groups from offers. Clear limits, transparency and regular review protect both customers and brand.

🤔 **PAUSE & THINK**

A forecasting model has MAPE of 8%; the naive 'same day last week' baseline has 9%. Is the model worth deploying? What else would you want to know?

💹 3. Building the business case

Return on investment compares net gain with cost: ROI = (gain - cost) ÷ cost. Costs include data preparation, development, infrastructure, licences, integration, training staff, monitoring and ongoing maintenance, not just the model. Payback period, the time for savings to cover costs, is often easier for managers to grasp.

Credible cases use conservative estimates, a pilot with a control group to measure actual impact, and sensitivity analysis showing how results change if assumptions are wrong. Overstated benefits are a common reason AI projects lose support after launch.

✏️ **WORKED EXAMPLE**

Cost ₹7 lakh a year (build ₹5 lakh + run ₹2 lakh); measured gain ₹12 lakh a year. ROI = (12 - 7) ÷ 7 ≈ 71%. If the gain is only ₹8 lakh, ROI falls to about 14%.

🛡️ 4. Model risk at business scale

When models act at scale, small errors multiply. Zillow Offers used price estimates to buy homes; in 2021, as the market shifted, it found itself holding many homes worth less than it paid, wrote down over $300 million and closed the business. The model was only one factor, but speed and scale turned forecast error into a major loss.

Model risk management, common in banking, includes independent validation, limits on automated actions, monitoring for drift, human review of large decisions, clear ownership and plans to pause a model quickly. These practices are now spreading to every sector that relies on AI for consequential decisions.

🖐️ **TRY IT**

For a business you know, write a one-page AI business case: use case, KPI, data, cost, expected gain, ROI, pilot design and three risks with controls.

✅ Key takeaways

- Good use cases have a KPI, a changeable decision, data and an owner.
- Forecasts must beat naive baselines and be trusted by planners.
- Personalisation and pricing need fairness limits and transparency.
- ROI and payback include all costs; pilots measure real impact.
- Manage model risk: validation, limits, monitoring and fast shutdown.

❓ Quick check: answer in one or two lines.

1. What makes a business problem a good AI use case?

2. Calculate ROI for gain ₹9 lakh and cost ₹6 lakh.

3. What lessons does Zillow Offers teach about model risk?

🔬 CONCEPT DEEP DIVE: AI for Smarter Business

🔷 Choose use cases by value × feasibility: clear KPI, available data, a decision the prediction will change.

🔷 Forecasting drives inventory, staffing and cash; error metrics such as MAPE must be compared with simple baselines.

🔷 Personalisation and dynamic pricing raise revenue but need fairness limits and customer trust.

🔷 Business cases use ROI, total cost of ownership and payback period; include running and maintenance costs.

🔷 Model risk management: validation, monitoring, limits on automated action and clear owners for each model.

🔄 How it works

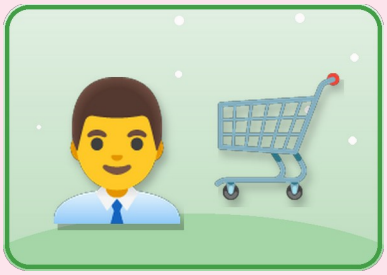
💻 ROI of an AI project

```
gain, build, run = 1200000, 500000, 200000   # ₹ per year
roi = (gain - build - run) / (build + run)
print(round(roi * 100), '%')   # 71 %
```

✍️ In your own words: explain this topic to a Class 10 student.

📚 AI STORY: Less Waste, More Trust

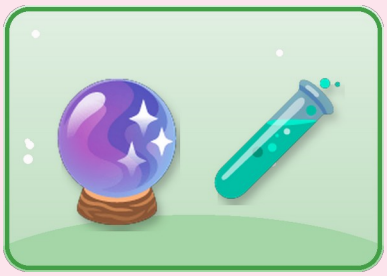
🎭 **Characters:** *Vikram (MBA Business Analytics), store manager Mrs. Fernandes, CFO Mr. Kapoor.*



SCENE 1

A grocery chain threw away nearly a quarter of its leafy vegetables. Vikram proposed a demand forecast per store. Mr. Kapoor asked: what will it cost, what will it save, and how will we know?

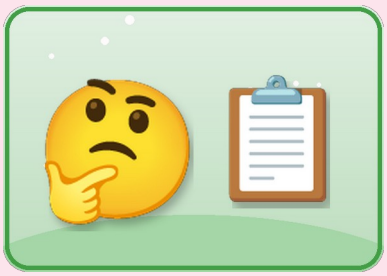
⏸ **Pause & predict:** What belongs in the answer?



SCENE 2

Vikram ran a 6-week pilot in 10 stores, with 10 similar stores as a control. Forecasts beat 'same day last week' and cut waste by 18% in pilot stores, while control stores barely changed.

⏸ **Pause & predict:** Why compare with control stores?



SCENE 3

Mrs. Fernandes ignored the forecast during a festival because it seemed too low. She was right: festival dates were missing. Vikram added a festival calendar and a manual override with a reason box.

⏸ **Pause & predict:** Why keep a human override?

SCENE 4

With full costs included, ROI was 71% and payback under a year. The rollout included weekly error monitoring and a rule that large overrides trigger a model review.

⏸ **Pause & predict:** Why monitor after a successful pilot?

🌟 The big idea

Business AI earns trust through measured pilots, honest costs and room for human judgement.

💬 Discuss with your classmates

1. Which business decision near you could a forecast improve?

2. How would you calculate its ROI?

SCENARIO: AI for Smarter Business

Read the scenario. Think about the choices made at each step.

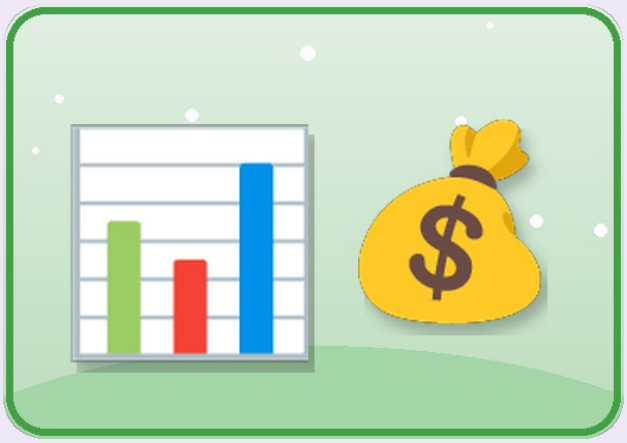
1

Vikram runs analytics for a grocery chain.

2

A demand forecast cuts wasted vegetables.

3



Waste falls 18%; savings beat the system cost.

4

He monitors errors weekly during festivals.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: AI across the business

Inventory

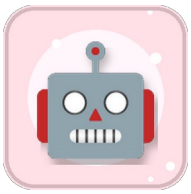
Customer support

Marketing

Pricing

Fraud checks

Logistics



BOLT SAYS:
I'm only smart for business if my predictions turn into profit, safely!

FUN FACT!
Zillow closed its AI-priced home-buying arm in 2021 after big losses on homes.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Zillow Offers: When the Model Bought the House

CASE FILE

Zillow Offers used algorithmic price estimates to buy homes, renovate them and resell them. In 2021, as the housing market shifted, Zillow found it had paid more for many homes than it could sell them for. In November 2021 it announced it would close the business, wrote down over $300 million of inventory and planned to cut about a quarter of its staff. Forecast errors, multiplied by fast buying at scale, became business risk.

🔍 Analyse the case

1. Why are forecast errors so costly when you buy at scale?

2. Which guardrails could have limited the losses?

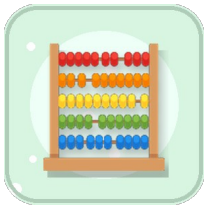
3. How should model risk be governed in a business?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: AI Business Pitch

You need:

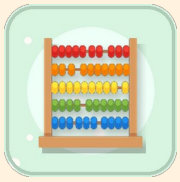
Case cards



Calculators

Teams of 4

Steps:

1.
   Pick a business problem card.
2.
   Name the AI use, data and KPI.
3. 
   Estimate yearly gain and total cost.
4.
   Pitch ROI and one key risk in 2 minutes.

THINK & TALK

Which pitch had the most believable numbers?

Record your observations & results:

💻 CODE LAB: Forecast Error

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
actual = [120, 135, 150, 160]
pred   = [110, 140, 145, 170]
mape = sum(abs(a - p) / a for a, p in zip(actual, pred)) /
len(____)
print(round(mape * 100, 1), '%')
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. What is printed?**

INTERPRET **T3. Is 5.4% good?**

EXTEND **T4. Compare with a last-value baseline.**

WORKSHEET: AI for Smarter Business (Part A)

Name: _____ Date: _____

1. Match the AI use to its purpose.

Forecasting ●

Personalisation ●

Fraud detection ●

Dynamic pricing ●

● Flag suspicious payments

● Adjust prices to demand

● Plan stock and staff

● Tailor offers to users

2. Mainly revenue or cost? Tick ✓.

	Revenue	Cost
Product recommendations		
Chatbot resolving tickets		
Upsell offers		
Route optimisation		

WORKSHEET: AI for Smarter Business (Part B)

Name: _____ Date: _____

1. Business maths. Use the word box.

WORD BOX: 150 ✿ 20 ✿ 6

Gain ₹5L on cost ₹2L → ROI ____%

Waste falls from 25% to 5%: ____ points

Cost ₹12L, saves ₹2L/month → payback ____ months

2. Order an AI business rollout. Write 1–4.

Scale with monitoring ☐

Choose a valuable use case ☐

Measure ROI ☐

Run a pilot ☐

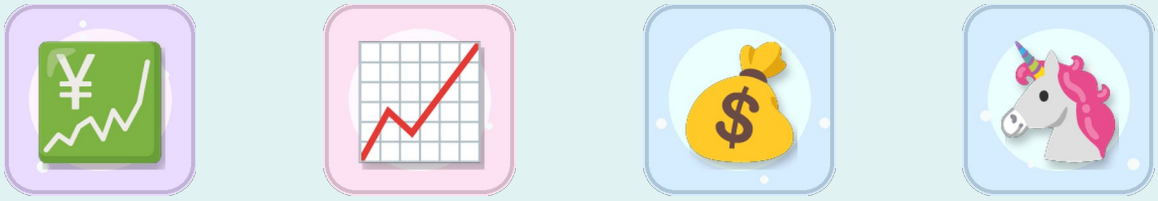
3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

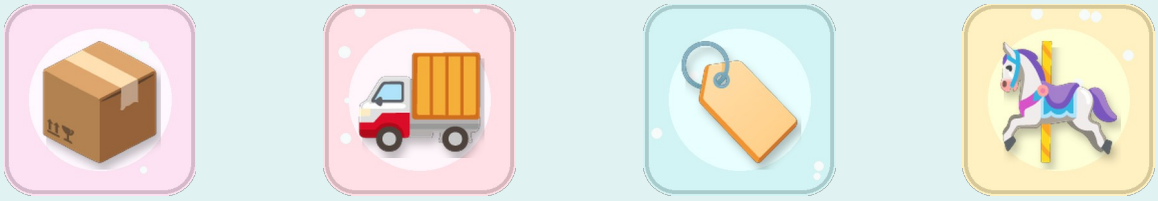
PUZZLE & DEFINE: AI for Smarter Business

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A 

B

C 

2. Define it! Write each term in your own words with an example.

ROI

Meaning: _____

Example: _____

Forecast

Meaning: _____

Example: _____

Churn

Meaning: _____

Example: _____

? QUIZ TIME: AI for Smarter Business

Circle the right answer!

Q1. ROI compares gain with...

a) Cost

b) Design

c) Time zone

Q2. Predicting next month's sales:

a) Branding

b) Forecasting

c) Cleaning

Q3. Customers leaving a service:

a) Growth

b) Loyalty

c) Churn

Q4. Zillow's iBuying shows the risk of...

a) Model error

b) Music

c) Weather

Q5. Start business AI with a...

a) Full rollout

b) Pilot

c) Advert

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI for Smarter Business

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
ROI compares gain with cost.		
Every AI project should skip the pilot.		
Forecasting helps plan inventory.		
Dynamic pricing can raise fairness concerns.		
Model errors don't matter in business.		
Zillow wound down Zillow Offers in 2021.		

HOME CONNECT

Visit a local shop. Which decision (stock, price, staff) could AI improve? Estimate a rough yearly gain.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: AI for Smarter Business

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why link every AI project to a KPI?**

COMPARE **Q2. Build vs buy an AI solution.**

APPLY **Q3. Gain ₹10 lakh, cost ₹4 lakh: ROI?**

EVALUATE **Q4. Is dynamic pricing always fair?**

CREATE **Q5. Pilot plan for AI customer support.**

📝 EXAM PREP: AI for Smarter Business

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Name three AI value levers.

2. Give the ROI formula.

3. Why pilot before scaling?

4. What is model risk?

PART B • Long answer (5 marks)

5. Build a business case for an AI system in a company of your choice.

TOPIC 6 • MONTH 3 • AI APPLICATIONS

Engineering Solutions

LET'S LEARN

- AI speeds design, monitoring and control in engineering.
- Digital twins and surrogates simulate faster than physics.
- Safety needs physical limits and human sign-off.

AI engineering toolkit

Predictive upkeep

Digital twins

Surrogate models

Physics-informed

KEY TERMS

PdM Predictive maintenance

Digital twin Live virtual copy

Surrogate Fast model stand-in

📖 CONCEPT EXPLAINED: Engineering Solutions

💡 In this chapter

Engineering has always used models, from beam equations to fluid simulations. AI adds models learned from data that can monitor machines, accelerate design and optimise control. But engineered systems interact with the physical world, where failures can injure people and destroy equipment. AI for engineering must therefore combine data-driven learning with physical understanding and rigorous safety practice.

📡 1. Sensors and industrial data

Modern plants, vehicles and buildings carry sensors for vibration, temperature, pressure, current, flow and acoustic signals, often sampled thousands of times per second. Industrial IoT platforms collect these streams, align them with maintenance logs and operating conditions, and store them for analysis.

Industrial data has particular challenges: failures are rare, so labelled examples are scarce; sensors drift or fail; operating conditions change with load and season; and data may be locked in proprietary systems. Data engineering, domain knowledge and careful labelling of past incidents are usually the hardest part of an industrial AI project.

✏️ WORKED EXAMPLE

Pump dataset: vibration at 10 kHz, motor current, inlet pressure and temperature every second, plus 3 years of maintenance tickets. Only 14 recorded bearing failures, so anomaly detection is used alongside supervised models.

🔧 2. Predictive maintenance

Reactive maintenance fixes machines after they fail; preventive maintenance follows a fixed schedule; predictive maintenance acts when data show that failure is approaching. Anomaly detection flags unusual patterns, while remaining useful life (RUL) models estimate how long a component can safely operate.

Value comes from avoiding unplanned downtime, extending component life and planning repairs for convenient times. Alerts must be precise enough that engineers trust them: too many false alarms and they are ignored, too few and failures are missed. Evaluation should count avoided failures, false alarms and lead time.

PAUSE & THINK

An anomaly model catches 90% of failures but raises 30 false alarms a month. How would you decide whether that is acceptable for a water-supply pump?

3. Digital twins and surrogate models

A digital twin is a virtual model of a physical asset, updated continuously with live data, used to monitor condition, test 'what-if' scenarios and optimise operation. Twins can range from simple statistical models of a pump to detailed simulations of an entire factory or power grid.

High-fidelity physics simulations such as computational fluid dynamics can take hours per run. Surrogate models learned from simulation results approximate them in milliseconds, enabling engineers to explore thousands of designs. Physics-informed neural networks add governing equations to the training loss, so predictions respect conservation laws even where data are sparse.

WORKED EXAMPLE

Wing design: 500 CFD runs train a surrogate that predicts lift and drag in 0.01 seconds. Engineers search 100,000 shapes with it, then verify the best 10 with full simulation.

4. Safety, verification and control

When AI moves from advising to controlling equipment, safety engineering becomes central. DeepMind's data-centre cooling system, which reduced cooling energy by up to 40%, operated within strict constraints, with checks on every action and human operators able to take over at any time.

Engineering practice requires hazard analysis, validation on test rigs and historical data, compliance with industry standards, fallback to safe states, logging for incident investigation and clear human authority. Data-only models should never be trusted outside the conditions they were trained on without physical checks.

TRY IT

Choose a machine on your campus. Draft a predictive-maintenance design: sensors, data, model type, alert rules, evaluation metrics, safety constraints and who acts on alerts.

✅ Key takeaways

- Industrial AI depends on sensor data, domain knowledge and labelled incidents.
- Predictive maintenance uses anomaly detection and remaining-useful-life models.
- Digital twins mirror assets; surrogates and PINNs speed up simulation.
- Alerts must balance missed failures and false alarms.
- Control needs constraints, validation, fallbacks and human authority.

❓ Quick check: answer in one or two lines.

1. Contrast reactive, preventive and predictive maintenance.

2. Why are surrogate models valuable in design?

3. List three safety practices for AI that controls equipment.

🔬 CONCEPT DEEP DIVE: Engineering Solutions

🔷 Industrial AI runs on sensor data (vibration, temperature, pressure, current) collected through IoT systems.

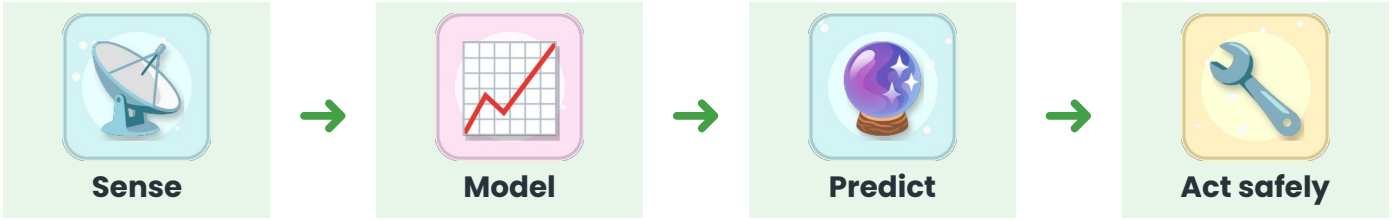
🔷 Predictive maintenance detects anomalies and estimates remaining useful life, so repairs are planned, not reactive.

🔷 Digital twins combine physics models with live data to simulate, monitor and optimise real assets.

🔷 Surrogate models and physics-informed neural networks approximate slow simulations, enabling fast design exploration.

🔷 Safety-critical use needs physical constraints, validation on test rigs, standards compliance and human override.

🔄 How it works



💻 Sensor anomaly z-score

```
import numpy as np
v = np.array([2.1, 2.0, 2.2, 2.1, 3.4])   # vibration, mm/s
z = (v[-1] - v[:-1].mean()) / v[:-1].std()
print(round(z, 1))   # 18.4: anomaly
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Pump Number Three

Characters: *Kavitha (MTech Mechanical), plant engineer Mr. Murugan, operator Lakshmi.*

SCENE 1

A town's water plant suffered surprise pump failures every few months, each causing a three-day outage. Kavitha installed vibration and temperature sensors and trained an anomaly model on normal operation.

Pause & predict: Why model normal behaviour rather than failures?

SCENE 2

In March, the model flagged pump 3. Mr. Murugan was sceptical: the pump sounded fine. But vibration in one frequency band had risen steadily for two weeks.

Pause & predict: Why might humans miss this signal?

SCENE 3

Engineers opened the pump during a planned low-demand night and found a worn bearing. The two-hour repair avoided an estimated three-day outage and ₹10 lakh in costs.

Pause & predict: How should this result affect trust in alerts?

SCENE 4

Kavitha added rules: alerts suggest, engineers decide; the model never shuts pumps automatically; every alert's outcome is logged to improve precision. Lakshmi now checks the dashboard each shift.

⏸ **Pause & predict:** Why should the model not act automatically?

🌟 The big idea

Engineering AI should warn early, explain clearly and leave safety decisions to accountable people.

💬 Discuss with your classmates

1. Which machine near you would benefit from predictive maintenance?

2. How would you handle false alarms?

SCENARIO: Engineering Solutions

Read the scenario. Think about the choices made at each step.

1

Kavitha monitors pumps at a water plant.

2

AI flags rising vibration on pump 3.

3

Engineers find a worn bearing early.

4

A planned 2-hour fix replaces a 3-day outage.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: AI in engineering

Factories

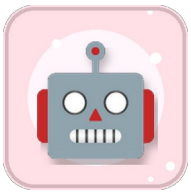
Power grids

Construction

Aircraft engines

Bridge monitoring

Railways



BOLT SAYS:

I learn physics from data, but I must still obey the laws of physics!

FUN FACT!

DeepMind's 2016 AI cut energy used to cool Google data centres by up to 40%.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: Cooling Google's Data Centres

CASE FILE

Data centres use a lot of energy for cooling. In 2016 DeepMind and Google reported that a machine-learning system, trained on data from thousands of sensors, recommended settings that cut cooling energy by up to 40%. Neural networks predicted future temperatures and pressures to keep the plant within safe limits. By 2018 AI directly controlled cooling at several data centres, with safety constraints and operators able to take over at any time.

🔍 Analyse the case

1. Why were safety constraints essential?

2. What data made the model possible?

3. Where else could similar AI control help?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Vibration Detective

You need:

Phone app

Table fan

Teams of 3

Steps:

1. Record fan vibration with a phone sensor app.
2. Tape a coin to one blade; record again.
3. Compare the two vibration patterns.
4. Decide when the fan needs maintenance.

THINK & TALK

How could AI spot faults before failure?

Record your observations & results:

💻 CODE LAB: Remaining Life Estimate

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
wear = [0.10, 0.18, 0.26, 0.34]   # monthly wear fraction
rate = (wear[-1] - wear[0]) / (len(wear) - ____)
months_left = (1.0 - wear[-1]) / rate
print(round(rate, 2), round(months_left, 1))
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. Rate and months left?**

EXPLAIN **T3. What does this assume?**

EXTEND **T4. Fit a trend with np.polyfit.**

WORKSHEET: Engineering Solutions (Part A)

Name: _____ Date: _____

1. Match the tool to its role.

Predictive upkeep ●

Digital twin ●

Surrogate model ●

Anomaly detection ●

● **Fast stand-in for simulation**

● **Spot unusual sensor readings**

● **Fix before failure**

● **Virtual copy with live data**

2. Reactive or predictive? Tick ✓.

	Reactive	Predictive
Repair after breakdown		
Replace when vibration trend rises		
Fixed yearly service only		
Alert from sensor anomaly		

WORKSHEET: Engineering Solutions (Part B)

Name: _____ Date: _____

1. Maintenance maths. Use the word box.

WORD BOX: **10** ✿ **72** ✿ **4**

₹50,000/hr × 20 hrs = ₹____ lakh

A 3-day outage = ____ hours

Surrogate 0.5 s vs simulation 2 s → ____× faster

2. Circle safe engineering-AI practices.

Physical limits

Ignore sensor faults

Human override

Untested updates

Validate on test rigs

No logging

PUZZLE & DEFINE: Engineering Solutions

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

PdM

Meaning: _____

Example: _____

Digital twin

Meaning: _____

Example: _____

Surrogate

Meaning: _____

Example: _____

? QUIZ TIME: Engineering Solutions

Circle the right answer!

Q1. Fixing machines before they fail:

a) Predictive

b) Reactive

c) Random

Q2. A live virtual copy of a system:

a) Photo

b) Digital twin

c) Manual

Q3. Fast stand-in for slow simulation:

a) Full CFD

b) Ruler

c) Surrogate

Q4. AI cut cooling energy by up to...

a) 40%

b) 4%

c) 90%

Q5. Safety-critical AI needs...

a) No checks

b) Human sign-off

c) Luck

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Engineering Solutions

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Predictive maintenance uses condition data.		
	Digital twins never use live data.		
	Surrogates approximate slow simulations.		
	Physics-informed ML adds physical laws to training.		
	Safety AI needs no human override.		
	DeepMind cut cooling energy by up to 40%.		

HOME CONNECT

List three machines at home or college. What sensor data could predict each one's failure?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Engineering Solutions

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why use a surrogate instead of a full simulation?**

COMPARE **Q2. Preventive vs predictive maintenance.**

APPLY **Q3. Downtime ₹50,000/hr; AI avoids 20 hrs/yr: value?**

EVALUATE **Q4. Trust a data-only model outside its training range?**

CREATE **Q5. Digital twin plan for a campus building.**

📝 EXAM PREP: Engineering Solutions

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What is predictive maintenance?

2. Define a digital twin.

3. Why use surrogate models?

4. Name two safety safeguards.

PART B • Long answer (5 marks)

5. Design an AI predictive-maintenance system for a factory, including safety and evaluation.

UNIT REVIEW: AI SOLVE

Name: _____ Date: _____

Circle the right answer!

Q1. Gain ₹10L, cost ₹4L → ROI

a) 150% | b) 60% | c) 250%

Q2. Zillow Offers was wound down in...

a) 2015 | b) 2021 | c) 2025

Q3. MAPE measures forecast...

a) Colour | b) Profit | c) Error

Q4. Live virtual copy of an asset:

a) Digital twin | b) Photo | c) Manual

Q5. Repair based on condition data:

a) Reactive | b) Predictive | c) Calendar

Q6. Data-centre cooling cut by up to...

a) 4% | b) 90% | c) 40%

WORD SEARCH: AI SOLVE

🔍 **Find these words. They go across → or down ↓.**

ROI

FORECAST

CHURN

PDM

SURROGATE

PROFIT

L	A	P	I	L	O	T	I	W	B
R	F	L	T	U	U	D	S	G	E
O	C	H	U	R	N	U	E	I	A
I	K	D	C	R	R	P	N	C	R
M	C	O	E	A	K	U	S	O	I
O	D	B	W	W	B	M	O	N	N
U	L	T	P	D	M	P	R	I	G
S	H	P	R	O	F	I	T	B	K
S	U	R	R	O	G	A	T	E	A
C	F	O	R	E	C	A	S	T	D

I finished AI SOLVE! Teacher's sign: _____

MONTH 4

AI RISE

Skills • Careers • Future

Area: AI & Career

In this unit, you will explore:

Topic 7

Pick Your AI Path

Topic 8

Portfolio to Profession

TOPIC 7 • MONTH 4 • AI & CAREER

Pick Your AI Path

LET'S LEARN

- AI careers span research, engineering, product and policy.
- Match roles to your strengths, interests and values.
- Test paths with internships, projects and mentors.

Main AI career families

Research

Engineering

Product

Policy & ethics

KEY TERMS

MLE ML engineer

AI PM AI product manager

MLOps Running ML in production

📖 CONCEPT EXPLAINED: Pick Your AI Path

💡 In this chapter

As you approach the end of your postgraduate programme, the question changes from 'what should I learn?' to 'what work do I want to do?'. AI offers an unusually wide range of careers, from inventing algorithms to shaping national policy. Choosing well means understanding the roles, testing your fit through real experience and building skills that will last as the field changes.

🗺️ 1. The landscape of AI roles

Research roles, such as research scientist or PhD researcher, create new methods and publish findings. Engineering roles, such as ML engineer, data engineer and MLOps engineer, build and run systems that serve real users. Data scientists and analysts turn data into insight and decisions. Product roles, including AI product manager and UX researcher, decide what to build and for whom.

A growing family of roles focuses on governance: AI policy analysts, responsible-AI specialists, auditors and legal experts on technology. Domain-AI roles combine AI with health, agriculture, finance, education or climate. Entrepreneurs build new companies. Most careers move between several of these over time.

✏️ WORKED EXAMPLE

One problem, many roles: for a crop-advisory app, a research scientist improves the model, an ML engineer deploys it, a data engineer builds pipelines, a PM talks to farmers, and a policy specialist checks data and liability rules.

🧩 2. Matching roles to your profile

Each role mixes four ingredients differently: mathematical depth, software engineering, domain knowledge and communication. Research rewards deep maths and curiosity about open problems; engineering rewards building reliable systems; product rewards understanding users and making trade-offs; policy rewards writing, law and stakeholder skills.

Values matter as much as skills. Some people want rapid, visible impact; others want depth and independence; others want stability or social purpose. A simple weighted score of skills, interest and values across candidate roles can make an unclear choice easier to discuss with mentors, though it should inform, not replace, judgement.

🤔 **PAUSE & THINK**

Rank what matters most to you: salary, learning, impact, location, independence, stability. How would your top two priorities point you towards different roles?

🧪 3. Testing paths cheaply

Career choices are easier to test than most students think. Internships, research assistantships, open-source contributions, hackathons and freelance projects let you experience real tasks before committing. Informational interviews, short conversations with people in the role, reveal what a normal week looks like, which job descriptions rarely do.

Treat each experience as an experiment with a question: do I enjoy debugging production systems? Do I like reading papers every week? Do I enjoy talking with customers? Record what energised and drained you. Two or three such experiments usually make a direction much clearer.

✏️ **WORKED EXAMPLE**

Six-month plan: month 1–2 build a small deployed model (engineering); month 3–4 reproduce a paper (research); month 5 interview three AI PMs (product); month 6 reflect and choose a direction for applications.

📈 4. Building career capital in a changing field

The World Economic Forum's Future of Jobs Report 2023 ranked AI and machine learning specialists as the fastest-growing job and expected around 44% of workers' core skills to change within five years. Tools and job titles will keep changing; fundamentals in mathematics, programming, data, evaluation and communication remain valuable.

In India, opportunities span global capability centres, product companies, start-ups, research institutions and public programmes in digital public infrastructure and AI. Whatever path you choose, keep evidence of impact, a network of mentors and peers, and the habit of learning something new every quarter.

🖐️ **TRY IT**

Create a fit matrix for four AI roles with your scores for skills, interest and values. Choose weights, compute totals and discuss the result with a mentor.

✅ Key takeaways

- AI careers span research, engineering, data, product, policy, domain and entrepreneurship.
- Roles differ in maths, engineering, domain and communication demands.
- Values and preferred ways of working matter as much as skills.
- Test paths with internships, projects and informational interviews.
- Fundamentals and learning habits outlast changing tools and titles.

❓ Quick check: answer in one or two lines.

1. Compare an ML engineer and a research scientist.

2. Describe one cheap experiment to test a career path.

3. Which skills are likely to stay valuable as AI changes?

🔬 CONCEPT DEEP DIVE: Pick Your AI Path

🔷 AI roles cluster into research, engineering (ML, data, MLOps), product and design, and policy, ethics and governance.

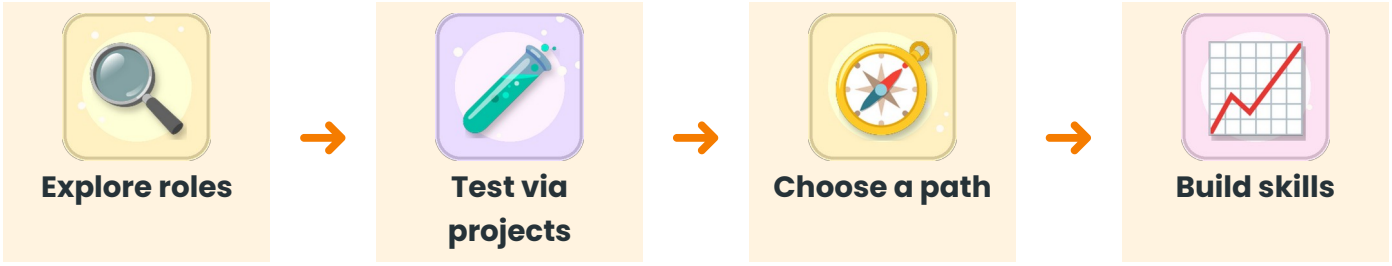
🔷 Each role mixes maths, coding, domain knowledge and communication in different proportions.

🔷 Paths include industry jobs, PhD or MS study, start-ups, public-sector AI programmes and freelancing.

🔷 Choose by experiment: internships, short projects and informational interviews reveal fit faster than reading.

🔷 Build career capital that transfers: fundamentals, evidence of impact, networks and the habit of learning.

🔄 How it works



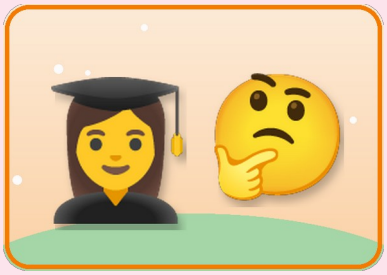
💻 Weighted role-fit score

```
fit = {'MLE': [5, 4, 3], 'Research': [3, 5, 4], 'AI PM': [4, 3, 5]}
w = [0.4, 0.3, 0.3]   # skills, interest, values
score = {r: round(sum(a * b for a, b in zip(v, w)), 1) for r, v in
fit.items()}
print(score)   # MLE 4.1, Research 3.9, AI PM 4.0
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Two Summers, Two Answers

Characters: *Fathima (MSc Computer Science), mentor Dr. Anand, team lead Sneha.*



SCENE 1

Fathima was torn between a PhD and an industry job. Her friends had strong opinions, but she had never done either kind of work for more than a class assignment.

Pause & predict: What information is Fathima missing?

SCENE 2

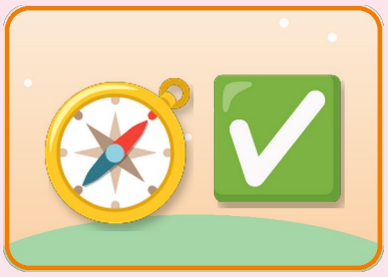
Dr. Anand suggested two experiments. First, she spent eight weeks reproducing a paper in his lab. She enjoyed the reading but felt restless waiting weeks for results.

Pause & predict: What did this experiment reveal?

SCENE 3

Next summer she interned with Sneha's ML team, shipping a feature that 50,000 users touched within a month. She loved the fast feedback and teamwork, and the messy data did not scare her.

Pause & predict: How did real tasks change her view?



SCENE 4

Fathima chose ML engineering, keeping research alive through open-source work. Her fit matrix agreed, but she said the real answer came from how she felt on Monday mornings.

⏸ **Pause & predict:** Why combine scores with lived experience?

The big idea

Test career paths with real work; your experience is the best data about your fit.

💬 Discuss with your classmates

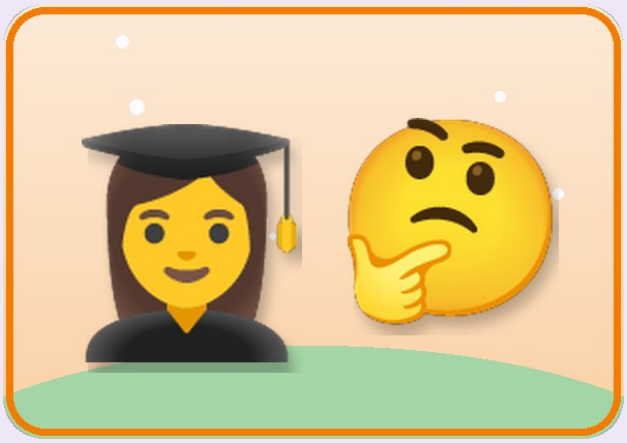
1. Which two AI roles would you like to test?

2. Whom could you interview this month?

SCENARIO: Pick Your AI Path

Read the scenario. Think about the choices made at each step.

1



Fathima is unsure: research or industry?

2

She takes a summer internship in an ML team.

3

She loves shipping features more than papers.

4

She aims for ML engineering, then AI product.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: AI roles in demand

ML engineer

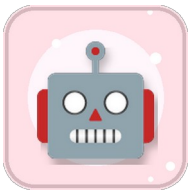
Data scientist

Research scientist

AI product manager

MLOps engineer

AI governance



BOLT SAYS:
There's no single AI job. Find the path where your strengths shine brightest!

FUN FACT!
The WEF Future of Jobs Report 2023 ranked AI and ML specialists fastest-growing.

👀 **Reflect: where have YOU seen this in real life? Note it down.**

CASE STUDY: The Fastest-Growing Jobs

CASE FILE

The World Economic Forum's Future of Jobs Report 2023 surveyed over 800 companies employing more than 11 million workers. AI and machine learning specialists topped its list of fastest-growing jobs, ahead of sustainability specialists and business intelligence analysts. Employers expected about 44% of workers' core skills to change within five years, with analytical and creative thinking ranked most important. AI roles are growing, but their skills keep shifting.

Analyse the case

1. Why might AI roles keep changing?

2. Which of your skills will stay valuable?

3. How would you check job-trend claims for India?

My verdict: Was AI used well here? Why?

LAB ACTIVITY: Career Speed-Dating

You need:

Role cards

Timer

Whole class

Steps:

1		Each student researches one AI role card.
2		Pairs swap role pitches every 3 minutes.
3		Rate each role on skills, interest, values.
4		Pick top 2 roles and a next step for each.

THINK & TALK

Which role surprised you most?

Record your observations & results:

💻 CODE LAB: Role Fit Ranker

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
fit = {'MLE': [5, 4, 3], 'DS': [4, 4, 4], 'PM': [3, 4, 5]}
w = [0.5, 0.3, 0.2]   # skills, interest, values
best = max(fit, key=lambda r: sum(a * b for a, b in zip(fit[r],
____)))
print(best)
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. Which role wins?**

MODIFY **T3. Use weights 0.2, 0.3, 0.5.**

EXTEND **T4. Print all scores, sorted.**

WORKSHEET: Pick Your AI Path (Part A)

Name: _____ Date: _____

1. Match the role to its main work.

ML engineer ●

● Defines what to build

Research scientist ●

● Runs models reliably

AI product manager ●

● Builds and ships models

MLOps engineer ●

● Invents new methods

2. Mostly building or leading? Tick ✓.

	Build	Lead
Optimise training code		
Write a product roadmap		
Deploy a model API		
Align teams on AI policy		

WORKSHEET: Pick Your AI Path (Part B)

Name: _____ Date: _____

1. Career maths. Use the word box.

WORD BOX: **4.1** ✿ **12** ✿ **3**

0.5×4 + 0.3×5 + 0.2×3 = ____

1 project a month for a year = ____

1 mentor per role, 3 roles = ____ mentors

2. Order a career decision. Write 1–4.

Build skills for the role ☐

Explore role options ☐

Choose a direction ☐

Test with projects ☐

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

PUZZLE & DEFINE: Pick Your AI Path

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

MLE

Meaning: _____

Example: _____

AI PM

Meaning: _____

Example: _____

MLOps

Meaning: _____

Example: _____

? QUIZ TIME: Pick Your AI Path

Circle the right answer!

Q1. Invents new ML methods:

a) Research sci.

b) Designer

c) Sales

Q2. Decides which AI product to build:

a) Tester

b) AI PM

c) Technician

Q3. Keeps models running in production:

a) Marketer

b) Artist

c) MLOps

Q4. A good way to test a career path:

a) Internship

b) Coin toss

c) Waiting

Q5. WEF 2023 fastest-growing job:

a) Chef

b) AI specialist

c) Fax operator

MY SCORE: _____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Pick Your AI Path

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
AI careers include policy roles.		
Only PhDs can work in AI.		
Internships help test career fit.		
MLOps engineers run models in production.		
AI skill needs never change.		
WEF 2023 ranked AI/ML specialists fastest-growing.		

HOME CONNECT

Interview someone working in tech for 15 minutes. What does a normal day in their role look like?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Pick Your AI Path

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why test a career with an internship?**

COMPARE **Q2. ML engineer vs data scientist.**

APPLY **Q3. Weights 0.5, 0.3, 0.2; scores 4, 5, 3: fit?**

EVALUATE **Q4. Choose a role only by salary?**

CREATE **Q5. A 6-month plan to test two AI roles.**

📝 EXAM PREP: Pick Your AI Path

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Name four AI career families.

2. What does an MLOps engineer do?

3. How can you test career fit?

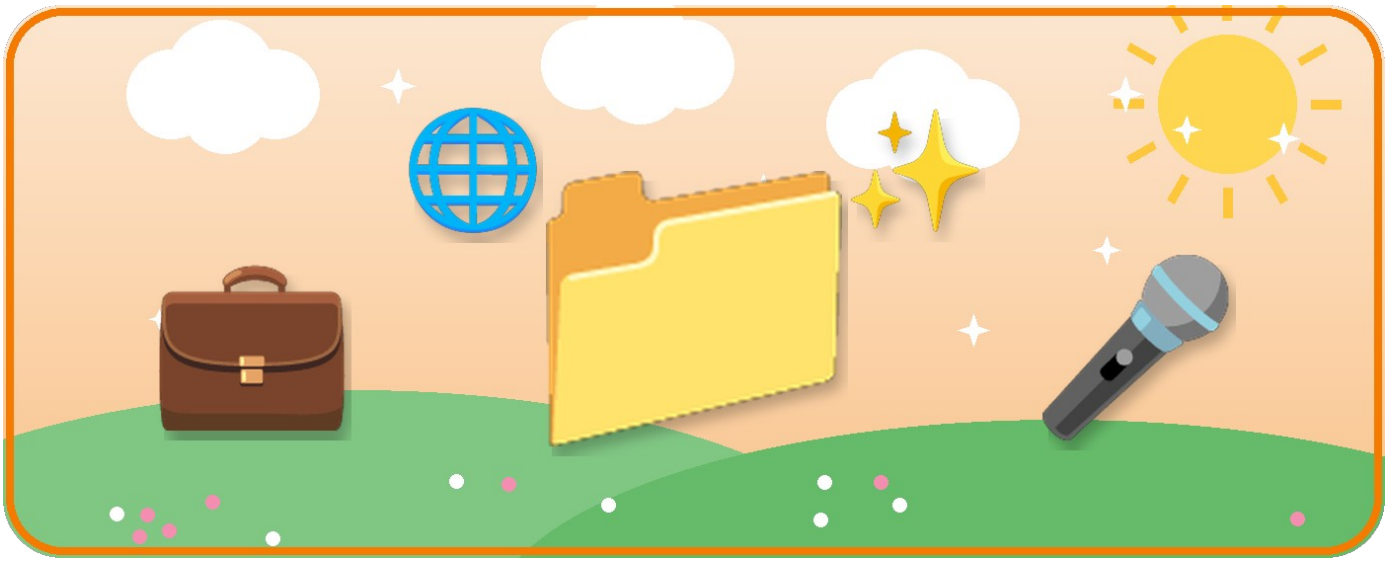
4. Name two durable skills.

PART B • Long answer (5 marks)

5. Compare three AI career paths and justify which suits you, with a one-year plan.

TOPIC 8 • MONTH 4 • AI & CAREER

Portfolio to Profession



LET'S LEARN

- A portfolio proves skill better than a list of courses.
- Show problem, method, result and impact for each project.
- Interviews test basics, projects and communication.

Portfolio essentials

Strong projects

Clear READMEs

Online presence

Storytelling

KEY TERMS

Portfolio Proof of work

README Project front page

STAR Situation-Task-Action-Result

📖 CONCEPT EXPLAINED: Portfolio to Profession

💡 In this chapter

Degrees and certificates open doors, but in AI hiring, demonstrated work is what keeps them open. A portfolio shows employers and admissions panels what you can actually build, how you think and how you communicate. This chapter covers how to choose and present projects, navigate interviews and conduct a professional job search.

📁 1. What makes a strong project

The best portfolio projects solve a real problem, use real or realistic data, run end to end from data collection to a working demo, and report a measured result against a sensible baseline. They also show judgement: why this approach, what failed, what you would do next and what ethical issues you considered.

Two or three such projects outweigh a dozen tutorial copies. Variety helps: perhaps one modelling project, one deployed application and one data or analysis project. Projects connected to a domain you care about, such as agriculture, health or education in your region, are memorable and show initiative.

✏️ WORKED EXAMPLE

Weak: 'Titanic survival notebook (tutorial)'. Strong: 'Tamil crop-advice chatbot: 2,000 farmer questions collected with consent, retrieval-based answers, 78% rated helpful by 40 farmers, deployed on WhatsApp for 3 villages.'

📝 2. Presenting your work

A project's README is often the only thing a recruiter reads. In the first screen it should state the problem, the data, the approach, the key result with a number, a demo link or screenshot, and how to run the code. Clean, tested code, sensible commit history and a short demo video add credibility.

Write about your projects: a short blog post explaining decisions and mistakes shows communication skills. In team projects, state your exact contribution. Keep online profiles consistent and professional, and never exaggerate results; interviewers will probe details.

🤔 PAUSE & THINK

Open your best project repository. Could a stranger understand the problem, result and how to run it in two minutes? What would you change first?

🎤 3. Interviews and assessments

AI hiring processes typically combine a coding round, questions on ML fundamentals such as bias–variance, metrics and evaluation, an ML system design discussion, a project deep-dive and behavioural questions. Some firms use take-home assignments or pair-programming sessions instead of puzzles.

Behavioural answers work best with the STAR structure: Situation, Task, Action, Result. Prepare stories about a technical challenge, a failure and what you learned, a disagreement in a team and a time you took initiative. Practise explaining a project at three levels: one sentence, two minutes and a full technical deep-dive.

✏️ **WORKED EXAMPLE**

STAR: Situation, our model scored 96% but failed in pilot. Task, find out why. Action, I discovered leakage from duplicate records and rebuilt the split by customer. Result, honest accuracy 84%, and the pilot succeeded.

🤝 4. The job search as a professional

Job searching is a funnel: applications, replies, interviews and offers. Referrals and genuine networking, at meetups, conferences, open-source communities and alumni groups, typically raise reply rates far above cold applications. Tailor each application to the role, and track your funnel to see where to improve.

Evaluate offers on learning, mentorship, the quality of the team, the problems you will work on and fair pay, not salary alone. Competitions like the Netflix Prize show that winning skill matters, but employers also value people who understand cost, simplicity and users. Integrity throughout, in claims, interviews and negotiations, builds a reputation that lasts.

🖐️ **TRY IT**

Write a STAR answer for your hardest technical problem and record yourself telling it in two minutes. Ask a friend which part was unclear.

✅ Key takeaways

- Strong projects: real problem, real data, end-to-end, measured against a baseline.
- READMEs, demos, clean code and blog posts make work visible.
- Interviews test coding, ML fundamentals, system design, projects and behaviour.
- Use STAR for behavioural answers; practise multi-level explanations.
- Network, track your funnel, evaluate offers broadly and act with integrity.

❓ Quick check: answer in one or two lines.

1. What should the first screen of a README contain?

2. Explain the STAR method with an example.

3. Why might referrals improve job-search results?

CONCEPT DEEP DIVE: Portfolio to Profession

🔷 Employers hire evidence: a few strong projects outweigh many certificates.

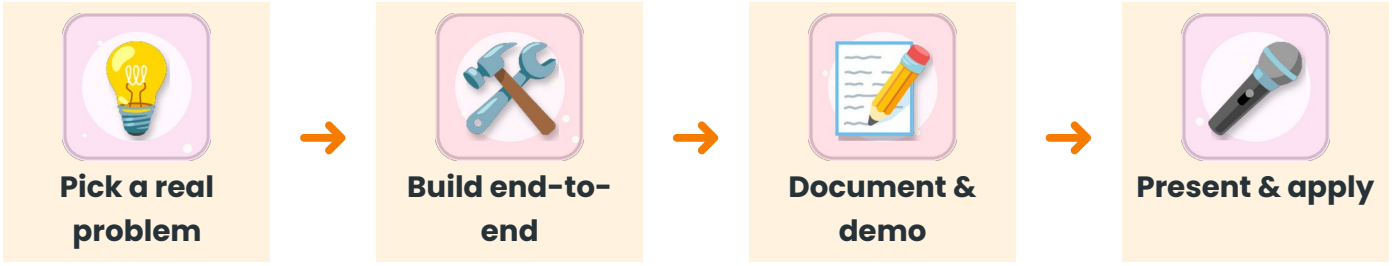
🔷 Pick projects with a real problem, real data, an end-to-end build and a measured result against a baseline.

🔷 Present them well: README, demo video, short write-up, clean code with tests, and your exact role in team work.

🔷 Interviews mix coding, ML fundamentals, ML system design, behavioural (STAR) questions and take-home tasks.

🔷 Job search is a funnel: referrals and networks raise reply rates; evaluate offers on learning, mentors and pay.

How it works



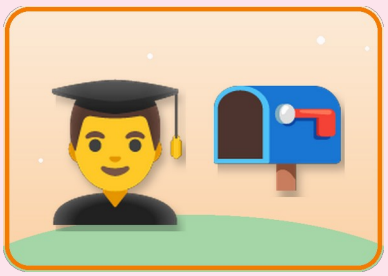
README in four lines

```
# Crop Yield Predictor
Problem: predict district yield 2 months early
Data: 10 yrs, 30 districts (public)   Model: XGBoost
Result: MAE 0.42 t/ha vs baseline 0.61   Demo: link
```

In your own words: explain this topic to a Class 10 student.

AI STORY: Forty Applications, Zero Replies

Characters: *Arun (MCA), senior alumnus Vignesh, recruiter Ms. Tara.*



SCENE 1

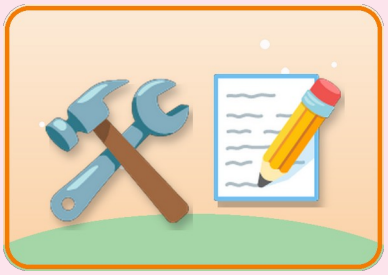
Arun had ten AI certificates and sent forty applications. Not one replied. He wondered whether the market had no jobs or whether something was wrong with him.

Pause & predict: What might his CV be missing?

SCENE 2

Vignesh reviewed his profile in five minutes: no projects with results, an empty GitHub and a generic CV. 'Show me one thing you built that someone used,' he said.

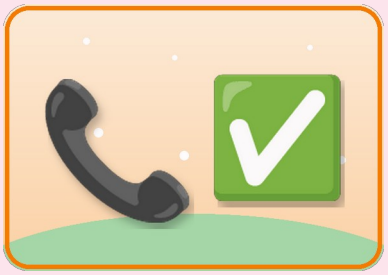
Pause & predict: Why do recruiters look for built work?



SCENE 3

Arun spent two months building a Tamil invoice reader for his uncle's shop, with a README, a demo video and a measured 92% field accuracy. He wrote a blog post about mistakes he fixed.

Pause & predict: What makes this project convincing?



SCENE 4

Vignesh referred him. Ms. Tara called within a week. In the interview Arun told the invoice story with STAR, including a data-leakage mistake. He received an offer.

⏸ **Pause & predict:** Why did admitting a mistake help?

🌟 The big idea

Evidence of real work, told honestly, turns applications into conversations.

💬 Discuss with your classmates

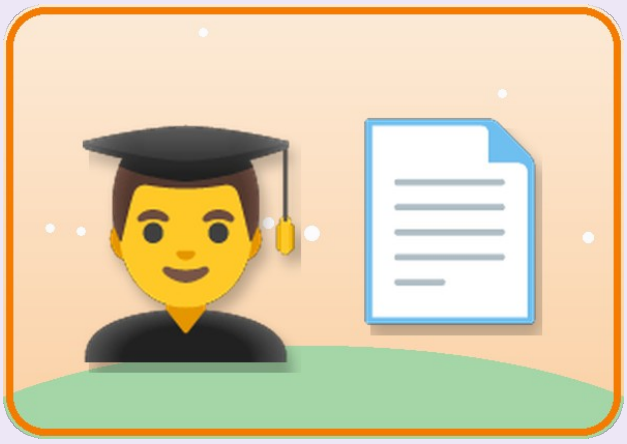
1. Which of your projects could become portfolio-ready?

2. What is your best STAR story?

SCENARIO: Portfolio to Profession

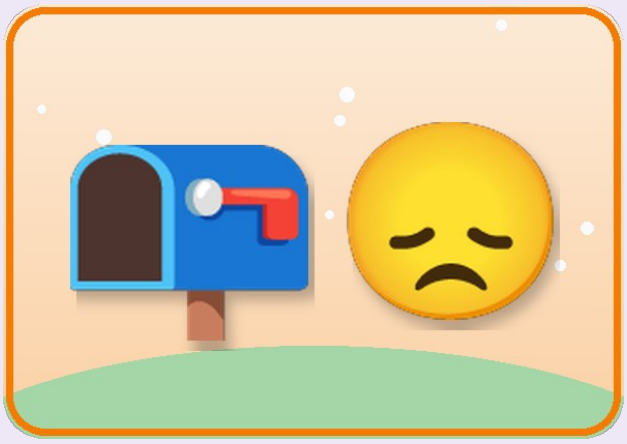
Read the scenario. Think about the choices made at each step.

1



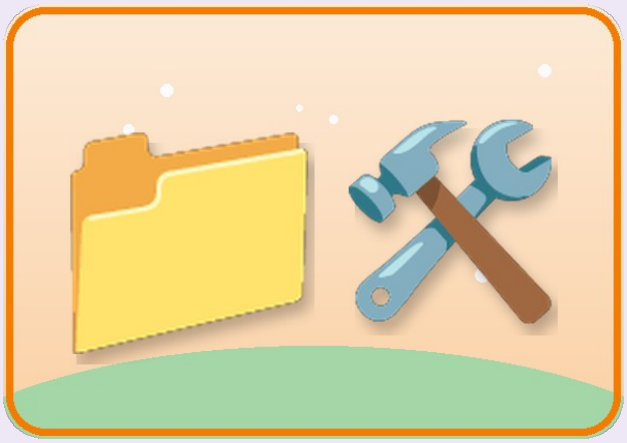
Arun's CV lists ten online AI courses.

2



He gets no replies to 40 applications.

3



He builds 3 projects with demos and READMEs.

4

Recruiters call; he explains results with STAR.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: Where portfolios live

GitHub

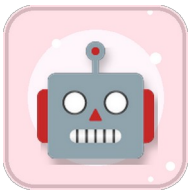
Kaggle

Blog posts

Papers & posters

Demo videos

LinkedIn



BOLT SAYS:

Recruiters can't see inside your brain. Show them what you've built!

FUN FACT!

The 2009 Netflix Prize winners earned $1 million for a 10% better recommender.

Reflect: where have YOU seen this in real life? Note it down.

CASE STUDY: The Netflix Prize

CASE FILE

In 2006 Netflix offered $1 million to any team that could improve its movie-rating predictions by 10%. Over 40,000 teams from more than 180 countries took part. In 2009 BellKor's Pragmatic Chaos won with a 10.06% improvement by blending over a hundred models. Netflix later said it did not deploy the full winning ensemble, as the extra accuracy did not justify the engineering cost and its business was moving to streaming.

Analyse the case

1. What did competitors gain besides prize money?

2. Why might the best model not be deployed?

3. How should competition work be shown in a portfolio?

My verdict: Was AI used well here? Why?

LAB ACTIVITY: Portfolio Gallery Walk

You need:

Laptops

Sticky notes

Whole class

Steps:

1.
   Open one project's README on a laptop.
2.
   Classmates walk around and read for 2 min.
3.
   They note: what is it, does it work, why care?
4.
   Rewrite your README using the feedback.

THINK & TALK

What made a README easy to understand?

Record your observations & results:

💻 CODE LAB: Application Funnel

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
apps, reply, interview, offer = 60, 0.10, 0.5, 0.33
replies = apps * reply
offers = replies * interview * ____
print(replies, round(offers, 1))
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. What is printed?**

MODIFY **T3. Reply rate 0.25 via referrals.**

EXTEND **T4. Plot the funnel.**

WORKSHEET: Portfolio to Profession (Part A)

Name: _____ Date: _____

1. Match the item to what it shows.

README

Demo video

Blog post

Tests & CI

Code quality

Explains the project

Tells the story

Shows it working

2. Strong or weak portfolio evidence? Tick ✓.

	Strong	Weak
Live demo + clear metrics		
Unlabelled notebook only		
Results vs a baseline		
Copied tutorial, no changes		

WORKSHEET: Portfolio to Profession (Part B)

Name: _____ Date: _____

1. Job-search maths. Use the word box.

WORD BOX: 2 ✿ 15 ✿ 3

40 applications × 5% reply = ____

Referrals: 30% of 50 reply = ____

1 project per skill, 3 skills = ____

2. Match the interview round to what it tests.

Coding round ●

ML system design ●

Behavioural ●

Take-home task ●

● **Teamwork and judgement**

● **Real work sample**

● **Algorithms and code**

● **Architecture trade-offs**

PUZZLE & DEFINE: Portfolio to Profession

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Portfolio

Meaning: _____

Example: _____

README

Meaning: _____

Example: _____

STAR

Meaning: _____

Example: _____

? QUIZ TIME: Portfolio to Profession

Circle the right answer!

Q1. A project's front page is its...

a) README

b) Zip file

c) Logo

Q2. STAR: Situation, Task, Action...

a) Reason

b) Result

c) Repeat

Q3. The strongest project shows...

a) Course list

b) Selfies

c) Measured impact

Q4. Netflix Prize winners improved by...

a) 10%

b) 1%

c) 50%

Q5. An unchanged copied tutorial is...

a) Strong

b) Weak evidence

c) Award-worthy

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Portfolio to Profession

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	A good README states problem, data and results.		
	Certificates alone prove job readiness.		
	STAR structures behavioural answers.		
	Referrals often raise reply rates.		
	Claiming solo credit for team work is fine.		
	The Netflix Prize was won in 2009.		

HOME CONNECT

Rewrite the README of your best project: problem, data, method, result, demo and your role.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Portfolio to Profession

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why do recruiters value projects over certificates?**

COMPARE **Q2. Kaggle project vs end-to-end product project.**

APPLY **Q3. 40 applications, 6% reply rate: replies?**

EVALUATE **Q4. List a team project as solo work?**

CREATE **Q5. STAR answer about a model that failed.**

EXAM PREP: Portfolio to Profession

Name: _____ Date: _____

Time: 30 min Max marks: 13 Answer all questions.

PART A • Short answers (2 marks each)

1. What makes a strong portfolio project?

2. What should a README include?

3. Explain the STAR method.

4. Name two interview round types.

PART B • Long answer (5 marks)

5. Plan a portfolio and job-search strategy to secure an AI role within six months.

UNIT REVIEW: AI RISE

Name: _____ Date: _____

Circle the right answer!

Q1. Keeps models running in production:

a) MLOps | b) Design | c) Sales

Q2. A cheap way to test a career:

a) Guess | b) Internship | c) Wait

Q3. 0.5×4 + 0.3×5 + 0.2×3 =

a) 3.6 | b) 4.5 | c) 4.1

Q4. STAR ends with...

a) Result | b) Repeat | c) Reason

Q5. Netflix Prize needed improvement of...

a) 1% | b) 10% | c) 50%

Q6. Best evidence for recruiters:

a) Certificates | b) Selfies | c) Projects

WORD SEARCH: AI RISE

🔍 **Find these words. They go across → or down ↓.**

MLE

AI PM

MLOPS

PORTFOLIO

README

STAR

T	G	R	E	A	D	M	E	L	D
B	P	O	R	T	F	O	L	I	O
H	M	E	N	T	O	R	I	U	W
A	I	P	M	H	S	T	A	R	D
O	F	F	E	R	L	F	K	P	A
B	M	N	E	T	W	O	R	K	C
K	L	A	L	R	E	S	U	M	E
K	M	L	O	P	S	L	E	O	W
C	I	N	T	E	R	V	I	E	W
K	W	G	P	K	M	L	E	E	I

I finished AI RISE! Teacher's sign: _____

MONTH 5

AI CREATE

Try • Build • Innovate

Area: AI Tools

In this unit, you will explore:

Topic 9

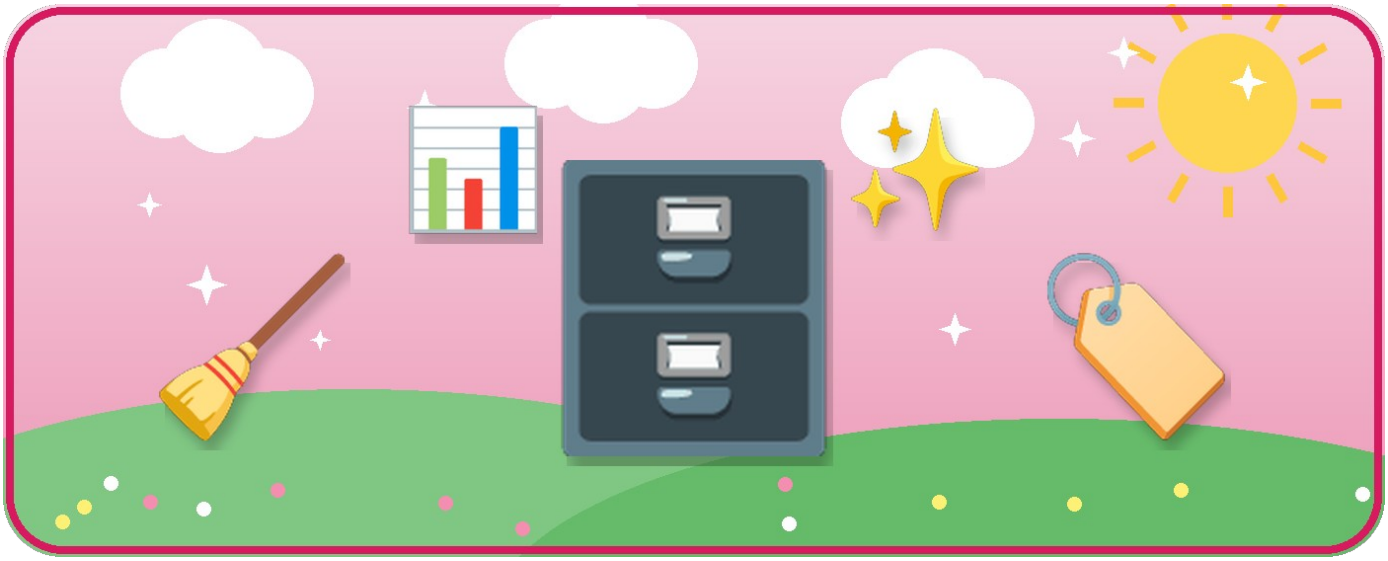
AI Data Studio

Topic 10

Smart Automation Tools

TOPIC 9 • MONTH 5 • AI TOOLS

AI Data Studio



LET'S LEARN

- Great AI starts with well-built, well-labelled data.
- Pipelines clean, transform and validate data automatically.
- Document data and use synthetic data with care.

Inside the data studio

Cleaning

Labelling

Pipelines

Documentation

KEY TERMS

ETL Extract, transform, load

Labelling Adding ground truth

Synthetic Generated data

📖 CONCEPT EXPLAINED: AI Data Studio

💡 In this chapter

Behind every reliable AI system is a data studio: the people, processes and tools that collect, clean, label, transform, document and version data. Data-centric AI recognises that improving data is often the fastest route to better models. Postgraduates leading AI projects need to design data work as carefully as they design models.

📥 1. Collecting data responsibly

Good data starts with a clear purpose: what decisions or predictions will this data support, and which populations and conditions must it represent? Collection plans should consider coverage across regions, languages, devices and groups, and how data will be refreshed as conditions change.

Personal data requires lawful grounds, informed consent where appropriate, minimisation and security. In India the Digital Personal Data Protection Act, 2023 sets duties for data fiduciaries. Record the source, licence and collection method of every dataset so that later users know what they may do with it.

✏️ WORKED EXAMPLE

Receipt dataset plan: 3,000 receipts from 60 shops in 4 districts, Tamil and English, phone photos in varied light; shop owners' consent recorded; customer names blurred before storage.

🧹 2. Cleaning and validation

Raw data contains duplicates, missing values, typos, inconsistent units, impossible values and outliers. Cleaning should follow explicit, documented rules: remove exact duplicates, standardise formats, impute or flag missing values, and reject values outside valid ranges. Every rule should be reproducible in code, not applied by hand in a spreadsheet.

Validation checks run automatically whenever new data arrives: schema checks for types and columns, range checks, uniqueness checks and distribution comparisons with past data. Tools such as Great Expectations or simple assertions in pandas catch problems before they silently corrupt a model.

🤔 **PAUSE & THINK**

Missing prices could be filled with the median, the item's average or left as 'unknown'. How would each choice affect a demand-forecasting model?

🏷️ 3. Labelling at quality

Labels are the model's definition of truth, so labelling deserves engineering discipline. Clear guidelines define each class with examples and edge cases, and include an 'unsure' option. Gold-standard items with known answers check labeller accuracy; multiple labellers on a sample measure agreement, for example with Cohen's kappa.

Disagreement is information: it often reveals ambiguous guidelines or genuinely hard cases. ImageNet showed both the power of crowd labelling at scale and its risks, when later audits found offensive labels in person categories. Fair pay, clear instructions and review of sensitive categories are part of responsible labelling.

✏️ **WORKED EXAMPLE**

Agreement check: two labellers agree on 160 of 200 items (80%). After clarifying three confusing categories in the guidelines, agreement on a fresh sample rises to 95%.

🔄 4. Pipelines, versioning and documentation

Data pipelines automate extract, transform and load (ETL) steps on a schedule, using tools from simple scripts and cron jobs to workflow engines such as Airflow. Each run should be logged, and datasets versioned so any model can be traced to the exact data that trained it.

Datasheets for datasets record motivation, composition, collection process, preprocessing, uses, distribution and maintenance. Synthetic data, generated to fill gaps or protect privacy, should be labelled as such, validated against real data and never assumed to be automatically private, since generators can memorise real records.

🖐️ **TRY IT**

Take a messy public dataset. Write a Python cleaning script with at least four rules and three validation checks, version the result, and write a one-page datasheet.

✅ Key takeaways

- Collect with purpose, coverage, consent and documented licences.
- Clean with reproducible rules; validate every new batch automatically.
- Labelling needs guidelines, gold items and agreement measurement.
- Pipelines automate ETL; versioning links models to exact data.
- Datasheets document data; synthetic data needs validation and privacy checks.

❓ Quick check: answer in one or two lines.

1. Why should cleaning rules be written in code?

2. How would you measure labelling quality?

3. What does a datasheet for a dataset contain?

🔬 CONCEPT DEEP DIVE: AI Data Studio

🔷 Data-centric AI improves results by improving data: coverage, cleanliness, labels and documentation.

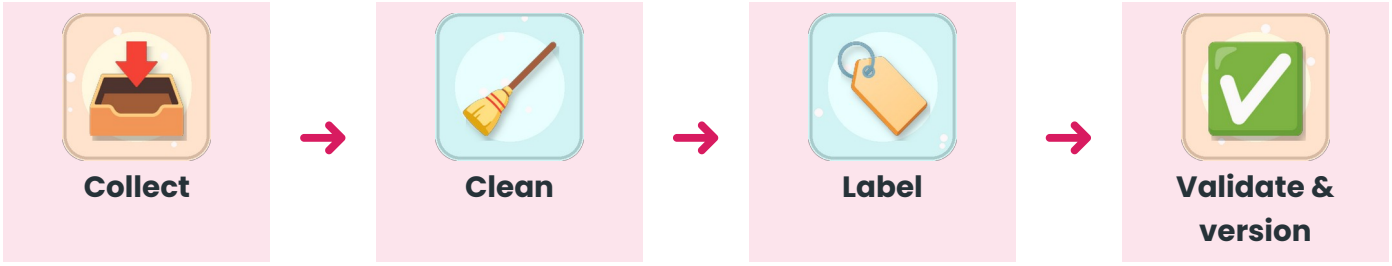
🔷 Collect with consent and purpose; record sources, licences and any personal data handling.

🔷 Clean with rules: deduplicate, handle missing values, validate ranges and formats, and log every change.

🔷 Labelling needs guidelines, gold examples and agreement checks such as Cohen's kappa between annotators.

🔷 Pipelines automate ETL and validation; version datasets, write datasheets, and treat synthetic data as a supplement.

🔄 How it works



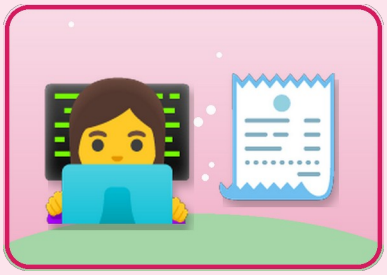
💻 Cleaning with pandas

```
import pandas as pd
df = pd.read_csv('receipts.csv').drop_duplicates()
df['price'] = df['price'].fillna(df['price'].median())
df = df[df['price'] > 0]   # validation rule
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Receipts That Disagreed

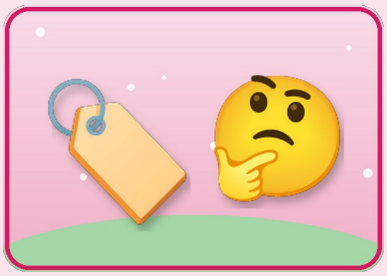
Characters: *Lavanya (MSc Data Science), labellers Gopi and Meena, shop owner Mr. Rafiq.*



SCENE 1

Lavanya collected 3,000 receipt photos from local shops, with owners' consent, to train an item reader. Gopi and Meena began labelling item categories.

Pause & predict: What should be decided before labelling starts?



SCENE 2

A check on 200 receipts showed they agreed only 80% of the time. Was 'coconut oil' cooking oil or personal care? Was 'rava' a grain or a snack ingredient?

Pause & predict: What does disagreement reveal?

SCENE 3

Lavanya wrote guidelines with examples and an 'unsure' option, added gold items and removed 140 duplicate photos. Mr. Rafiq explained local product names she had misunderstood.

Pause & predict: Why involve a domain expert?

SCENE 4

Agreement rose to 95%, and model accuracy jumped more than any model change had achieved. Lavanya published a datasheet describing sources, consent and limits.

⏸ **Pause & predict:** Why did better labels beat a better model?

🌟 The big idea

Clear guidelines, clean data and local knowledge often improve AI more than new algorithms.

💬 Discuss with your classmates

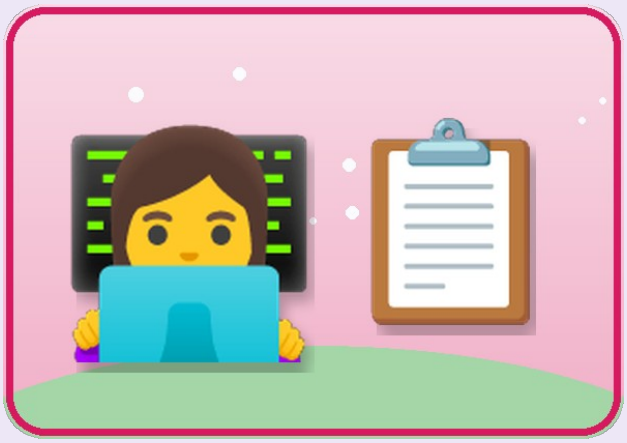
1. What ambiguous labels exist in your field?

2. Who are the domain experts for your data?

SCENARIO: AI Data Studio

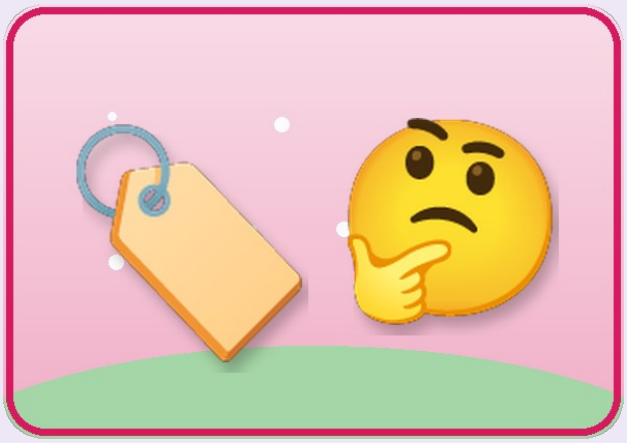
Read the scenario. Think about the choices made at each step.

1



Lavanya builds a dataset of Tamil shop receipts.

2



Two labellers disagree on 20% of items.

3

She writes clear guidelines and cleans duplicates.

4

Agreement rises to 95% and accuracy improves.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Data studio tools

pandas

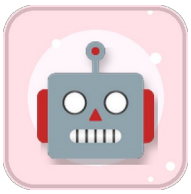
SQL

Labelling tools

Data lakes

Schedulers

BI dashboards



BOLT SAYS:

Feed me messy data and I learn messy lessons. Clean data, clear answers!

FUN FACT!

ImageNet's 14 million crowd-labelled images fuelled the 2012 deep-learning leap.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: ImageNet: The Dataset That Changed AI

CASE FILE

Led by Fei-Fei Li, the ImageNet project, presented in 2009, labelled around 14 million images in over 20,000 categories using crowd workers with quality checks. Its annual challenge drove fast progress, and in 2012 the AlexNet deep network sharply cut error rates, sparking the deep-learning boom. Later audits found offensive labels in some 'person' categories, and in 2019 many of those images were removed: quality and ethics matter as much as scale.

🔍 Analyse the case

1. Why did a large labelled dataset speed up progress?

2. How can labelling at scale introduce harm?

3. What checks would you add to a crowd-labelling project?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Messy Data Clinic

You need:

Messy sheets

Highlighters

Teams of 3

Steps:

1.
   Get a printed table with 20 messy rows.
2.
   Mark duplicates, blanks, typos and outliers.
3.
   Agree a rule to fix each problem type.
4.
   Count how many rows survive cleaning.

THINK & TALK

Which fixes could change the model's answer?

Record your observations & results:

💻 CODE LAB: Data Quality Report

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
 1  import pandas as pd
 2  df = pd.DataFrame({'price': [40, None, 40, -5, 60], 'item':
['rice', 'dal', 'rice', 'oil', 'ghee']})
 3  print(df.isna().sum().sum(), df.duplicated().sum(), (df['price']
____ 0).sum())
```

COMPLETE **T1. Count negative prices.**

PREDICT **T2. Three numbers printed?**

FIX **T3. Clean all three issues.**

EXTEND **T4. Write a validate(df) function.**

WORKSHEET: AI Data Studio (Part A)

Name: _____ Date: _____

1. Match the cleaning step to its job.

Deduplication ●

Imputation ●

Normalisation ●

Validation rule ●

● Reject impossible values

● Remove repeated rows

● Put values on one scale

● Fill missing values

2. Keep or fix? Tick ✓.

	Keep	Fix
Age = 34		
Age = -5		
Pincode 629001		
Same order listed twice		

WORKSHEET: AI Data Studio (Part B)

Name: _____ Date: _____

1. Data maths. Use the word box.

WORD BOX: 920 ✿ 80 ✿ 5

1,000 - 50 dupes - 30 invalid = ____

Labellers agree on 160 of 200 → ____%

100 GB in 20 GB parts → ____ parts

2. Order a data pipeline. Write 1–4.

Validate & version

☐

Collect with consent

☐

Label with guidelines

☐

Clean & deduplicate

☐

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

PUZZLE & DEFINE: AI Data Studio

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

ETL

Meaning: _____

Example: _____

Labelling

Meaning: _____

Example: _____

Synthetic

Meaning: _____

Example: _____

? QUIZ TIME: AI Data Studio

Circle the right answer!

Q1. Extract, transform, load is...

a) ETL

b) CSS

c) VoIP

Q2. Filling missing values is...

a) Pruning

b) Imputation

c) Hashing

Q3. ImageNet has about...

a) 14 thousand

b) 1 billion

c) 14 million

Q4. A dataset is documented in a...

a) Datasheet

b) Playlist

c) Map

Q5. Many labellers' labels need...

a) Guessing

b) Agreement checks

c) No checks

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: AI Data Studio

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Duplicates can leak across train and test.		
	Synthetic data can never reveal private details.		
	Labelling guidelines improve agreement.		
	Datasheets document how data was made.		
	Rows with missing values must always be deleted.		
	ImageNet has about 14 million images.		

HOME CONNECT

Open any public dataset. Find three quality problems and write how you would fix each one.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: AI Data Studio

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why does label noise hurt models?**

COMPARE **Q2. Real vs synthetic data.**

APPLY **Q3. 1,000 rows, 5% duplicates, 3% invalid: rows left?**

EVALUATE **Q4. Is more data always better than cleaner data?**

CREATE **Q5. Labelling guide for crop-disease photos.**

📝 EXAM PREP: AI Data Studio

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What is data-centric AI?

2. Define inter-annotator agreement.

3. Name two cleaning steps.

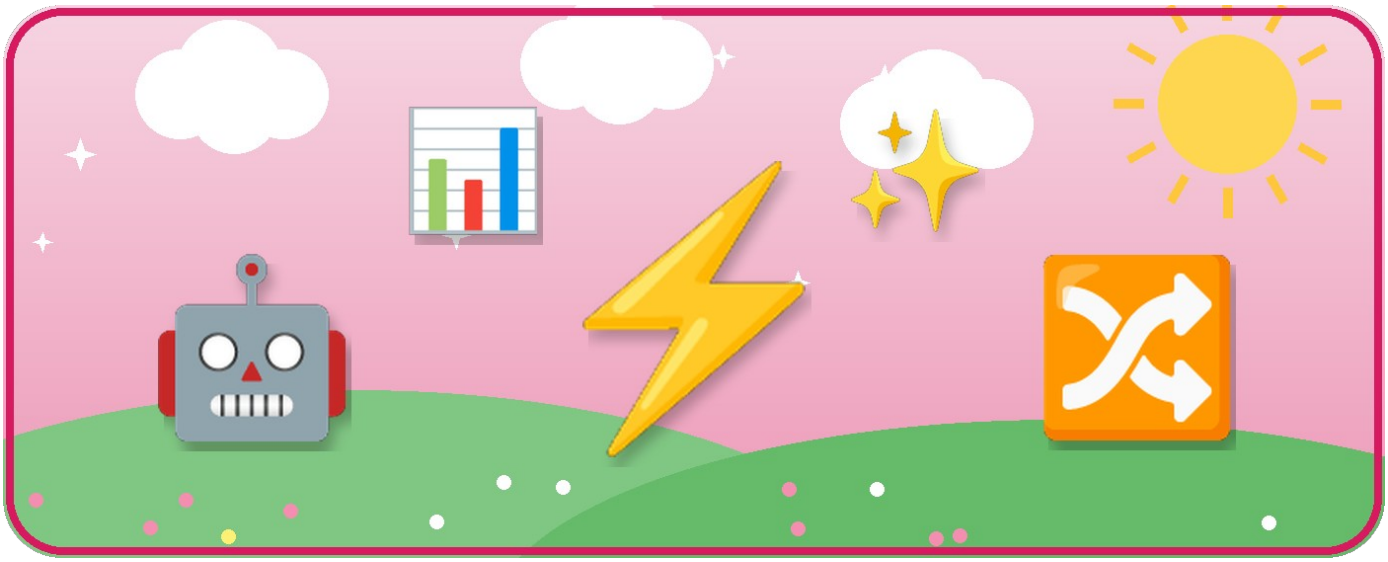
4. What is a datasheet for datasets?

PART B • Long answer (5 marks)

5. Design a data pipeline and labelling process for a new dataset in an Indian language.

TOPIC 10 • MONTH 5 • AI TOOLS

Smart Automation Tools



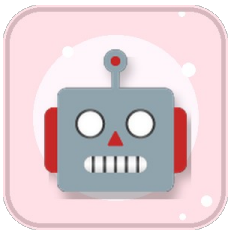
LET'S LEARN

- Automation tools link apps, data and AI into workflows.
- AI handles judgement; rules handle the routine.
- Keep humans in the loop and monitor every bot.

Automation building blocks

Triggers

Rules



AI steps

Human approval

KEY TERMS

RPA Software robots

Trigger Event starting a flow

HITL Human in the loop

📖 CONCEPT EXPLAINED: Smart Automation Tools

💡 In this chapter

Much office work is repetitive: copying data between systems, sorting emails, compiling reports, sending reminders. Automation tools, now enhanced with AI, can take over these tasks and free people for work needing judgement. Designed carelessly, though, automations fail silently, frustrate customers and create legal liability. This chapter covers how to automate wisely.

⚡ 1. Workflow automation basics

An automated workflow starts with a trigger, such as a new form submission, an incoming email or a scheduled time. It then runs actions: reading and transforming data, applying rules, calling other applications through their APIs and sending notifications. Low-code tools let people without programming experience build such flows visually.

Before automating, map the current process step by step, measure how long each step takes and where errors occur. Automating a broken process only produces errors faster. Often the best first step is simplifying the process itself, then automating what remains.

✏️ WORKED EXAMPLE

Daily staff report: trigger at 6 pm → collect form entries into a sheet → flag missing submissions → AI drafts a summary → manager reviews → email sent to the director → run logged.

🤖 2. RPA, APIs and AI steps

Robotic process automation (RPA) mimics human clicks and typing on screens, useful for legacy systems without APIs, but fragile when screens change. API integrations exchange data directly between systems and are far more robust where available. Many organisations combine both during modernisation.

AI steps handle tasks rules cannot: classifying free-text emails, extracting fields from invoices and IDs, summarising long reports, drafting replies or planning multi-step agent actions. Because AI outputs can be wrong, they work best when paired with validation rules, confidence thresholds and human review for uncertain or important cases.

🤔 **PAUSE & THINK**

An invoice-extraction model is 97% accurate per field. With 10 fields per invoice, roughly what share of invoices will have at least one error? What does that imply for review?

👤 3. Humans in the loop

Human-in-the-loop design decides where people review, approve or handle exceptions. Typical rules route high-value transactions, low-confidence AI outputs, complaints and anything legally sensitive to a person. Clear escalation paths matter: customers should always be able to reach a human.

The Air Canada case shows why. In 2024 a Canadian tribunal held the airline responsible for wrong bereavement-fare advice from its website chatbot, rejecting the idea that the bot was a separate entity. Organisations are accountable for what their automations say and do.

✏️ **WORKED EXAMPLE**

Routing rule set: refund over ₹5,000 → human; FAQ question with confidence above 0.9 → automatic reply; everything else → AI draft reviewed by staff within 2 hours.

📊 4. Monitoring and governance

Every automation needs an owner, logs of each run, alerts when steps fail, and periodic review of outputs. Silent failures are the biggest risk: a flow that stops sending reminders may go unnoticed for weeks. Dashboards should show volumes, error rates, time saved and items waiting for human review.

Governance covers access rights, data protection, testing changes before going live, version control for workflows and a list of all automations in the organisation. Measure benefits honestly, including time spent maintaining automations, so that decisions to expand are based on evidence.

🖐️ **TRY IT**

Map a weekly task in your department. Design an automation with trigger, steps, one AI step, two human checkpoints, logging and an error alert. Estimate weekly time saved.

✅ Key takeaways

- Workflows run on triggers and actions; map and simplify before automating.
- RPA mimics screens and is fragile; APIs are robust; AI handles messy inputs.
- Pair AI steps with validation, confidence thresholds and human review.
- Organisations stay accountable for automated outputs (Air Canada, 2024).
- Own, log, alert, review and measure every automation.

❓ Quick check: answer in one or two lines.

1. Why map a process before automating it?

2. Compare RPA and API integration.

3. Where should humans stay in an automated refund process?

🔬 CONCEPT DEEP DIVE: Smart Automation Tools

🔷 Workflow automation links apps through triggers and actions; low-code tools let non-programmers build flows.

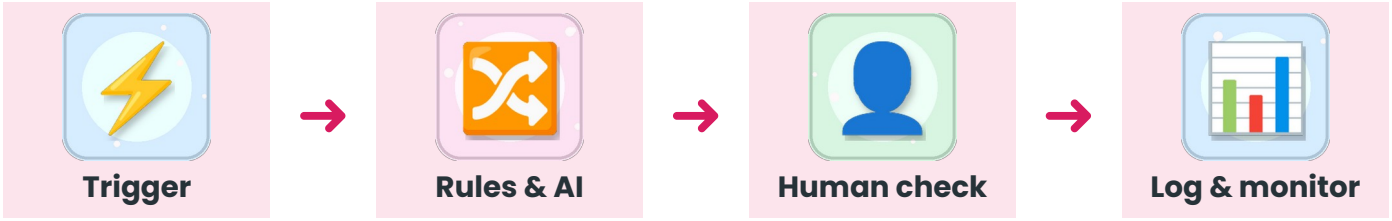
🔷 RPA mimics clicks on screens and breaks when screens change; API integrations are sturdier where available.

🔷 AI steps classify messages, extract fields from documents, draft replies or plan multi-step agent actions.

🔷 Human-in-the-loop design sets approval points, limits and escalation paths for risky or unusual cases.

🔷 Every bot needs logging, error alerts, an owner and periodic review; the organisation stays accountable.

🔄 How it works



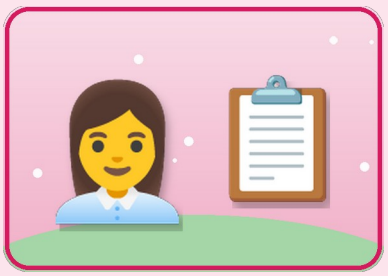
💻 Routing with rules and AI

```
def route(ticket):
    if ticket['amount'] > 5000: return 'human'
    if ticket['type'] == 'faq': return 'auto_reply'
    return 'ai_draft_then_review'
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: The Report That Wrote Itself

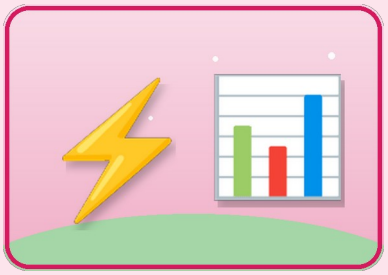
Characters: *Shalini (office administrator), director Dr. Mohan, IT volunteer Kiran.*



SCENE 1

Every evening Shalini spent an hour copying staff updates from messages into a report for Dr. Mohan. Mistakes crept in when she was rushed.

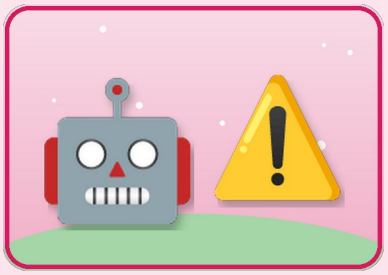
Pause & predict: Which steps are good candidates for automation?



SCENE 2

Kiran built a flow: staff submit a short form; entries land in a sheet; missing names are flagged at 6 pm. An AI step drafts a summary of highlights and issues.

Pause & predict: Why flag missing submissions?



SCENE 3

One evening the AI summary said a project was 'completed' when a staff member had written 'almost completed'. Shalini caught it during her review before the email went out.

Pause & predict: Why keep a human review step?

SCENE 4

They added a rule: the AI must quote the exact status words. Shalini now spends 10 minutes reviewing instead of an hour typing, and the log shows every report sent.

⏸ **Pause & predict:** How did the team improve the AI step?

🌟 The big idea

Automate the routine, keep humans on the judgement, and log everything.

💬 Discuss with your classmates

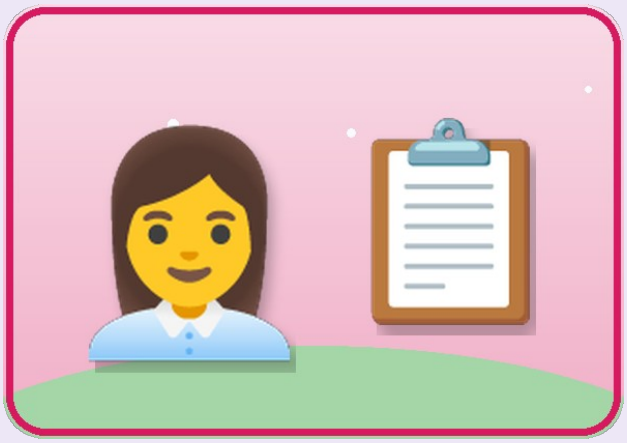
1. Which daily task in your work could follow this pattern?

2. Where must a human always check?

SCENARIO: Smart Automation Tools

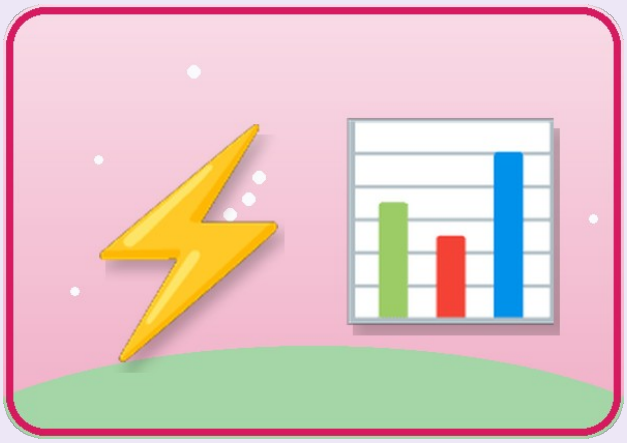
Read the scenario. Think about the choices made at each step.

1



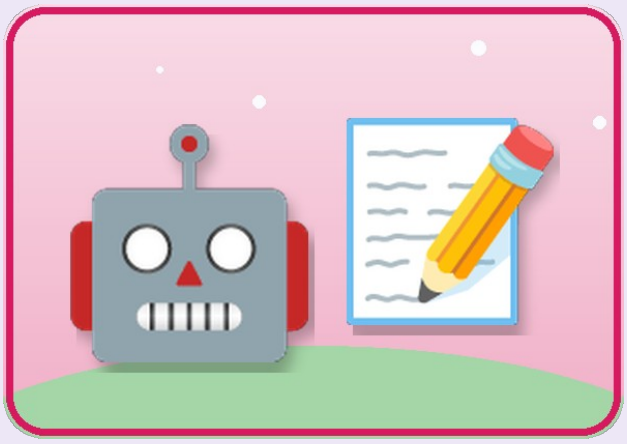
Shalini compiles staff reports by hand every day.

2



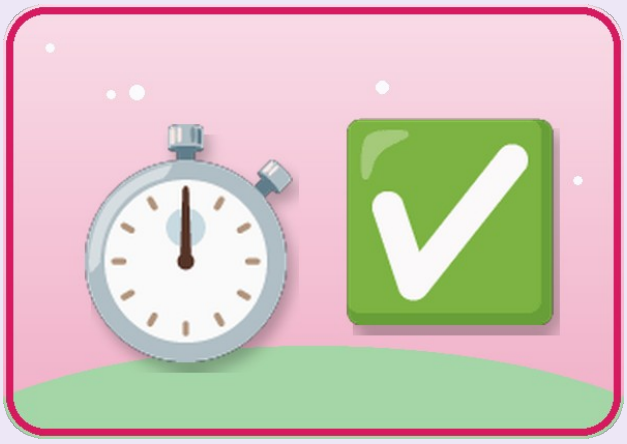
A form now feeds reports into a sheet automatically.

3



AI drafts a summary; Shalini reviews before sending.

4



She saves an hour daily and catches two errors.

Talk about it:

What decision was made? What would YOU have done differently?

What happens next? Write or sketch scene 5.

LOOK AROUND YOU: What teams automate

Email triage

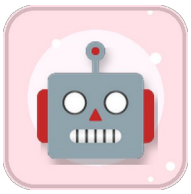
Invoice entry

Scheduling

Support replies

Weekly reports

Form processing



BOLT SAYS:

Give me the boring, repeatable work. Keep the tricky decisions for humans!

FUN FACT!

In 2024 a tribunal held Air Canada liable for wrong refund advice from its chatbot.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: The Chatbot's Refund Promise

CASE FILE

In 2022 a grieving passenger asked Air Canada's website chatbot about bereavement fares. It said he could buy a full-price ticket and claim the reduced fare within 90 days, which the airline's real policy did not allow. When Air Canada refused the refund, he went to British Columbia's Civil Resolution Tribunal. In February 2024 the tribunal rejected the claim that the chatbot was responsible for its own words and held the airline liable.

🔍 Analyse the case

1. Who is responsible for an automated answer?

2. Which safeguard could have prevented this?

3. How should chatbots handle policy questions?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Paper Workflow

You need:

Sticky notes

String

Teams of 4

Steps:

1.
   Pick a weekly office task, e.g. a report.
2.
   Map each step on notes linked by string.
3.
   Mark steps for rules, AI or a human.
4.
   Estimate minutes saved per week.

THINK & TALK

Which step must always stay human?

Record your observations & results:

💻 CODE LAB: Ticket Router

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
def route(amount, kind):
    if amount > 5000: return 'human'
    return 'auto' if kind == ____ else 'ai_draft'
print(route(8000, 'faq'), route(200, 'faq'), route(200,
'refund'))
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. Three outputs?**

EXPLAIN **T3. Why check amount first?**

EXTEND **T4. Log each decision.**

WORKSHEET: Smart Automation Tools (Part A)

Name: _____ Date: _____

1. Match the block to its role.

Trigger ●

Rule ●

AI step ●

Human approval ●

● Reads or drafts content

● Checks risky actions

● Starts the workflow

● Fixed if-then step

2. Automate or keep human? Tick ✓.

	Automate	Human
Copy form data into a sheet		
Approve a large refund		
Send meeting reminders		
Handle an upset customer		

WORKSHEET: Smart Automation Tools (Part B)

Name: _____ Date: _____

1. Automation maths. Use the word box.

WORD BOX: 75 ✿ 15 ✿ 260

Save 15 min a day for 5 days = ____ min

20 min task, 75% automated → saves ____ min

1 hour a day × 260 workdays = ____ hours

2. Circle good automation practices.

Log every run

No error alerts

Approve big actions

Shared admin passwords

Test before going live

Unlimited retries

PUZZLE & DEFINE: Smart Automation Tools

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

2. Define it! Write each term in your own words with an example.

RPA

Meaning: _____

Example: _____

Trigger

Meaning: _____

Example: _____

HITL

Meaning: _____

Example: _____

? QUIZ TIME: Smart Automation Tools

Circle the right answer!

Q1. An event that starts a flow:

a) Trigger

b) Stop

c) Theme

Q2. Software robots copying UI steps:

a) CNN

b) RPA

c) GPS

Q3. Human checks before risky actions:

a) Random

b) Blind

c) HITL

Q4. Air Canada was liable for its...

a) Chatbot

b) Pilot

c) Meals

Q5. Bots need ongoing...

a) Rest

b) Monitoring

c) Parties

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Smart Automation Tools

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Triggers start automated workflows.		
	RPA is always sturdier than APIs.		
	AI can extract fields from documents.		
	Risky actions should need human approval.		
	A chatbot is legally responsible for itself.		
	Air Canada was held liable in 2024.		

HOME CONNECT

List three repetitive tasks in your week. Sketch a trigger, the steps and one human check for each.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Smart Automation Tools

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why mix rules and AI in one workflow?**

COMPARE **Q2. RPA vs API integration.**

APPLY **Q3. Task 20 min/day, 75% automated: weekly saving?**

EVALUATE **Q4. Should a bot send refunds without review?**

CREATE **Q5. Automation for daily staff reports.**

📝 EXAM PREP: Smart Automation Tools

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. What is a workflow trigger?

2. Define RPA.

3. What is human-in-the-loop?

4. Why log automations?

PART B • Long answer (5 marks)

5. Design an AI-assisted automation for an office process, with safeguards.

UNIT REVIEW: AI CREATE

Name: _____ Date: _____

Circle the right answer!

Q1. Filling missing values:

a) Imputation | b) Pruning | c) Hashing

Q2. ImageNet labels came from...

a) Robots | b) Crowd workers | c) Dice

Q3. 160 of 200 labels agree =

a) 60% | b) 16% | c) 80%

Q4. An event that starts a flow:

a) Trigger | b) Stop | c) Theme

Q5. Mimics clicks on screens:

a) API | b) RPA | c) BI

Q6. Air Canada's chatbot case shows...

a) Bots are free | b) No one's liable | c) Firm is liable

WORD SEARCH: AI CREATE

🔍 **Find these words. They go across → or down ↓.**

ETL	LABELLING	SYNTHETIC	RPA	TRIGGER	HITL

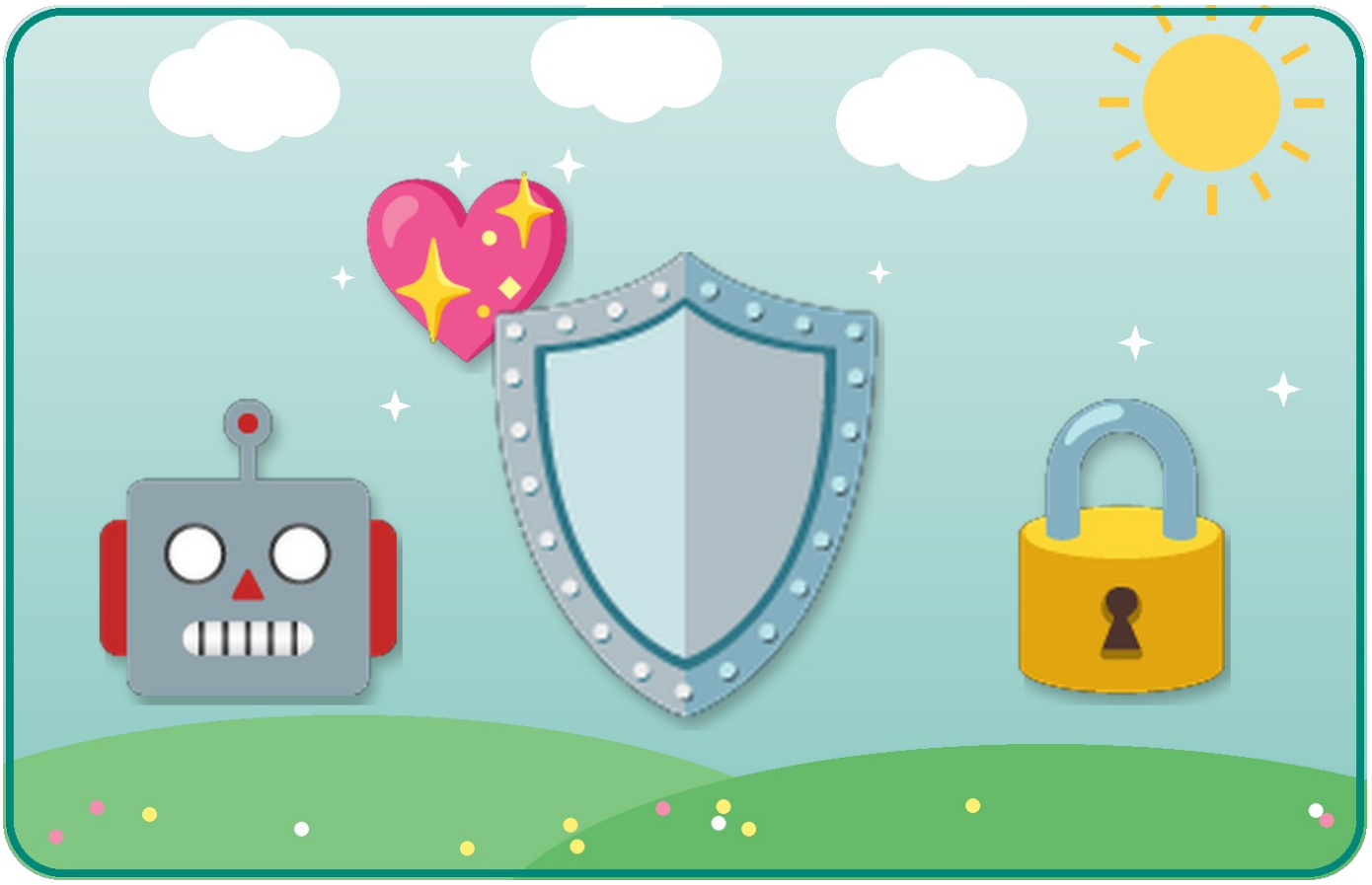
C S A W T N B H W H
M Y P R P A I P L I
B N P A N D A S A T
O T R I G G E R B L
T H O O E B B R E L
G E V E E U E O L D
F T A O T M E B L O
K I L E L P W F I S
P C S C H E M A N R
W O R K F L O W G L

🏅 **I finished AI CREATE! Teacher's sign: _____**

MONTH 6

AI CARE

Verify • Protect • Respect



Area: Responsible AI

In this unit, you will explore:

Topic 11

Fair AI, Fair Future

Topic 12

Ownership in the AI Age

TOPIC 11 • MONTH 6 • RESPONSIBLE AI

Fair AI, Fair Future

LET'S LEARN

- Bias enters through data, labels, design and use.
- Fairness metrics can conflict; choose them with people.
- Audit, mitigate and monitor fairness over time.

Where bias creeps in

Data

Labels

Objectives

Deployment

KEY TERMS

Parity Equal positive rates

Eq. odds Equal error rates

Bias audit Test across groups

📖 CONCEPT EXPLAINED: Fair AI, Fair Future

💡 In this chapter

AI systems increasingly influence who gets loans, jobs, medical attention, admission and even bail. If they reproduce or amplify unfair patterns, they can harm millions quietly and at scale. Fairness in AI is both a technical and a social challenge: it requires measurement and mitigation, but also choices about values that affected people should help make.

📥 1. Where bias comes from

Historical bias arises when past decisions encoded in labels were themselves unfair, such as loans denied to certain communities. Representation bias occurs when some groups are under-sampled, as with rural users, darker skin tones or regional accents. Measurement bias comes from proxies that mean different things for different groups, such as using healthcare spending as a proxy for health need.

Bias can also arise from aggregation, when one model serves very different populations, and from deployment, when a tool is used for a purpose or population it was not designed for. Removing a sensitive column rarely solves the problem, because names, locations, schools and purchase histories can act as proxies.

✏️ WORKED EXAMPLE

Gender Shades (2018): commercial gender-classification systems had error rates up to 34.7% for darker-skinned women but at most 0.8% for lighter-skinned men, largely reflecting unbalanced training and evaluation data.

⚖️ 2. Defining fairness

Group fairness definitions compare outcomes across groups. Demographic parity requires equal rates of positive decisions. Equal opportunity requires equal true-positive rates, so qualified people are recognised equally. Equalised odds also equalises false-positive rates. Calibration within groups requires a given score to mean the same risk in every group.

Individual fairness asks that similar individuals receive similar outcomes, which requires a fair way to measure similarity. A practical screen used in employment is the four-fifths rule: if one group's selection rate is below 80% of another's, the gap deserves investigation. No single metric is right for every context.

🤔 **PAUSE & THINK**

For a scholarship screening tool, which matters more: equal selection rates or equal chances for qualified students? Who should decide?

⚔️ 3. Trade-offs and the COMPAS debate

In 2016 ProPublica found that the COMPAS risk tool labelled Black defendants who did not reoffend as higher risk almost twice as often as white defendants. The developer argued the scores were equally calibrated. Both statements could be true, because researchers including Kleinberg and colleagues and Chouldechova proved that when base rates differ, calibration and equal error rates generally cannot hold together.

Fairness therefore involves choices, not just optimisation. Teams must decide which errors are most harmful in their context, document the trade-offs, and involve affected communities, domain experts and legal advisors in those decisions.

✏️ **WORKED EXAMPLE**

Loan model: enforcing equal approval rates may increase defaults in one group; enforcing equal error rates may leave approval rates unequal. The bank must justify which harm it prioritises and why.

🔧 4. Mitigation and auditing

Mitigation can happen before training (collecting more representative data, reweighting, fixing labels), during training (fairness constraints or adversarial debiasing) and after training (group-aware thresholds, human review of borderline cases). Each approach has costs and legal considerations that must be assessed.

Fairness audits should be repeated: before launch, after deployment and whenever data or use changes. They report metrics by group with confidence intervals, examine errors qualitatively and check accessibility for people with disabilities and speakers of different languages. Affected people need explanations and a way to appeal.

🖐️ **TRY IT**

Using a public dataset, train a classifier and report selection rate, true-positive rate and false-positive rate for two groups. Apply one mitigation and describe what improved and what got worse.

✅ Key takeaways

- Bias enters via history, representation, measurement, aggregation and deployment.
- Fairness metrics include parity, equal opportunity, equalised odds and calibration.
- With different base rates, some fairness criteria cannot hold together.
- Mitigate before, during or after training; each has trade-offs.
- Audit repeatedly, involve affected people and provide appeals.

❓ Quick check: answer in one or two lines.

1. Why doesn't removing a sensitive column guarantee fairness?

2. Explain the four-fifths rule.

3. What did the COMPAS debate show about fairness metrics?

🔬 CONCEPT DEEP DIVE: Fair AI, Fair Future

🔷 Bias enters through historical labels, unrepresentative samples, flawed measurements, aggregation and deployment context.

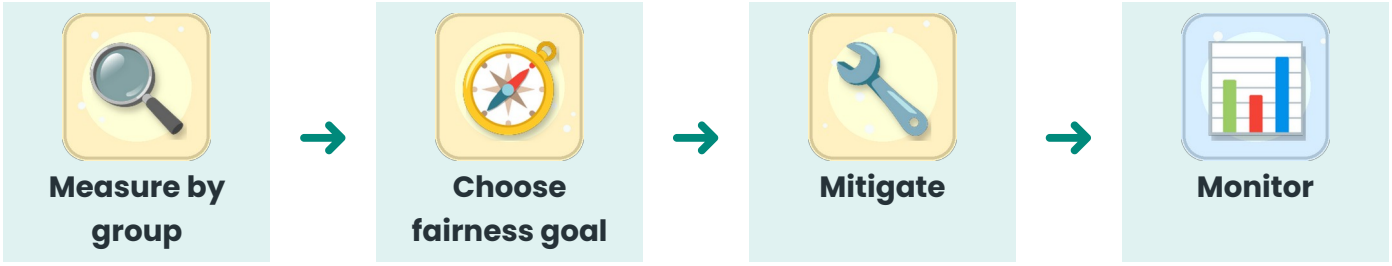
🔷 Group fairness metrics include demographic parity, equal opportunity, equalised odds and calibration within groups.

🔷 When base rates differ, calibration and equal error rates generally cannot all hold at once, so trade-offs must be chosen.

🔷 Mitigation can happen before training (data), during (constraints) or after (thresholds), each with costs.

🔷 Fairness is socio-technical: involve affected people, audit regularly, document trade-offs and provide appeals.

🔄 How it works



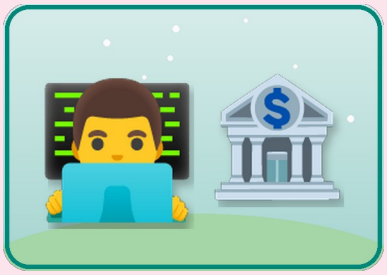
💻 Selection rates by group

```
approved = {'men': 60, 'women': 30}
applied  = {'men': 100, 'women': 75}
rates = {g: approved[g] / applied[g] for g in applied}
print(rates, round(rates['women'] / rates['men'], 2))   # 0.67
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Twice as Likely

Characters: *Imran (MTech AI), bank manager Mrs. Rao, community leader Ms. Kalaivani.*



SCENE 1

Imran built a loan-approval model for a rural cooperative bank. Overall accuracy was 88%. Mrs. Rao was ready to launch until Ms. Kalaivani asked how women applicants fared.

Pause & predict: Why check results by group?

SCENE 2

Imran found women were approved at 40% versus 60% for men, a ratio of 0.67. Among people who repaid, women had been rejected far more often.

Pause & predict: Which fairness metrics does this reveal?

SCENE 3

Past loan labels reflected older lending practices, and 'land ownership' acted as a proxy because land was usually registered in men's names. Imran added repayment history from self-help groups.

Pause & predict: How can a proxy create bias?

SCENE 4

With new features, an agreed fairness goal and human review of borderline cases, the gap narrowed sharply while defaults stayed stable. The bank now publishes a yearly fairness audit.

⏸ **Pause & predict:** Why audit every year?

🌟 The big idea

Fairness must be measured by group, discussed with affected people and checked again and again.

💬 Discuss with your classmates

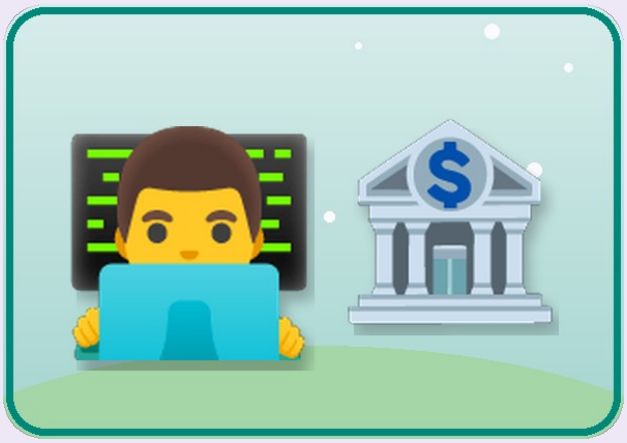
1. Which proxies might hide bias in your field?

2. Who should help choose fairness goals?

🎞️ SCENARIO: Fair AI, Fair Future

Read the scenario. Think about the choices made at each step.

1



Imran builds a loan model for a rural bank.

2

Women applicants are rejected twice as often.

3

He finds past loan labels reflect old biases.

4

New features, audits and reviews narrow the gap.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Where fairness matters

Loans

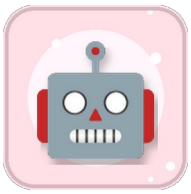
Hiring

Health care

Admissions

Justice

Speech tech



BOLT SAYS:

I copy patterns from data, including unfair ones. Check me for every group!

FUN FACT!

Gender Shades (2018) found face-tool errors up to 34.7% for darker-skinned women.

Reflect: where have YOU seen this in real life? Note it down.

📁 CASE STUDY: COMPAS and the Fairness Debate

CASE FILE

In 2016 ProPublica analysed COMPAS, a risk tool used in US courts, for about 7,000 people in Broward County, Florida. Among defendants who did not reoffend, Black defendants were almost twice as likely as white defendants to be labelled higher risk (about 45% vs 23%). The developer said scores were equally calibrated. Researchers then proved that, when base rates differ, calibration and equal error rates generally cannot both hold.

🔍 Analyse the case

1. Why could both sides claim the tool was 'fair'?

2. Which fairness definition would you choose for bail decisions, and why?

3. Who should decide which trade-off is acceptable?

⚖️ My verdict: Was AI used well here? Why?

LAB ACTIVITY: Fair Share Game

You need:

Applicant cards

Score sheet

Teams of 4

Steps:

1

Score 12 applicant cards with a given rule.

2

Group results by each card's region colour.

3

Compare approval and error rates by group.

4

Change one rule and re-check fairness.

THINK & TALK

Can a rule be fair one way and unfair another?

Record your observations & results:

💻 CODE LAB: Four-Fifths Check

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
1  rate = {'A': 45 / 90, 'B': 24 / 80}
2  ratio = min(rate.values()) / ____(rate.values())
3  print(round(ratio, 2), 'pass' if ratio >= 0.8 else 'flag')
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. What is printed?**

EXPLAIN **T3. What does 'flag' mean?**

EXTEND **T4. Compare true-positive rates.**

WORKSHEET: Fair AI, Fair Future (Part A)

Name: _____ Date: _____

1. Match the fairness idea to its meaning.

Demographic parity ●

● Similar people, similar outcomes

Equal opportunity ●

● Equal positive rates

Calibration ●

● Equal true-positive rates

Individual fairness ●

● Scores mean the same per group

2. Data bias or use bias? Tick ✓.

	Data	Use
Few rural samples in training		
Tool used for a new purpose		
Past labels reflect discrimination		
Staff ignore AI for some groups		

WORKSHEET: Fair AI, Fair Future (Part B)

Name: _____ Date: _____

1. Fairness maths. Use the word box.

WORD BOX: 0.6 ✿ 40 ✿ 20

Approval 30% ÷ 50% = ____

30 approved of 75 → ____%

False positives 45% vs 25% → gap of ____ points

2. Order a fairness audit. Write 1–4.

Monitor after launch ☐

Agree fairness goals ☐

Mitigate and retest ☐

Measure outcomes by group ☐

3. Bonus: Sketch a mind-map of this topic.

Key takeaway: _____

PUZZLE & DEFINE: Fair AI, Fair Future

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Parity

Meaning: _____

Example: _____

Eq. odds

Meaning: _____

Example: _____

Bias audit

Meaning: _____

Example: _____

? QUIZ TIME: Fair AI, Fair Future

Circle the right answer!

Q1. Gender Shades audited...

a) Face analysis

b) Music apps

c) Maps

Q2. Equal positive rates across groups:

a) Accuracy

b) Parity

c) Recall

Q3. ProPublica examined the tool...

a) WhatsApp

b) Search

c) COMPAS

Q4. Bias can enter via historical...

a) Labels

b) Colours

c) Fonts

Q5. Fairness metrics can...

a) Always agree

b) Conflict

c) Be ignored

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Fair AI, Fair Future

Look at the picture. Tick ✓ TRUE or FALSE.

		👍 TRUE	👎 FALSE
	Bias can come from historical labels.		
	Dropping the gender column guarantees fairness.		
	Fairness metrics can conflict.		
	Audits should be repeated after launch.		
	One fairness metric suits every context.		
	Gender Shades found the highest errors for darker-skinned women.		

HOME CONNECT

Test a voice assistant with friends who have different accents. Record the error rate for each accent.

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Fair AI, Fair Future

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. How can a model be biased with no gender column?**

COMPARE **Q2. Demographic parity vs equal opportunity.**

APPLY **Q3. Approval 30% vs 50%: ratio? Passes four-fifths?**

EVALUATE **Q4. Remove all sensitive data to ensure fairness?**

CREATE **Q5. Fairness audit plan for a hiring screen.**

📝 EXAM PREP: Fair AI, Fair Future

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Name three sources of bias.

2. Define equal opportunity.

3. Why can fairness metrics conflict?

4. What is a proxy variable?

PART B • Long answer (5 marks)

5. Audit a hypothetical loan model for fairness and propose mitigations.

TOPIC 12 • MONTH 6 • RESPONSIBLE AI

Ownership in the AI Age

LET'S LEARN

- AI raises new questions of authorship and ownership.
- Copyright, licences and consent govern data and outputs.
- Creators and builders need provenance and accountability.

Who owns what?

Authorship

Licences

Training data

Provenance

KEY TERMS

Copyright
Creator's legal right

Licence Permission to use

Provenance
Record of origin

📖 CONCEPT EXPLAINED: Ownership in the AI Age

💡 In this chapter

Generative AI has unsettled long-standing ideas of authorship and ownership. Who owns an image produced from a prompt? Can creators object to their work being used for training? What happens when an AI copies a voice or a style? These questions affect students, artists, programmers, companies and policymakers. This chapter explains the main principles and live debates, so you can create and build responsibly.

✍️ 1. Authorship and copyright

Copyright protects original creative expression, such as text, images, music and code, and gives authors exclusive rights to copy, adapt and distribute their work for a limited period. India's Copyright Act, 1957, and the laws of most countries, are built around human authors, although some provisions for computer-generated works exist in certain jurisdictions.

In Thaler v. Perlmutter, US courts refused copyright for an image Stephen Thaler said his AI created autonomously; in March 2025 the D.C. Circuit Court of Appeals upheld that human authorship is required. Works in which people make substantial creative choices, selecting, arranging and editing AI material, may still be protected for those human contributions.

✏️ WORKED EXAMPLE

Design student workflow: generates 40 drafts, selects 3, redraws layout by hand, adds original illustrations and typography. Her human creative contributions are the strongest basis for claiming ownership.

🗂️ 2. Training data disputes

Generative models are trained on huge collections of text, images, code and audio, much of it copyrighted. Developers argue that training is a transformative, lawful use; many creators argue it copies their work without consent or payment and produces competing outputs. In 2023 The New York Times sued OpenAI and Microsoft, and Getty Images sued Stability AI, among many other cases.

Rules differ by country: the US relies on fair use, the EU and UK have text-and-data-mining exceptions with opt-out options, and India's position is still developing. Responses include licensing deals with publishers, opt-out registries, datasets of licensed or public-domain content, and calls for transparency about training data.

🤔 **PAUSE & THINK**

Should artists be paid when their work trains a commercial model, even if no output copies their work directly? What would a fair system look like?

📜 3. Licences, consent and personality rights

Licences set what others may do with a work. Creative Commons licences combine conditions such as BY (credit the author), NC (non-commercial only), SA (share alike) and ND (no derivatives). Open-source software licences range from permissive (MIT, Apache) to copyleft (GPL), and AI models often come with their own licences restricting certain uses.

Voices, faces and names raise questions beyond copyright. Courts in India, including the Delhi High Court, have protected celebrities' personality rights against unauthorised AI-generated use of their likeness and voice. For anyone, cloning a voice or creating a deepfake without consent can enable fraud and harassment.

✏️ **WORKED EXAMPLE**

Licence check: a CC-BY-NC photo can be used in a free college newsletter with credit, but not in a paid advertisement without separate permission.

🏷️ 4. Provenance and accountability

As synthetic content spreads, knowing where content came from becomes vital. Provenance techniques include invisible watermarks, signed metadata such as C2PA content credentials, and records of the tools and edits used. Platforms and laws increasingly require labelling of AI-generated media, especially in political and news contexts.

Ownership also brings responsibility. Organisations deploying generative AI are accountable for infringing, defamatory or harmful outputs, and should set policies: approved tools, disclosure of AI use, human review, record-keeping and processes to respond to complaints from creators.

🖐️ **TRY IT**

Read the terms of service of two AI tools you use. Summarise who owns outputs, whether your inputs may be used for training and what uses are prohibited. Draft three rules for using them in your coursework.

✅ Key takeaways

- Copyright protects original human expression; AI-only works are generally not protected.
- Human creative contributions to AI-assisted works can be protected.
- Training-data use is contested in courts; rules differ by country.
- Licences, consent and personality rights govern reuse, voice and likeness.
- Provenance, disclosure and policies support accountable creation.

❓ Quick check: answer in one or two lines.

1. Why was copyright refused in Thaler v. Perlmutter?

2. What does CC-BY-NC allow and forbid?

3. Name two tools or practices that support content provenance.

🔬 CONCEPT DEEP DIVE: Ownership in the AI Age

🔷 Copyright protects original expression by human authors; India's Copyright Act, 1957, like US law, centres human creativity.

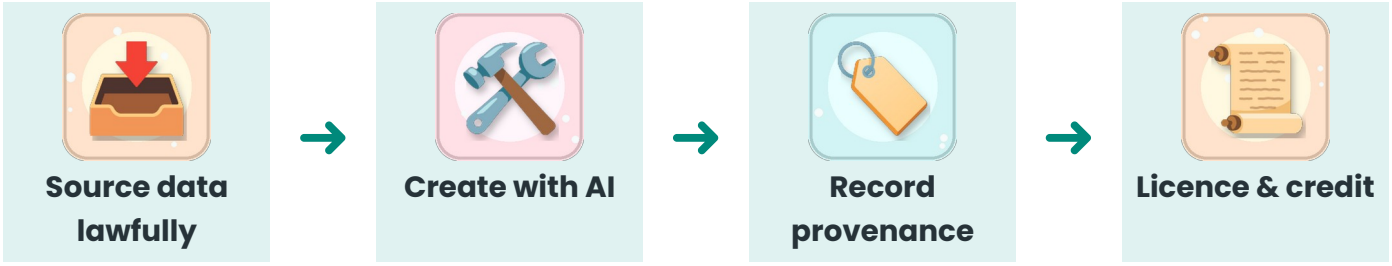
🔷 AI-only outputs may not be protected; works with meaningful human creative choices often can be, and tool terms also matter.

🔷 Training-data disputes are live: The New York Times sued OpenAI and Microsoft in 2023; Getty Images sued Stability AI.

🔷 Licences (Creative Commons, open-source, model licences) set what others may do; consent and personality rights cover voice and likeness.

🔷 Provenance tools such as watermarks and C2PA content credentials record origin; organisations stay accountable for outputs.

🔄 How it works



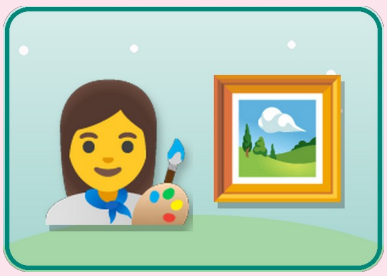
💻 A simple provenance record

```
import hashlib, json
meta = {'creator': 'Riya', 'tool': 'AI image model', 'edits':
'human'}
h = hashlib.sha256(open('poster.png', 'rb').read()).hexdigest()
print(json.dumps({**meta, 'sha256': h[:12]}))
```

✍️ In your own words: explain this topic to a Class 10 student.

AI STORY: Who Owns the Poster?

Characters: *Riya (MDes), client Mr. Desai, college legal adviser Ms. Nair.*



SCENE 1

Riya designed a festival poster series using an AI image tool plus her own illustrations. Mr. Desai's company loved it and asked her to sign over 'full exclusive copyright'.

Pause & predict: What questions should Riya ask first?

SCENE 2

Ms. Nair explained that AI-only elements might not be protected by copyright at all, and the tool's terms set conditions on commercial use. Riya's hand-drawn parts were clearly hers.

Pause & predict: Why does the human contribution matter?

SCENE 3

Mr. Desai then asked her to add a famous singer's AI-cloned voice to the promotional video. Ms. Nair warned that personality rights and consent made this risky and wrong.

Pause & predict: Why is voice cloning without consent a problem?

SCENE 4

Riya's contract listed AI tools used, transferred rights in her original work, and promised no use of anyone's likeness without consent. Provenance data was embedded in every file.

⏸ **Pause & predict:** How does provenance protect both parties?

🌟 The big idea

In the AI age, clear authorship, consent and records protect creators and clients alike.

💬 Discuss with your classmates

1. How would you disclose AI use in your own work?

2. Should AI-assisted work get the same protection as human work?

SCENARIO: Ownership in the AI Age

Read the scenario. Think about the choices made at each step.

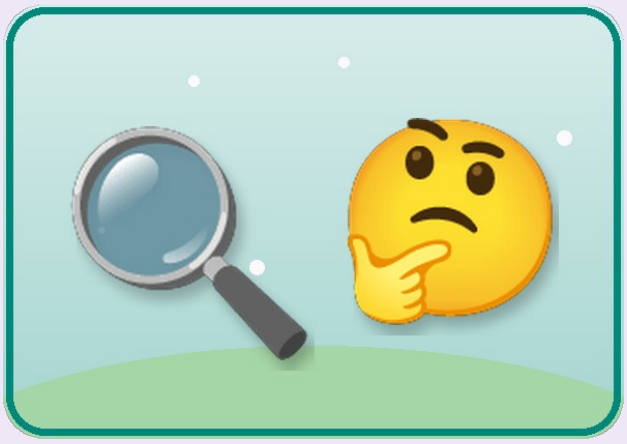
1

Riya, a design student, makes posters with AI tools.

2

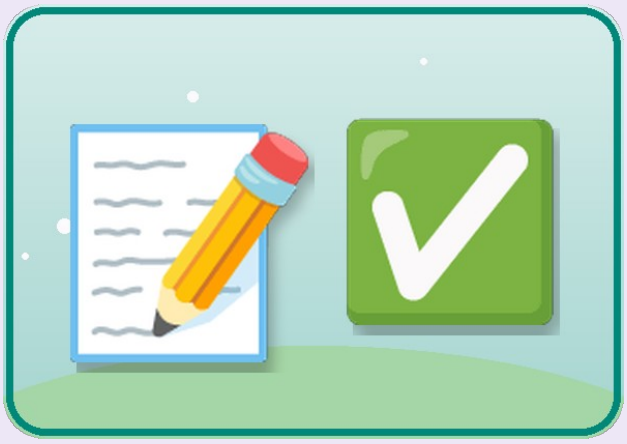
A client wants full ownership of the designs.

3



She checks the tool's terms and her own edits.

4



Her contract states AI use, rights and credits.

💬 **Talk about it:**

What decision was made? What would YOU have done differently?

🖍️ **What happens next? Write or sketch scene 5.**

LOOK AROUND YOU: Ownership questions today

AI art

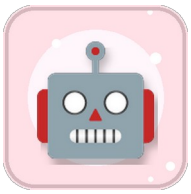
AI music

Book datasets

Generated code

News content

Voice clones



BOLT SAYS:

I can help you create, but credit, consent and copyright still matter!

FUN FACT!

In 2025 a US appeals court ruled a work made by AI alone can't be copyrighted.

Reflect: where have YOU seen this in real life? Note it down.

CASE STUDY: A Recent Entrance to Paradise

CASE FILE

Stephen Thaler sought US copyright for an image, 'A Recent Entrance to Paradise', which he said his AI system created on its own, listing the machine as author. The US Copyright Office refused because there was no human author. A federal court agreed in 2023, and in March 2025 the D.C. Circuit Court of Appeals upheld the decision. Works where people make creative choices with AI help may still be protected; rules differ between countries.

Analyse the case

1. Why do copyright laws focus on human authors?

2. How much human input should be needed for protection?

3. How might Indian law approach a similar case?

My verdict: Was AI used well here? Why?

LAB ACTIVITY: Who Made This?

You need:

Image prints

Label cards

Teams of 3

Steps:

1. Study 6 images: human, AI or mixed.
2. Guess the maker and the likely owner.
3. Check each image's stated licence.
4. Debate who should get credit for mixed work.

THINK & TALK

What makes a person an author?

Record your observations & results:

💻 CODE LAB: Licence Checker

Name: _____ Date: _____

Read the code carefully, then complete the tasks. Blanks are marked ____.

```
 1  rules = {'CC-BY': (True, True), 'CC-BY-NC': (True, False), 'ARR': (False, False)}
 2  ok = lambda lic, paid: rules[lic][0] and (rules[lic][1] or not ____)
 3  print(ok('CC-BY', True), ok('CC-BY-NC', True), ok('CC-BY-NC', False), ok('ARR', False))
```

COMPLETE **T1. Fill the blank.**

PREDICT **T2. Four outputs?**

EXPLAIN **T3. What does ARR mean?**

EXTEND **T4. Add CC0 to the rules.**

WORKSHEET: Ownership in the AI Age (Part A)

Name: _____ Date: _____

1. Match the term to its meaning.

Copyright ●

● Marks content origin

Open licence ●

● Permission to use data or likeness

Watermark ●

● Protects original expression

Consent ●

● Reuse allowed with terms

2. Respectful or problematic? Tick ✓.

	Respectful	Problem
Credit dataset sources		
Clone a singer's voice, no consent		
Use CC-BY images with credit		
Scrape private photos to train		

WORKSHEET: Ownership in the AI Age (Part B)

Name: _____ Date: _____

1. Licence codes. Use the word box.

WORD BOX: BY ✿ NC ✿ SA

CC-____ means give attribution

CC-____ means non-commercial only

CC-____ means share alike

2. Circle responsible ownership practices.

Credit sources

Hide AI use

Get consent

Scrape private data

Read licences

Strip watermarks

PUZZLE & DEFINE: Ownership in the AI Age

Name: _____ Date: _____

1. Odd one out! Circle it and justify your choice.

A				
B				
C				

2. Define it! Write each term in your own words with an example.

Copyright

Meaning: _____

Example: _____

Licence

Meaning: _____

Example: _____

Provenance

Meaning: _____

Example: _____

? QUIZ TIME: Ownership in the AI Age

Circle the right answer!

Q1. US copyright needs an author who is...

a) Human

b) Any AI

c) Any animal

Q2. Permission to reuse a work is a...

a) Theme

b) Licence

c) Password

Q3. Record of where content came from:

a) Random

b) Trash

c) Provenance

Q4. Cloning someone's voice needs...

a) Consent

b) Speed

c) A mic only

Q5. A CC-BY licence requires...

a) Payment

b) Attribution

c) Hiding

MY SCORE: ____ / 5. Shade one star per correct answer.

☆ ☆ ☆ ☆ ☆

✅ TRUE OR FALSE: Ownership in the AI Age

Look at the picture. Tick ✓ TRUE or FALSE.

	👍 TRUE	👎 FALSE
US law requires human authorship for copyright.		
Everything online is free to reuse.		
CC-BY requires attribution.		
Provenance tools record content origin.		
Voice cloning never needs consent.		
Thaler's AI-only image was denied copyright.		

HOME CONNECT

Check the terms of one AI tool you use. Who owns the outputs, and can your inputs be used for training?

🔢 Self-assessment: tick one.

I know it well!

I know some.

I need practice.

CHALLENGE PROBLEMS: Ownership in the AI Age

Name: _____ Date: _____

Higher-order thinking: answer in 2–4 sentences. Use examples!

EXPLAIN **Q1. Why is the use of training data contested?**

COMPARE **Q2. Copyright vs licence.**

APPLY **Q3. CC-BY-NC image in a paid advert: allowed?**

EVALUATE **Q4. Should AI voice clones of actors need consent?**

CREATE **Q5. AI-use clause for a design contract.**

📝 EXAM PREP: Ownership in the AI Age

Name: _____ Date: _____

⏱️ **Time: 30 min** **Max marks: 13** **Answer all questions.**

PART A • Short answers (2 marks each)

1. Why was Thaler's AI image refused copyright?

2. Define a licence.

3. What is provenance?

4. Name two personality-rights concerns.

PART B • Long answer (5 marks)

5. Discuss ownership and accountability questions raised by generative AI, with policy suggestions for India.

UNIT REVIEW: AI CARE

Name: _____ Date: _____

Circle the right answer!

Q1. Bias can enter through past...

a) Labels | b) Colours | c) Fonts

Q2. Approval 30% vs 50% ratio =

a) 0.8 | b) 0.6 | c) 1.6

Q3. ProPublica's 2016 study was on...

a) Cameras | b) Songs | c) COMPAS

Q4. AI-only works in US copyright:

a) Not protected | b) Protected | c) Taxed

Q5. CC-NC forbids...

a) Credit | b) Commercial use | c) Viewing

Q6. Record of content origin:

a) Random | b) Trash | c) Provenance

WORD SEARCH: AI CARE

🔍 **Find these words. They go across → or down ↓.**

PARITY

BIAS AUDIT

COPYRIGHT

LICENCE

PROVENANCE

EQUITY

B	P	L	M	F	W	N	P	W	I
I	L	N	R	F	C	K	R	T	A
A	W	P	B	F	O	C	O	A	J
S	P	R	W	C	P	O	V	L	U
A	A	I	E	R	Y	N	E	I	S
U	R	G	Q	E	R	S	N	C	T
D	I	H	U	D	I	E	A	E	I
I	T	T	I	I	G	N	N	N	C
T	Y	S	T	T	H	T	C	C	E
U	D	M	Y	M	T	R	E	E	U

I finished AI CARE! Teacher's sign: _____

AI GLOSSARY (1/2)

A/B test
Randomised comparison

Active learn
Label the uncertain

AI PM
AI product manager

Bias audit
Test across groups

Churn
Customers leaving

Code review
Peer checks code

Confounder
Hidden common cause

Copyright
Creator's legal right

CV
Rotating validation

Digital twin
Live virtual copy

Distillation
Big teaches small

Eq. odds
Equal error rates

ETL
Extract, transform, load

Forecast
Predict future demand

HITL
Human in the loop

Labelling
Adding ground truth

Licence
Permission to use

MLE
ML engineer

AI GLOSSARY (2/2)

MLOps
Running ML in production

Overfit
Memorises noise

Parity
Equal positive rates

PdM
Predictive maintenance

Portfolio
Proof of work

Provenance
Record of origin

README
Project front page

Regularise
Penalise complexity

RLHF
Learning from preferences

ROI
Return on investment

RPA
Software robots

STAR
Situation-Task-Action-Result

Surrogate
Fast model stand-in

Synthetic
Generated data

Trigger
Event starting a flow

Unit test
Checks one function

Uplift
Extra effect of action

Vulnerability
Exploitable flaw

? OLYMPIAD PRACTICE TEST • Part 1 of 4

Read each question. Circle the right answer. 20 questions in all!

Q1. 5-fold CV on 1,000 rows → rows per fold

a) 200

b) 500

c) 50

Q2. Simple model on complex data...

a) Overfits

b) Underfits

c) Is ideal

Q3. 32-bit to 8-bit weights → smaller by

a) 2×

b) 8×

c) 4×

Q4. Big model teaching a small model:

a) Distillation

b) Cleaning

c) Styling

Q5. Pasting API keys into a public AI is...

a) Safe

b) Risky

c) Faster

? OLYMPIAD PRACTICE TEST • Part 2 of 4

Q6. safe_div(5, 0) should return...

a) 5

b) 0

c) None

Q7. A 156 vs 120 conversions → lift

a) 30%

b) 3%

c) 36%

Q8. A variable causing both X and Y:

a) Target

b) Confounder

c) Metric

Q9. Cost ₹12L, saves ₹2L/month → payback

a) 2 months

b) 12 months

c) 6 months

Q10. Fast stand-in for a slow simulation:

a) Surrogate

b) Full CFD

c) Ruler

? OLYMPIAD PRACTICE TEST • Part 3 of 4

Q11. Wear 0.34, +0.08 a month → months to 1.0 ≈

a) 4

b) 8

c) 12

Q12. Defines what AI product to build:

a) Technician

b) Tester

c) AI PM

Q13. A project's front page is its...

a) README

b) Logo

c) Zip

Q14. 60 applications × 10% reply =

a) 10

b) 6

c) 60

Q15. 1,000 − 50 − 30 =

a) 950

b) 970

c) 920

? OLYMPIAD PRACTICE TEST • Part 4 of 4

Q16. Rows repeated in the data:

a) Duplicates

b) Colours

c) Echoes

Q17. Save 15 min a day for 5 days =

a) 15 min

b) 75 min

c) 150 min

Q18. Gender Shades: highest errors for...

a) Lighter men

b) Lighter women

c) Darker women

Q19. Fairness metrics can...

a) Conflict

b) Always agree

c) Be skipped

Q20. Thaler's case was decided on appeal in...

a) 2016

b) 2025

c) 2009

MY SCORE: I got ____ / 20. Colour a star for every 4 right!

CERTIFICATE

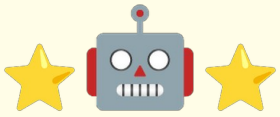
of AI Explorer

This is to say that

_____

of PG 2nd Year finished the 6-month AI journey:

DISCOVER → CONNECT → SOLVE → RISE → CREATE → CARE



Teacher: _____ Date: _____

? ANSWER KEY (for teachers & parents)

1. Learning from Data

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-F 5-T 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Underfitting - High bias; Overfitting - High variance; Regularisation - Penalise complexity; Cross-validation - Rotate folds
Part A 2: Train 60%, test 58%: Underfit; Train 99%, test 70%: Overfit; Straight line on curved data: Underfit; Deep tree on 50 rows: Overfit
Part B 1: 200, 800, 5
Part B 2: 1-Split train/val/test, 2-Train a baseline, 3-Tune with CV, 4-Final test, once
Challenge hints: Q1: It fits noise; high variance on new samples. • Q2: L1 drives weights to zero; L2 shrinks them smoothly. • Q3: 200 per fold; train on 800 each round. • Q4: No: leaks the future; use a time-based split. • Q5: Group k-fold by city; test on unseen cities.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: test; [0.02, 0.02, 0.28]; degree 3 (high test, small gap); matplotlib line plot of both lists.
Exam prep key points: 1: Fitting noise; good on train, poor on test. • 2: Simple models err by bias; complex ones by variance. • 3: Keep class ratios equal in every fold. • 4: L1, L2, dropout, early stopping. • 5: Bias–variance, regularisation, CV types, grouped/time splits, shift checks.

2. Making Machines Smarter

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Transfer - Reuse features; Augmentation - Varied copies; Active learning - Useful labels; Distillation - Small imitates large
Part A 2: Add rotated images: Data fix; Use a pre-trained backbone: Model fix; Relabel noisy examples: Data fix; Prune unneeded weights: Model fix
Part B 1: 4, 25, 2
Part B 2: Circle: Transfer learning, Active learning, Augmentation
Challenge hints: Q1: Features are pre-learned; only adapt top layers. • Q2: Augment: modify real samples; synthetic: generate new ones. • Q3: Often yes: 3 points for big speed and cost savings. • Q4: Yes, if it changes labels, e.g. flipping digits like 6 and 9. • Q5: Train, score uncertainty, label top 50, retrain, repeat.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: True; [0, 2]; probabilities nearest 0.5 are most uncertain; loop: label, retrain, rescore.
Exam prep key points: 1: Reusing knowledge from one task for another. • 2: The model chooses which examples to label. • 3: A reward model from human rankings guides fine-tuning. • 4: Distillation, pruning, quantisation. • 5: Transfer, augmentation, active learning, evaluation, distillation, quantisation.

3. Program with AI

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Unit test - One function; Linting - Style and errors; Code review - Peer reads; Fuzzing - Random inputs
Part A 2: Merge code you have tested: Safe; Paste API keys into prompts: Risky; Use a vetted library: Safe; Run unknown generated commands: Risky
Part B 1: 30, 70, 6
Part B 2: 1-Write spec & tests, 2-Generate code, 3-Run tests & scans, 4-Merge after review
Challenge hints: Q1: They predict plausible text, not verified facts. • Q2: Completion suggests lines; an agent edits, runs and iterates. • Q3: 3 fewer buggy PRs; a 25% relative drop. • Q4: Yes, with disclosure, understanding and own tests. • Q5: Inputs, outputs, edge cases, examples, tests, no extra deps.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: 0; 5.0, None, -3.0; avoids ZeroDivisionError; assert safe_div(5, 0) is None.
Exam prep key points: 1: A language model that suggests or writes code. • 2: Vulnerabilities, logic errors, fake packages. • 3: An unusual input at the boundary of valid use. • 4: Tests define correct behaviour to verify output. • 5: Uses, disclosure, testing, review, security scans, secrets, measurement.

4. Data into Decisions

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-F 2-T 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Correlation - Move together; Causation - One changes the other; Confounder - Hidden cause; Randomisation - Balances groups
Part A 2: Ice-cream sales and drownings rise: Correlation; Randomised fertiliser trial: Causal test; Email openers buy more: Correlation; A/B test of two checkouts: Causal test
Part B 1: 100, 60, 40
Part B 2: A/B - Randomised; Diff-in-diff - Changes over time; Propensity - Matching; DAG - Map causes
Challenge hints: Q1: It predicts risk, not who responds to the offer. • Q2: A/B randomises; observational must adjust for confounders. • Q3: 0.4 × 0.5 × 2000 - 300 = ₹100; positive. • Q4: Not necessarily: festivals, trends; need a control. • Q5: Randomise students, pre-register metric, compare scores.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: 1; 30.0 % lift; z ≈ 2.25, p ≈ 0.02, so likely real; statsmodels proportions_ztest.
Exam prep key points: 1: What will happen vs what an action changes. • 2: A variable affecting both treatment and outcome. • 3: Balances known and unknown confounders. • 4: The extra outcome caused by an action. • 5: Prediction, uplift, A/B test, sample size, expected value, monitoring.

5. AI for Smarter Business	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Forecasting - Stock and staff; Personalisation - Tailor offers; Fraud - Flag payments; Pricing - Adjust to demand **Part A 2:** Product recommendations: Revenue; Chatbot resolving tickets: Cost; Upsell offers: Revenue; Route optimisation: Cost **Part B 1:** 150, 20, 6 **Part B 2:** 1-Choose a valuable use case, 2-Run a pilot, 3-Measure ROI, 4-Scale with monitoring **Challenge hints:** Q1: Shows value; guides decisions to scale or stop. • Q2: Build: control, fit, time; buy: speed, less customisation. • Q3: (10 - 4) ÷ 4 = 150%. • Q4: No: may exploit urgency or groups; set limits. • Q5: Scope, KPIs, human handoff, 4-week test, review. **Case study:** Open answers. Reward clear reasoning and examples. **Code lab:** Blank: actual; 5.4 %; judge against a naive baseline and business tolerance; pred = previous actual. **Exam prep key points:** 1: Revenue, cost, risk, customer experience. • 2: (Gain - cost) ÷ cost. • 3: Test value and risk cheaply. • 4: Loss from wrong or misused models. • 5: Use case, data, KPIs, ROI, risks, pilot, governance.
6. Engineering Solutions	**Quiz:** 1-a 2-b 3-c 4-a 5-b **True/False:** 1-T 2-F 3-T 4-T 5-F 6-T **Odd one out:** A: picture 4; B: picture 3; C: picture 4 **Part A 1:** Predictive - Fix before failure; Twin - Virtual copy; Surrogate - Fast stand-in; Anomaly - Unusual readings **Part A 2:** Repair after breakdown: Reactive; Replace when vibration trend rises: Predictive; Fixed yearly service only: Reactive; Alert from sensor anomaly: Predictive **Part B 1:** 10, 72, 4 **Part B 2:** Circle: Physical limits, Human override, Validate on test rigs **Challenge hints:** Q1: Much faster; explores many designs quickly. • Q2: Preventive: fixed schedule; predictive: based on condition. • Q3: ₹10 lakh per year. • Q4: No: may break physics; add constraints and tests. • Q5: Sensors, data link, model, dashboard, safe controls. **Case study:** Open answers. Reward clear reasoning and examples. **Code lab:** Blank: 1; 0.08 and about 8.2 months; linear wear; np.polyfit(range(4), wear, 1). **Exam prep key points:** 1: Repair based on predicted failure from condition data. • 2: A virtual model updated with live data. • 3: To speed up costly simulations. • 4: Physical limits, human override, validation. • 5: Sensors, data, model, alerts, remaining life, validation, override, ROI.

7. Pick Your AI Path

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: ML engineer - Builds and ships; Research scientist - New methods; AI PM - What to build; MLOps - Runs models
Part A 2: Optimise training code: Build; Write a product roadmap: Lead; Deploy a model API: Build; Align teams on AI policy: Lead
Part B 1: 4.1, 12, 3
Part B 2: 1-Explore role options, 2-Test with projects, 3-Choose a direction, 4-Build skills for the role
Challenge hints: Q1: Real tasks reveal fit better than job titles. • Q2: MLE builds systems; DS analyses and models data. • Q3: 2 + 1.5 + 0.6 = 4.1. • Q4: No: fit, growth, values and stability matter too. • Q5: One project per role, mentor talks, monthly reflection.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: w; MLE (4.3 vs 4.0 and 3.7); with 0.2/0.3/0.5 PM wins (4.3); sort scores with sorted(..., reverse=True).
Exam prep key points: 1: Research, engineering, product, policy. • 2: Deploys, monitors and maintains models. • 3: Internships, projects, informational interviews. • 4: Problem-solving, communication, maths, coding. • 5: Roles, skills, fit scoring, evidence, plan, risks.

8. Portfolio to Profession

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: README - Explains; Demo video - Shows it working; Blog post - Tells the story; Tests & CI - Code quality
Part A 2: Live demo + clear metrics: Strong; Unlabelled notebook only: Weak; Results vs a baseline: Strong; Copied tutorial, no changes: Weak
Part B 1: 2, 15, 3
Part B 2: Coding - Algorithms; System design - Architecture; Behavioural - Teamwork; Take-home - Work sample
Challenge hints: Q1: Projects show applied skill, judgement and results. • Q2: Kaggle: modelling; product: data, deployment, users. • Q3: 40 × 0.06 = 2.4, so about 2 replies. • Q4: No: misrepresentation; state your role clearly. • Q5: Situation, task, action fixing leakage, improved result.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: offer; 6.0 and 1.0; with referrals 15.0 and 2.5; matplotlib bar chart of each stage.
Exam prep key points: 1: Real problem, end-to-end, measured result, clear docs. • 2: Problem, data, method, results, how to run. • 3: Situation, Task, Action, Result. • 4: Coding, ML system design, behavioural, take-home. • 5: Projects, documentation, presence, networking, interview prep, metrics.

9. AI Data Studio

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Deduplication - Remove repeats; Imputation - Fill missing; Normalisation - One scale; Validation - Reject impossible
Part A 2: Age = 34: Keep; Age = -5: Fix; Pincode 629001: Keep; Same order listed twice: Fix
Part B 1: 920, 80, 5
Part B 2: 1-Collect with consent, 2-Clean & deduplicate, 3-Label with guidelines, 4-Validate & version
Challenge hints: Q1: Model learns wrong patterns; caps achievable accuracy. • Q2: Real: true patterns, privacy limits; synthetic: scalable, may miss reality. • Q3: 1000 - 50 - 30 = 920. • Q4: No: clean, relevant data often beats noisy volume. • Q5: Classes, examples, edge cases, 'unsure' option, gold checks.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: <; 1 1 1; drop_duplicates, fillna(median), keep price > 0; validate() asserts each rule.
Exam prep key points: 1: Improving data, not only models, to improve results. • 2: How consistently labellers assign the same labels. • 3: Deduplicate, impute, validate, normalise. • 4: Documentation of data origin, use and limits. • 5: Collection, consent, cleaning, guidelines, agreement, versioning, datasheet.

10. Smart Automation Tools

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Trigger - Starts; Rule - If-then; AI step - Reads or drafts; Human approval - Checks risky actions
Part A 2: Copy form data into a sheet: Automate; Approve a large refund: Human; Send meeting reminders: Automate; Handle an upset customer: Human
Part B 1: 75, 15, 260
Part B 2: Circle: Log every run, Approve big actions, Test before going live
Challenge hints: Q1: Rules are reliable for routine; AI handles messy input. • Q2: RPA mimics clicks, fragile; APIs are stable but need access. • Q3: 15 min × 5 days = 75 minutes. • Q4: Only small, low-risk ones within strict limits. • Q5: Form trigger, sheet, AI summary, review, email, log.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: 'faq'; human auto ai_draft; high-value cases always need a person; append (amount, kind, result) to a log list.
Exam prep key points: 1: An event that starts an automation. • 2: Software robots that mimic user actions. • 3: A person reviews or approves key steps. • 4: Audit, debugging and accountability. • 5: Process map, triggers, rules, AI steps, approvals, logging, metrics.

11. Fair AI, Fair Future

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Parity - Equal positive rates; Equal opportunity - Equal TPR; Calibration - Same meaning; Individual - Similar people
Part A 2: Few rural samples in training: Data; Tool used for a new purpose: Use; Past labels reflect discrimination: Data; Staff ignore AI for some groups: Use
Part B 1: 0.6, 40, 20
Part B 2: 1-Agree fairness goals, 2-Measure outcomes by group, 3-Mitigate and retest, 4-Monitor after launch
Challenge hints: Q1: Proxies like name, area or job history encode it. • Q2: Parity: equal approvals; opportunity: equal true-positive rates. • Q3: 0.6; fails the 0.8 rule of thumb. • Q4: No: proxies remain and you can't audit; measure instead. • Q5: Groups, metrics, data checks, thresholds, reviews, appeals.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: max; 0.6 flag; a gap to investigate, not automatic proof of bias; compute TPR per group with true labels.
Exam prep key points: 1: Historical, representation, measurement, deployment. • 2: Equal true-positive rates across groups. • 3: Different base rates make them incompatible. • 4: A feature correlated with a sensitive attribute. • 5: Groups, metrics, data analysis, trade-offs, mitigation, governance, monitoring.

12. Ownership in the AI Age

Quiz: 1-a 2-b 3-c 4-a 5-b
True/False: 1-T 2-F 3-T 4-T 5-F 6-T
Odd one out: A: picture 4; B: picture 3; C: picture 4
Part A 1: Copyright - Protects expression; Open licence - Reuse with terms; Watermark - Marks origin; Consent - Permission
Part A 2: Credit dataset sources: Respectful; Clone a singer's voice, no consent: Problem; Use CC-BY images with credit: Respectful; Scrape private photos to train: Problem
Part B 1: BY, NC, SA
Part B 2: Circle: Credit sources, Get consent, Read licences
Challenge hints: Q1: Creators' works used without consent or payment. • Q2: Copyright is the right; a licence grants permissions. • Q3: No: NC forbids commercial use without permission. • Q4: Yes: likeness and personality rights, fraud risk. • Q5: Disclose tools, human edits, rights transferred, credits.
Case study: Open answers. Reward clear reasoning and examples.
Code lab: Blank: paid; True False True False; all rights reserved, so permission is needed; 'CC0': (True, True).
Exam prep key points: 1: It had no human author. • 2: Permission to use a work on stated terms. • 3: A record of content origin and edits. • 4: Voice clones, deepfakes, misuse of likeness. • 5: Authorship, training data, licences, consent, provenance, liability, policy.

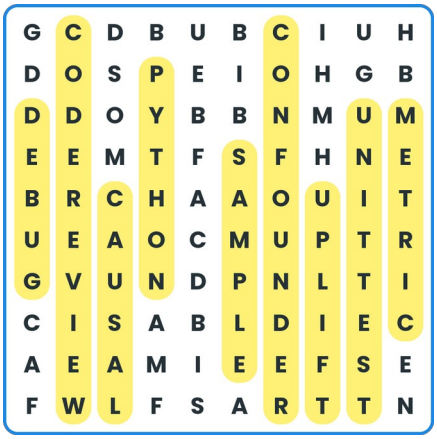
Unit reviews & practice test

AI DISCOVER: 1-a 2-b 3-c 4-a 5-b 6-c • AI CONNECT: 1-a 2-b 3-c 4-a 5-b 6-c • AI SOLVE: 1-a 2-b 3-c 4-a 5-b 6-c • AI RISE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CREATE: 1-a 2-b 3-c 4-a 5-b 6-c • AI CARE: 1-a 2-b 3-c 4-a 5-b 6-c

Olympiad Practice Test: 1-a 2-b 3-c 4-a 5-b 6-c 7-a 8-b 9-c 10-a 11-b 12-c 13-a 14-b 15-c 16-a 17-b 18-c 19-a 20-b

WORD SEARCH ANSWERS

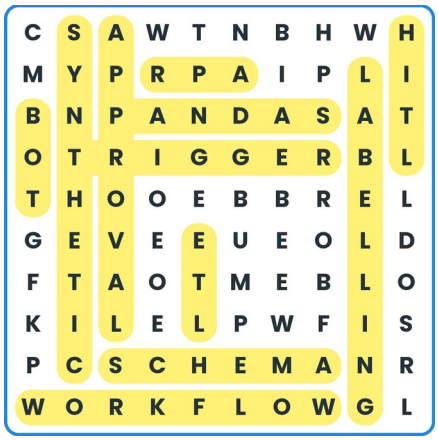
AI DISCOVER



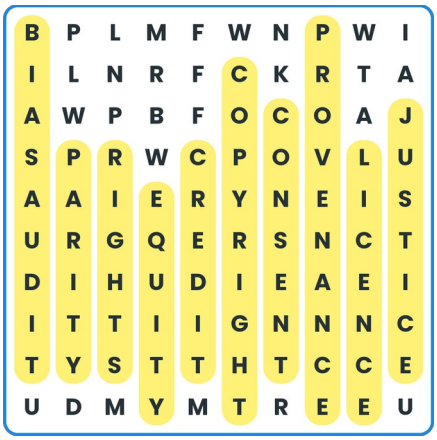
AI CONNECT

AI SOLVE

AI RISE



AI CREATE

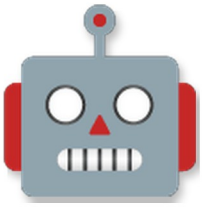


AI CARE

MAMMA ADA CENTRE

GAIO • Global AI Olympiad

From Data to Decisions, From Ideas to Impact



About this book

A postgraduate AI book for PG 2nd Year students: concepts, data labs, stories & career prep.

- 📖 Concept Explained: in-depth chapters
- 📊 Data, decisions & business labs
- 📚 AI Stories & real-world case studies
- 💼 Career paths & portfolio building
- ⚖️ Fair AI & ownership in the AI age

AI Basics | **Communicate with AI** | **AI Applications**

AI & Career | **AI Tools** | **Responsible AI**

Think | Explore | Create | A Better Tomorrow

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